Hierarchical and Multivariate Data

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Part I

Introductory Material

Chapter 1
Related References

- Aerts, M., Geys, H., Molenberghs, G., and Ryan, L.M. (2002). Topics in Modelling of Clustered Data. London: CRC/ Chapman & Hall.

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Chapter 2
Introduction

- Levels of complexity
- Multivariate analysis
- Longitudinal data
- Clustered data
- Spatially organized data

2.1 Levels of Complexity

2.1.1 One-Sample Problem

- The simplest statistical analysis is concerned with a single outcome variable, recorded for a sample of an homogeneous population.

$$Y_i, \qquad i = 1, \ldots, N$$

- Standard procedures include:
  - ▷ the computation of means or medians (location parameters)
  - ▷ the computation of standard errors or interquartile ranges (dispersion parameters).
  - ▷ For example, the height of a number of human subjects might be recorded.

2.1.2 Two-Sample Problem

- A first level of complexity arises when a variable is recorded for a sample out of two subgroups (subpopulations) of a larger population (treated and untreated patients, two species, boys and girls): the two-sample problems:

$$Y_{gi}, \qquad g = 1, 2, \qquad i = 1, \ldots, N$$

- A question of interest is whether the means are different in the two populations.

- The outcome variable might still be height, but we would have an explanatory variable: treatment allocation, or sex. For example, the height of boys can be compared to the height of girls.

- The outcome variable is often called *dependent variable.*

- The predictor is often called *covariate* or *independent variable*.
- The statistical tools for this data setting include Analysis of Variance (ANOVA), t test, Wilcoxon test.
- In the previous situation, the dependent variable had only two levels: a binary or dichotomous variable. This is the simplest case.

2.1.3 Regression

- Alternatively, the predictor itself could be a variable with several levels.
- Examples: dose administered in a clinical trial; one of several species of a plant; race,. . .). In addition, it could have an infinite number of levels, just as is the case with height.
- For example, a baseline height at 7 years of age can be compared to the height at 10 years.
- This leads to a family of models that is frequently referred to as *regression models*.

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \ldots, N$$

- When the dependent variable Y_i is continuous (height) one often uses linear regression.

- The independent variable x_i can be continuous, binary, categorical, or discrete.

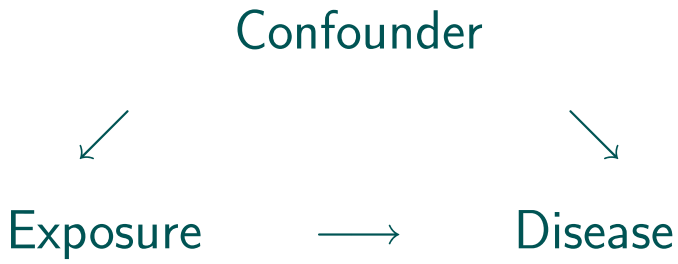
- The choice of the statistical analysis method is driven by the outcome or dependent variables, rather than by the predictor variables.

- Should the dependent variable be binary (diseased/non diseased; death/alive,...), then one would choose logistic regression rather than linear regression.

2.1.4 Several Predictors

- Up to now, there was only one predictor variable. However, this need not be the case.
- For instance, both treatment allocation and sex of the human subject might be of interest.
- Most of the well-known methods extend easily.
- One-way ANOVA extends to two-way or even multi-way ANOVA.
- Simple linear regression or single linear regression extends to **multiple** regression.
- Most other techniques, such as logistic regression are easily extended to encompass multiple covariates.

- It has to be noted that, while simple in theory, methods for multiple covariates require great care.
  - ▷ Indeed, issues such as collinearity arise only for multiple covariate models.
  - ▷ Often, not all predictors are on equal footing.
  - ▷ Often, the relation between an EXPOSURE and a DISEASE is of interest, while another variable is merely a CONFOUNDER.



- Thus, model building and interpretation of (regression) coefficients require both expertise as well as subject matter knowledge.

2.1.5 Several Outcome Variables

- The final extension is concerned with the fact that sometimes **several dependent variables** are recorded and studied simultaneously.

- In statistics, this is commonly termed **multivariate analysis**, in contrast to **multiple . . . .**

- The medical and epidemiological literature uses "multivariate" when the statistician would talk about "multiple". Danger for confusion!

- In conclusion:

  multiple refers to several independent variables;

  multivariate refers to several dependent variables.

2.2 Multivariate Analysis

- Multivariate analysis refers to a set of techniques which allow for the presence of **more than one outcome variable**.

- For example, height and weight might be recorded simultaneously for a group of boys and girls. Arguably, sex will influence height as well as weight. At the same time, height and weight are likely to be **correlated** or **associated**.

- Remarks:
  - ▷ association refers to the concept of dependence between two or more variables.
  - ▷ In contrast, correlation refers to a family of measures that can be computed to capture association (Pearson & Spearman correlation).
  - ▷ Especially for categorical data, a million measures of association have been proposed as alternatives to the correlation (including the odds ratio, concordance, Kendall's τ, the κ coefficient,...).

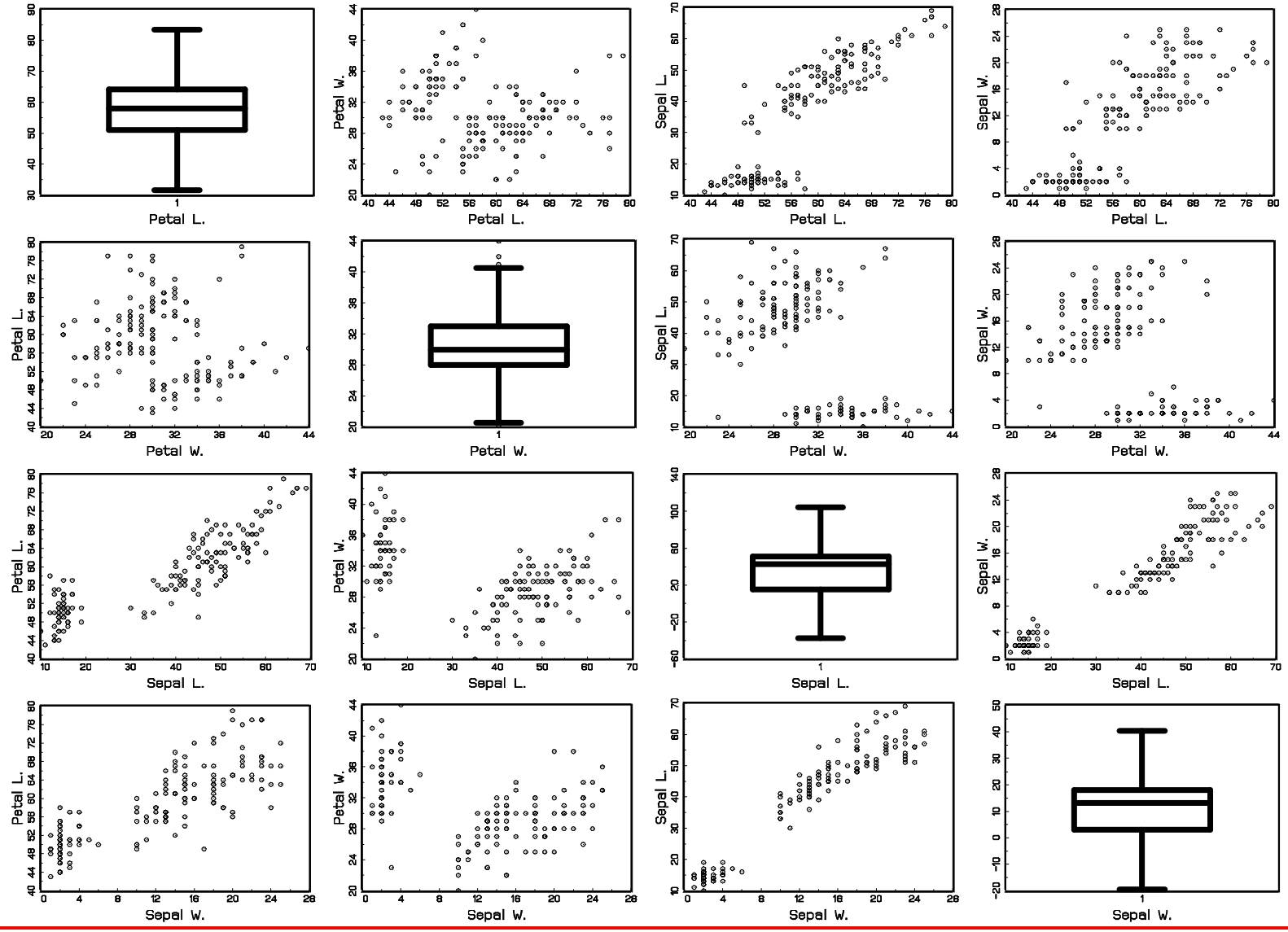
2.3 Multivariate Data: The Iris Data Set

- A classical example is provided by the well known Iris Data Set:
  - ▷ Outcome variables:
    - ∗ Petal Length
    - ∗ Petal Width
    - ∗ Sepal Length
    - ∗ Sepal Width
  - ▷ Sample size: $N = 150$
  - ▷ Three species:
    - ∗ setosa (50)
    - ∗ versicolor (50)
    - ∗ virginica (50)
  - ▷ Summary: 4 continuous outcome variables, one 3-level categorical covariate

2.3.1 Iris Data Set

Obs.	Species	Sepal		Petal	
		Length	Width	Length	Width
i	X_i	Y_{i1}	Y_{i2}	Y_{i3}	Y_{i4}
1	1	50	33	14	2
2	1	46	34	14	3
3	1	46	36	10	2
4	1	51	33	17	5
5	1	55	35	13	2
⋮	⋮	⋮	⋮	⋮	⋮
50	1	53	37	15	2
51	2	65	28	46	15
52	2	62	22	45	15
53	2	59	32	48	18
54	2	61	30	46	14
55	2	60	27	51	16
⋮	⋮	⋮	⋮	⋮	⋮
100	2	67	30	50	17
101	3	64	28	56	22
102	3	67	31	56	24
103	3	63	28	51	15
104	3	69	31	51	23
105	3	65	30	52	20
⋮	⋮	⋮	⋮	⋮	⋮
150	3	63	33	60	25

2.3.2 Graphical Representation



2.4 Questions of Interest

- A number of questions can be asked:

  1. Are there differences between the species?

  2. Given a species, do we need all four variables to have an idea about the characteristics?

  3. What is the relation between the Sepal variables on the one hand, and the Petal variables on the other? Formulated differently, is the sepal shape and/or size related to the petal shape and/or size?

- Let us take a closer look at these questions.
  - ▷ For a univariate sample, a difference in the mean parameter is usually investigated. One can test for equality of the mean between the different species. Here, differences can be spread out over 4 outcomes:
    - ∗ Is only one variable responsible for a difference?
    - ∗ Is the difference spread out over all four?

- For one outcome, the information contained in the data for a species is summarized by 1 mean and 1 variance. Here, we have 4 means, 4 variances, and 6 correlations ! (Think of 10 outcomes...). Thus, in order to gain insight, a data reduction is needed. This is the topic of **principal components** analysis.

- Can I reduce the number of variables and/or organize it differently, while taking the covariate structure (three species) into account? This is one of the questions behind **factor analysis**.

- Suppose I were to classify a new observation. If only one outcome is responsible for the difference, I could confine attention to the value of that one to classify the new observation. However, if it is spread out over all four it is less clear how to do it. Might it be possible to define a **new variable**, based on the four others which would make the job easier? This is the subject of **discriminant analysis**.

- ▷ The four variables have been divided into two groups of two variables. In other words, we are interested in the correlation between two *sets* of variables, instead of simply between two variables. Not only are there several correlations between the sets (four in this case), but also the correlation between the variables within each set (one correlation within each set) is present. Rather than having a single correlation, we have six numbers. To answer the question what the correlation between the two groups is will be answered in the chapter on **canonical correlation**.

- Principal component analysis & discriminant analysis: treated in this course.

2.5 General Multivariate Setting

- In general, one might have:
  - ▷ a set of dependent variables, some of which are continuous, discrete, categorical, binary,...
  - ▷ a set of independent variables, some of which are continuous, discrete, categorical, binary,...
- The most general setting is very hard to study. During the last century, a multitude of sub-problems of the general problem have been studied.

2.6 Other Correlated Data Settings

- The multivariate setting is but one of the correlated data settings.
- In all of the following situations, there are "many" outcome variables:
  - ▷ repeated measures
  - ▷ longitudinal data
  - ▷ spatial data
  - ▷ clustered data

  Are they related to multivariate analysis? Or special cases?
- The answer is: Yes and No!

- In all of the examples it is not true that an independent and identically distributed sample is recorded. Thus, they are different from univariate analysis.

- However, in all of the examples one usually records only one outcome variable. Thus, they are different from multivariate analysis.

- In conclusion, aspects of both are combined.

2.6.1 Example Situations

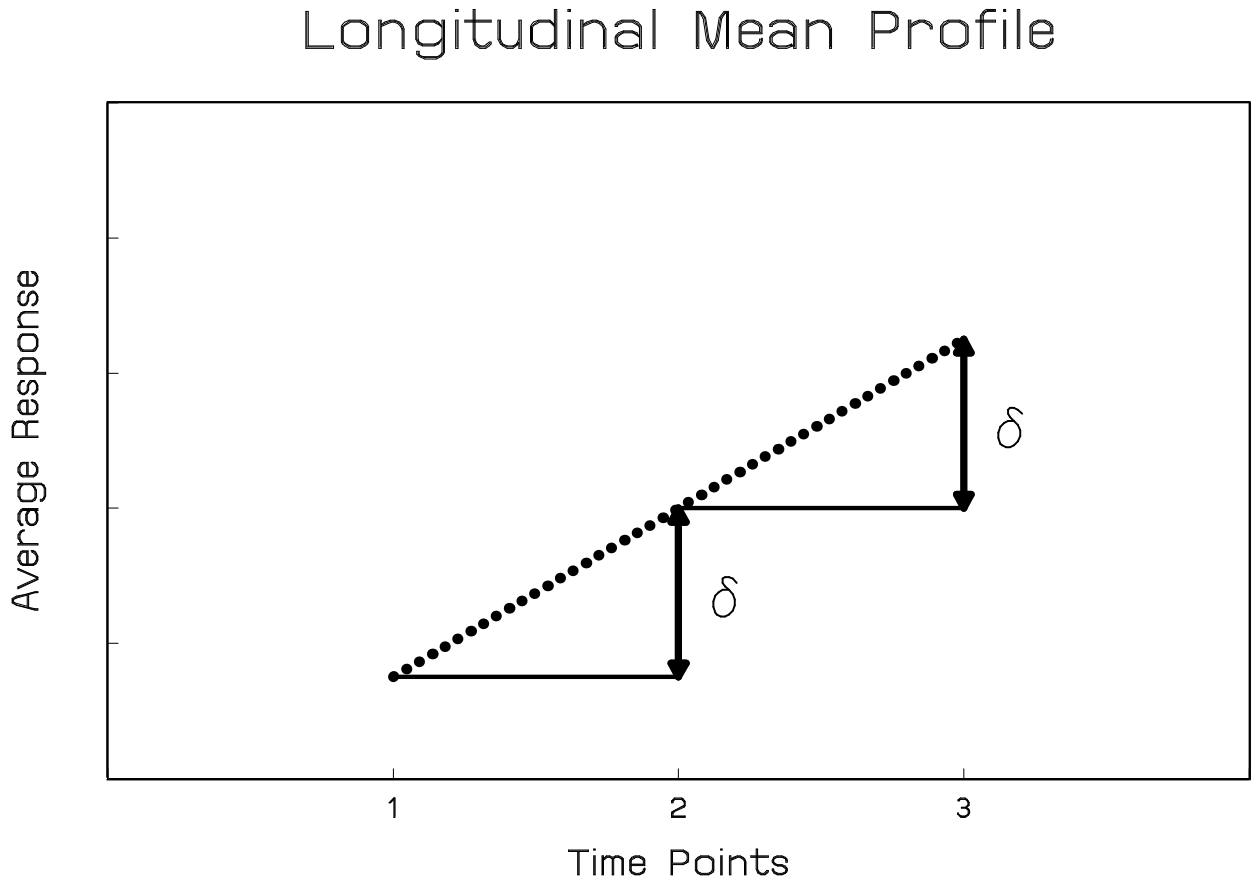
1	For each subject in a study, height is recorded.
2	For each family in a study, the height of all sibs is recorded.
3	For each subject in a study, height and weight are recorded.

- Example 1 is clearly univariate and example 3 is clearly multivariate.
- Example 2 is ambiguous:
  - ▷ There is only one outcome variable: height.
  - ▷ Each unit (family) yields several values.

- Thus, a multivariate study records many variables for *simple* subjects, such as individuals. Repeated measures, longitudinal data, etc. usually involve compound units (families, litters) or elaborate sampling mechanism (extending over time: longitudinal; or extending over space: spatial).

- Although height and weight are typically correlated when they are measured for the same subject, they have their own specific units (m and kg), implying that there is probably no simple relationship between their means: it is appropriate to consider one mean parameter for height and one mean parameter for weight.

- In contrast, when height is measured at months 1, 2 and 3, a connection between the corresponding means might be envisaged. Probably, the average at month 1, μ say, is incremented with an amount δ at month 2, and with an amount 2δ at month 3.



- While there are usually as many mean parameters as variables in a multivariate study, there may be more structure (and thus less parameters), in any of the other data settings.

- Similarly, it might be expected that the correlation between sepal length and sepal width in the iris data set is different from the correlation between sepal length and petal length.

- On the other hand, the correlation between the weight of littermates in a single litter is probably the same. Thus, in a multivariate model, there are typically as many correlation parameters as there are pairs of measurements, while in a repeated measures setting it might be just one! (or at least only a few).

- In conclusion, methods for multivariate analysis will be considerably different from methods for repeated measures, longitudinal data, clustered data, and spatial data.

- We will focus on:

  - ▷ multivariate methods
  - ▷ longitudinal data (repeated measures)
  - ▷ clustered data
  - ▷ spatial data

2.7 Longitudinal Data: Growth Curves

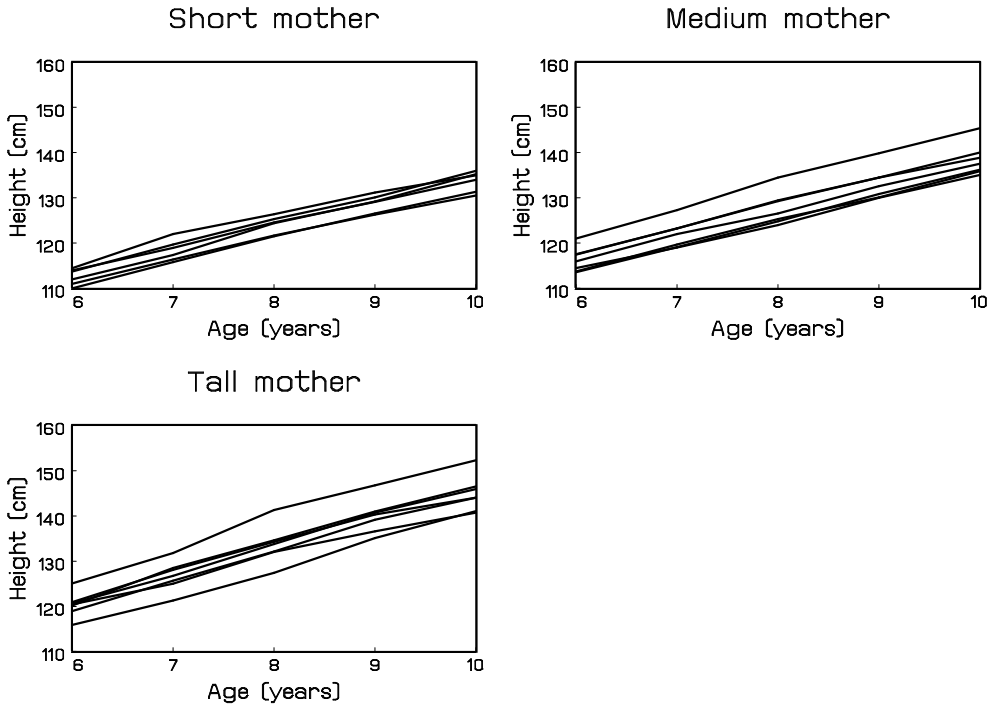
- Taken from Goldstein (1979)
- Research question:

> Is growth related to height of mother ?

- The height of 20 schoolgirls, with small, medium, or tall mothers, was measured over a 4-year period:

	Mothers height	Children numbers
Small mothers	< 155 cm	$1 \longrightarrow 6$
Medium mothers	$[155\text{cm};\ 164\text{cm}]$	$7 \longrightarrow 13$
Tall mothers	> 164 cm	$14 \longrightarrow 20$

- Individual profiles:



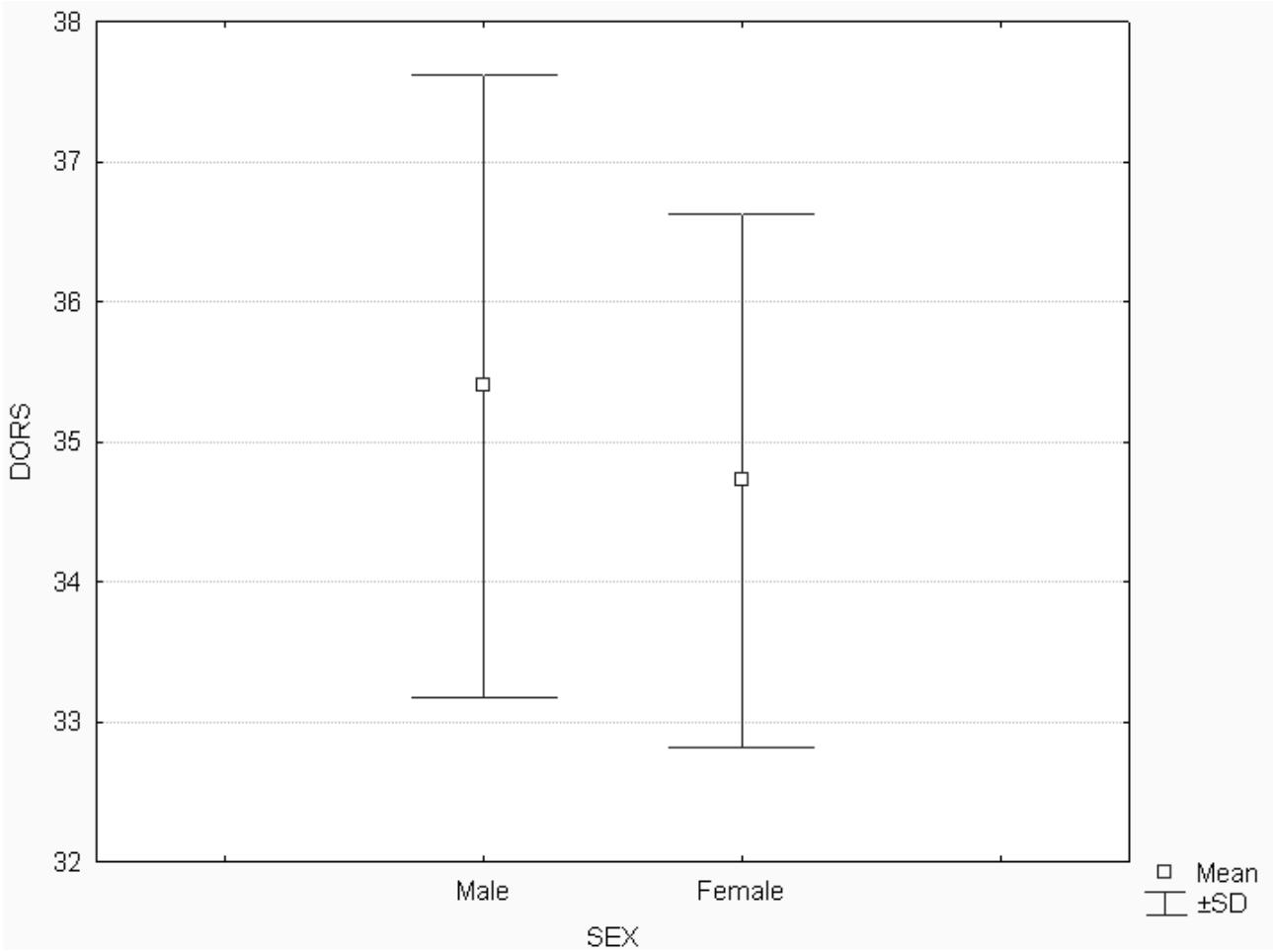
- Remarks:
  - ▷ Almost perfect linear relation between Age and Height
  - ▷ Much variability between girls
  - ▷ Little variability within girls
  - ▷ Fixed number of measurements per subject
  - ▷ Measurements taken at fixed time points

2.8 Clustered Data: Lizard Example

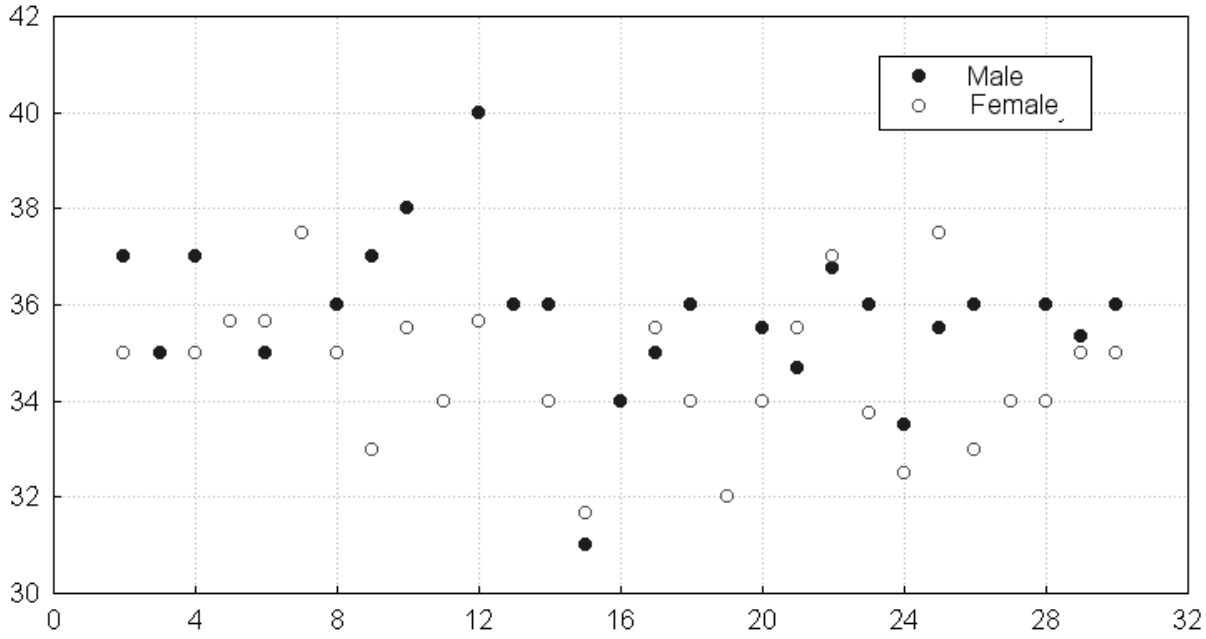
- Data on 102 lizards
- Response of interest: Number of dorsal shells
- Research question:

> Is number of dorsal shells gender-related ?

- Graphically:



- A typical aspect of the data is that some animals have the same mother.
- We have 102 lizards from 30 mothers
- Graphically:



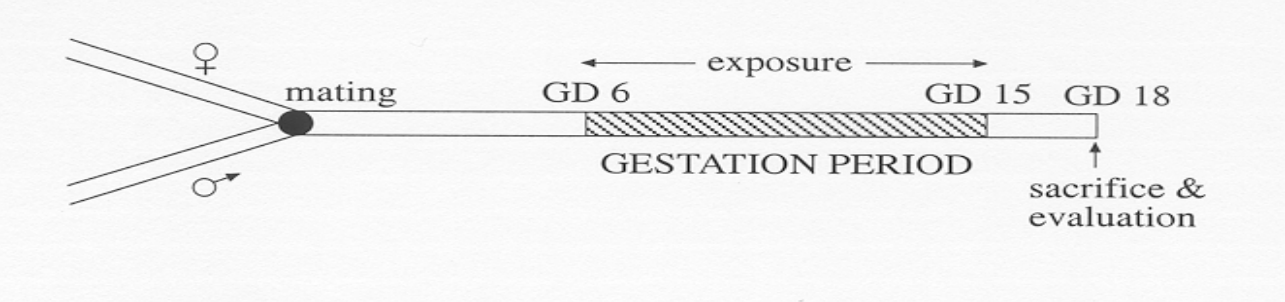
- Observations:
  - ▷ Much between-mother variability
  - ▷ Often, males (considerably) higher than females
  - ▷ In cases where females higher than males, small differences
- This is an example of **clustered data**: observations are clustered within mothers
- It is to be expected that measurements within mothers are more alike than measurements from different mothers.
- We expect correlated observations within mothers and independent observations between mothers.
- How to correct for differences between mothers ?

2.9 Correlated Data: Developmental Toxicity Studies

- Conducted in selected animal species, for ethical reasons
- There are three standard designs:
  - ▷ **Segment I:** Fertility studies, to assess male and female reproductive ability. Exposure of males (60 days) and females (14 days) of an animal species, prior to mating.
  - ▷ **Segment II:** Teratology studies: historically designed to assess malformations ("tera" is Greek for monster).
  - ▷ **Segment III:** Effects later in gestation (from 15th day of gestation through lactation).

2.10 Segment II (Teratology) Studies

- Time line for a typical Segment II study:



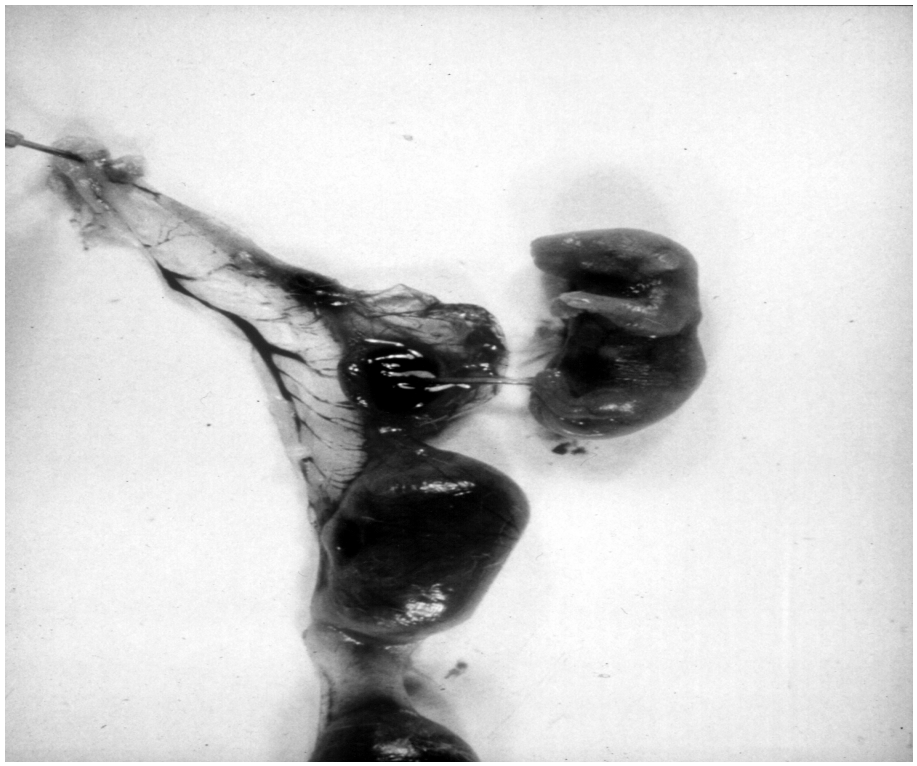
- exposure of timed-pregnant animals (rats, mice, rabbits) during major organogenesis (days 6–15 for mice and rats)

- exposure through clinical or environmental routes most relevant for human exposure

- each group: 20 to 30 dams
- offspring per litter: 2 to 17 foetuses
- control group and 3 or 4 dose groups
  dose is *cluster level covariate*
- dams sacrificed just prior to normal delivery
- uterus removed and thoroughly examined

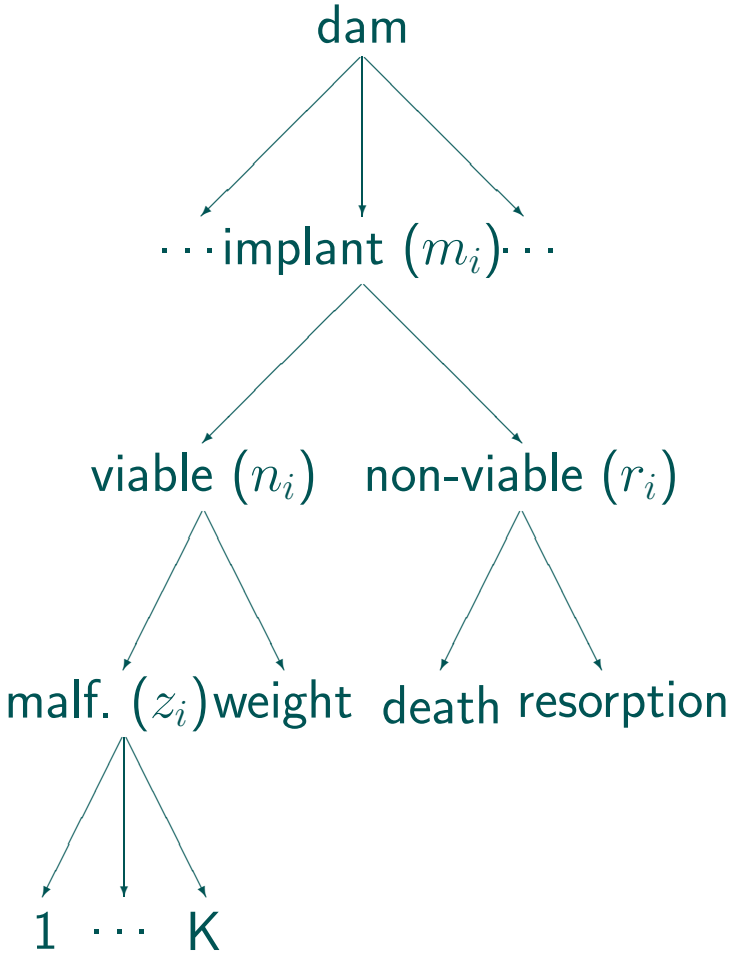


- Implanted foetuses:
  - ▷ death/resorbed
  - ▷ viable



- For the viables, record:
  - ▷ **malformations:**
    - ∗ **visceral:** affect the organs (heart, brain, lungs)
    - ∗ **skeletal:** might include missing or malformed bones
    - ∗ **external:** visible by naked eye (e.g., in the limbs)
    - ∗ collapsed: any
  - ▷ **weight**

2.10.1 Data Structure of Developmental Toxicity Studies



2.10.2 Goals

- Describe dose-response relationship
- Test for dose effect
- Estimation of dose effect
- Determine a benchmark dose:

$$\text{BD} = \text{effective dose} = \text{virtually safe dose}$$

- Use a fully quantitative (modeling-based) approach

2.10.3 Issues

- Account for **clustering** (litter effect)
- Analysis of **weight** outcome
- Univariate versus multivariate approach:
  - ▷ **individual** malformation types
  - ▷ trivariate model
  - ▷ **collapsed** outcome
- **Fetus** *versus* dam to mimick processes in humans
- Method of **estimation**?
- Family of **models**?

2.11 The NTP Studies

Developmental Toxicity Studies

- Research Triangle Institute US National Toxicology Program
- Segment II studies
- The effect in mice of 3 chemicals:
  - ▷ DEHP: di(2-ethyhexyl)-phtalate
  - ▷ EG: ethylene glycol
  - ▷ DYME: diethylene glycol dimethyl ether

2.11.1 Ethylene Glycol

- Ethylene glycol (EG) is also called 1,2-ethanediol
- Chemical formula $HOCH_2CH_2OH$.
- A high-volume industrial chemical with many applications.
- EG is used:
  - ▷ as an antifreeze in cooling and heating systems
  - ▷ as one of the components of hydraulic brake fluids
  - ▷ as an ingredient of electrolytic condensers
  - ▷ as a solvent in the paint and plastics industries
  - ▷ in the formulation of several types of inks
  - ▷ as a softening agent for cellophane

- ▷ as a stabilizer for soybean foam used to extinguish oil and gasoline fires
- ▷ in the synthesis of various chemical products, such as plasticizers, synthetic fibers and waxes

- EG represents little hazard to human health in normal industrial handling, except possibly when used as an aerosol or at elevated temperatures.

- EG at ambient temperatures has a low vapor pressure and is not very irritating to the eyes or skin.

- Accidental or intentional ingestion of antifreeze products, of which approximately 95% is EG, is toxic and may result in death

2.11.2 EG Study in Mice

- Timed-pregnant CD-1 mice were dosed by gavage with EG in distilled water.
- Dosing occurred during the period of organogenesis and structural development of the foetuses (gestational days 8 through 15).
- Doses 0, 750, 1500, and 3000 mg/kg/day.
- Clusters consisting of 10–15 implants occur frequently.

2.11.3 Summary Data by NTP Studies in Mice

		# dams, ≥ 1			Litter Size	Malformations		
Exposure	Dose	impl.	viab.	Live	(mean)	Ext.	Visc.	Skel.
EG	0	25	25	297	11.9	0.0	0.0	0.3
	750	24	24	276	11.5	1.1	0.0	8.7
	1500	23	22	229	10.4	1.7	0.9	36.7
	3000	23	23	226	9.8	7.1	4.0	55.8
DEHP	0	30	30	330	13.2	0.0	1.5	1.2
	44	26	26	288	11.1	1.0	0.4	0.4
	91	26	26	277	10.7	5.4	7.2	4.3
	191	24	17	137	8.1	17.5	15.3	18.3
	292	25	9	50	5.6	54.0	50.0	48.0
DYME	0	21	21	282	13.4	0.0	0.0	0.0
	62.5	20	20	225	11.3	0.0	0.0	0.0
	125	24	24	290	12.1	1.0	0.0	1.0
	250	23	23	261	11.3	2.7	0.1	20.0
	500	22	22	141	6.1	66.0	19.9	79.4
TGDM	0	27	26	319	12.3	0.003	0.000	0.000
	250	26	26	275	10.6	0.000	0.000	0.000
	500	26	24	262	10.9	0.004	0.000	0.004
	1000	28	26	286	11.0	0.042	0.003	0.073
THEO	0	26	25	296	11.8	0.003	0.000	0.000
	282	26	25	278	11.1	0.007	0.000	0.000
	372	33	29	300	10.3	0.017	0.003	0.003
	396	23	17	197	11.6	0.020	0.005	0.000

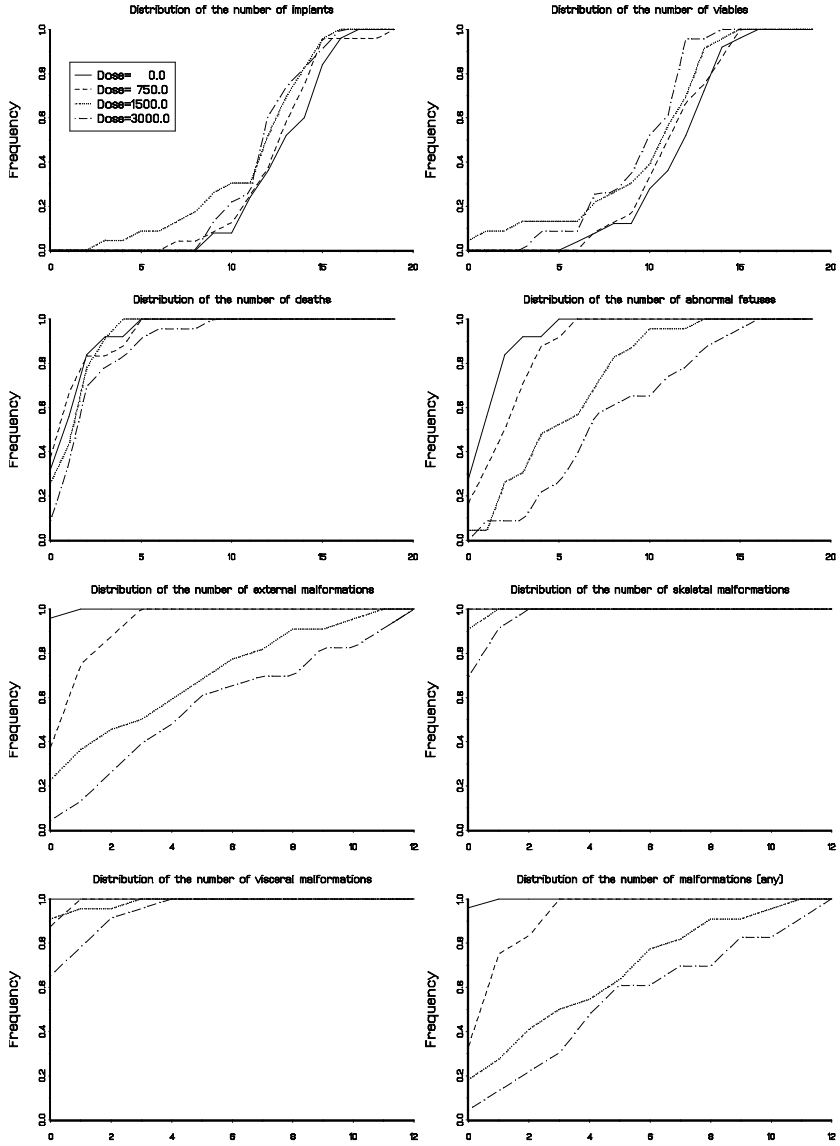
2.11.4 Focus on the EG Study

Dose	Dams	Live	Litter Size		Malformation								Weight	
			Mean	SD	Ext.		Visc.		Skel.		Coll.		Mean	SD
					Nr.	%	Nr.	%	Nr.	%	Nr.	%		
0.00	25	297	11.9	2.45	0	0.0	0	0.0	1	0.0	1	0.3	0.97	0.09
0.75	24	276	11.5	2.38	3	1.0	0	0.0	24	8.7	26	9.4	0.88	0.10
1.50	22	229	10.4	3.46	4	1.7	2	0.8	84	36.7	89	38.9	0.76	0.10
3.00	23	226	9.8	2.69	16	7.1	9	3.9	126	55.8	129	57.1	0.70	0.12

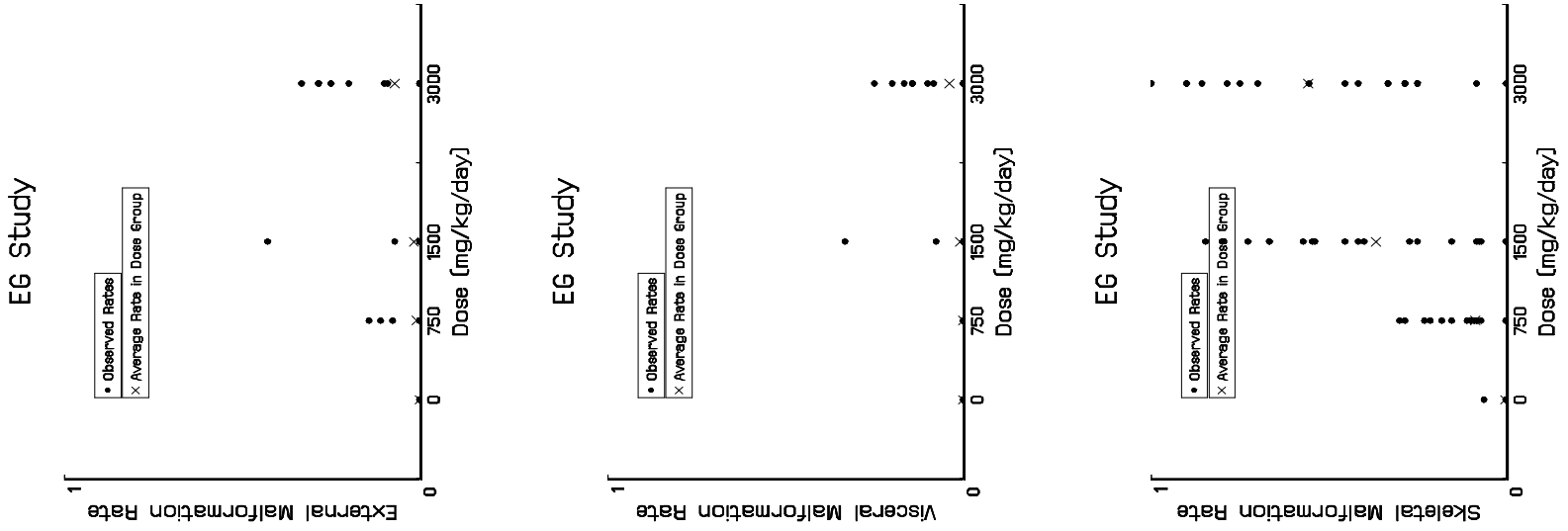
2.11.5 NTP Data in Mice: Number of Implants.

Number of implants	EG	TGDM	DYME	DEHP	THEO
1	0	1	0	1	2
2	0	0	0	1	2
3	1	1	1	0	1
4	0	3	1	2	1
5	1	1	0	0	0
6	0	1	0	2	3
7	2	2	2	0	0
8	1	0	2	4	0
9	8	2	2	5	6
10	4	7	7	7	4
11	8	21	10	18	14
12	19	26	15	21	17
13	16	19	27	26	21
14	11	10	19	21	19
15	16	8	9	10	12
16	6	4	10	8	3
17	1	1	5	2	2
18	0	0	0	2	1
19	1	0	0	1	0
	95	107	110	131	108

2.11.6 EG Study in Mice: Cumulative Relative Frequencies



2.11.7 EG Study in Mice: Observed and Averaged Malformation Rates



EG Study in Rats

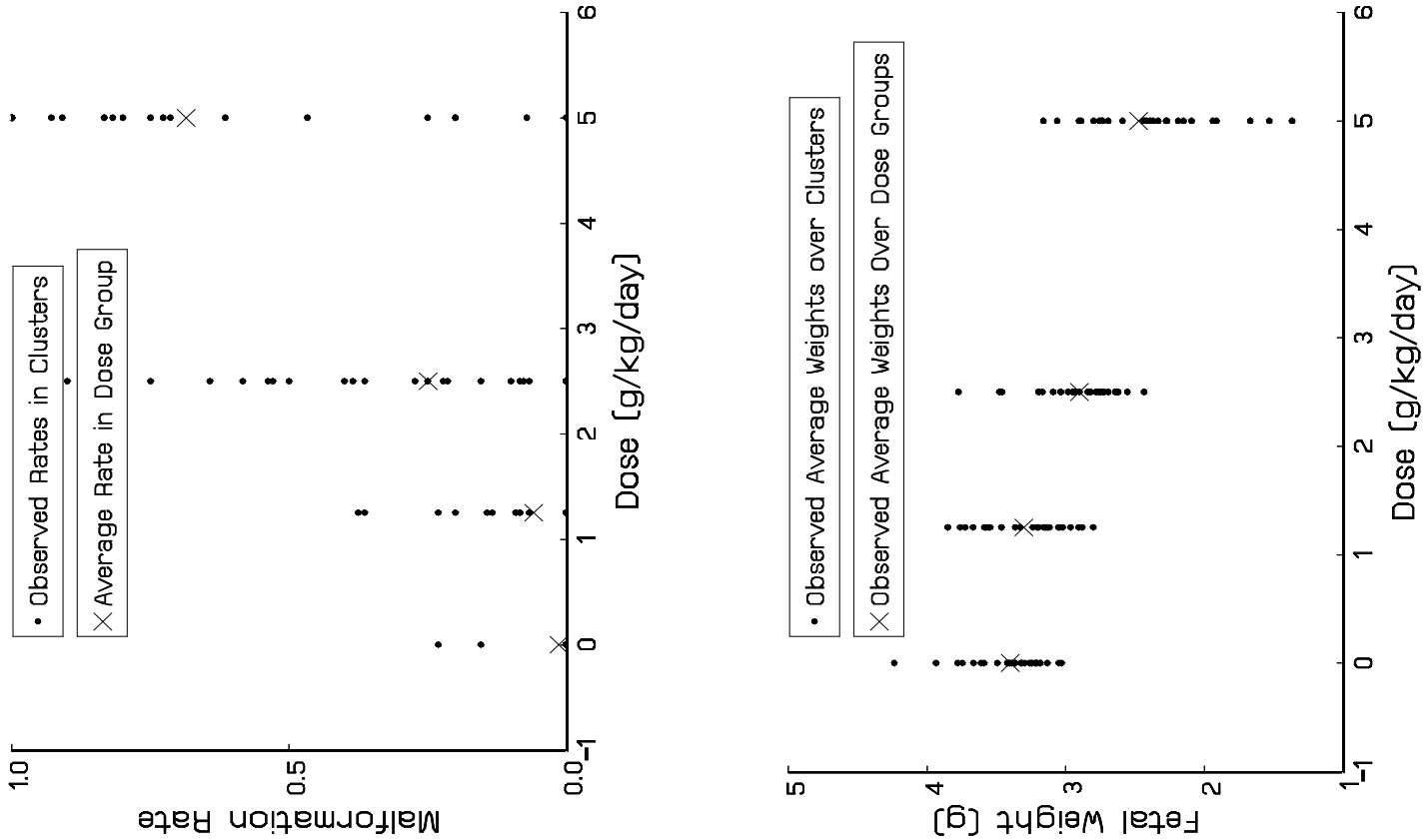
- Another experiment, investigating the effect of EG in rats.
- Doses selected: 0, 1.25, 2.50, and 5.0 g/kg/day.
- $N = 1368$ live rat foetuses were examined for low birth weight (continuous) or defects (binary).
- This joint occurrence of continuous and binary outcomes will provide additional challenges in model development.

Ethylene Glycol Study in Rats

Dose	Dams	Live	Litter Size		Malf.		Weight		ρ
			Mean	SD	#	%	Mean	SD	
0.00	28	379	13.50	1.50	5	1.3	3.40	0.38	0.07
1.25	28	357	12.75	2.03	21	5.8	3.30	0.37	0.00
2.50	29	345	11.89	3.26	86	24.9	2.90	0.36	-0.29
5.00	26	287	11.04	4.01	197	68.6	2.48	0.46	-0.70

2.11.8 EG Study in Rats

Observed malformation rates and average weights for all clusters.



2.12 Spatially Organized Data: The Heat Shock Study

- Effects of temperature and duration of exposure on viable foetuses:
  - ▷ embryos are explanted from the uterus of maternal dam
  - ▷ cultured in vitro
  - ▷ each embryo exposed to short period of heat stress:
    - ∗ 3 to 5C
    - ∗ 5 to 60 minutes
  - ▷ these are *individual-level* covariates
  - ▷ examined 24 hours later for impaired/accelerated development

- Alternative Segment II study
- Advantages:
  - ▷ Exposure directly administered to the embryo
  - ▷ Exposure patterns easy to control
  - ▷ Effects of exposure are quickly obtained $\leftrightarrow$ more than a week in the standard Segment II design
  - ▷ Mechanism to assess the joint effects of **duration of exposure** & **exposure level**

2.12.1 Design and Modelling

- 327 embryos:
  - ▷ 50 did not survive heat stress
  - ▷ 277 embryos with complete data
- Haber's Law: duration$\times$exposure
- Binary Outcomes:
  - ▷ MBN: Midbrain
  - ▷ FBN: Forebrain
  - ▷ OLF: Olfactory System
  - ▷ BRB: Branchial Bars

  Coded as affected versus normal

- Continuous outcomes:
  - ▷ YSD: yolk sack
  - ▷ CRL: crown rump length
  - ▷ HEAD: head circumference

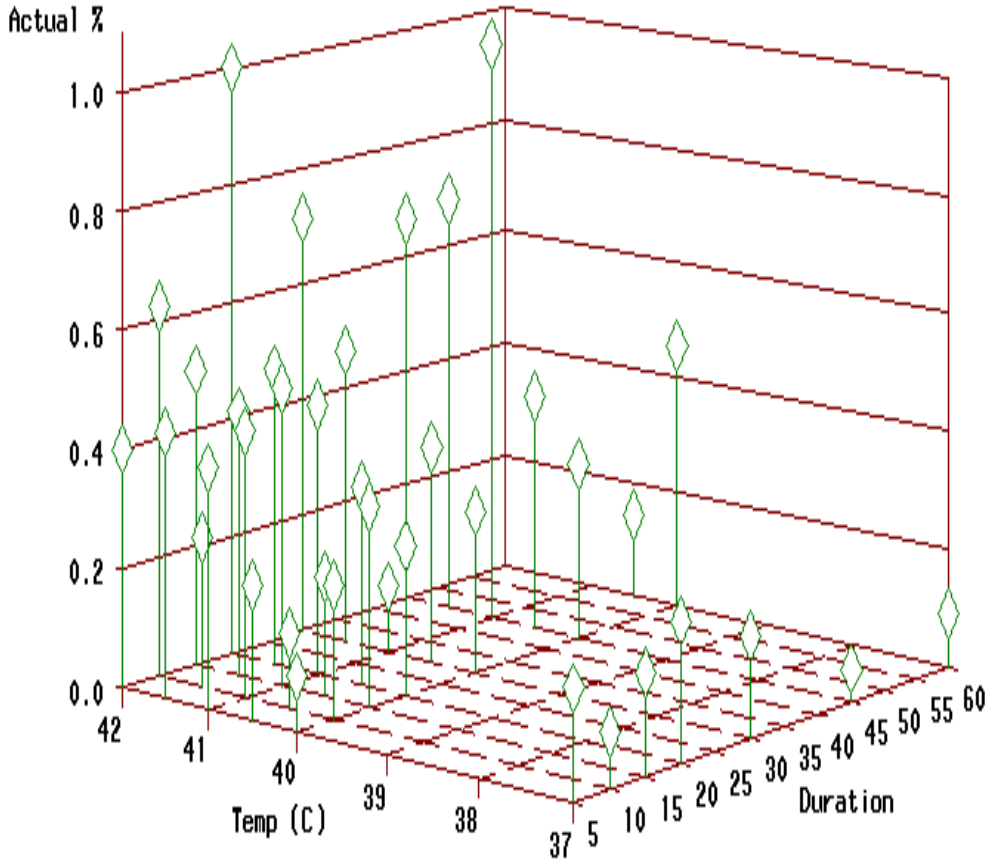
Study Design for Heat Shock Experiments: Number of Viable Embryos

Temperature	Duration of Exposure							Total
	5	10	15	20	30	45	60	
37.0	11	11	12	13	12	18	11	88
40.0	11	9	9	8	11	10	11	69
40.5	9	8	10	9	11	10	7	64
41.0	10	9	10	11	9	6	0	55
41.5	9	8	9	10	10	7	0	53
42.0	10	8	10	5	7	6	0	46
Total	60	53	60	56	60	57	29	375

Distribution of Cluster Sizes for the Heat Shock Data

cluster size n_i	1	2	3	4	5	6	7	8	9	10	11
number of clusters of size n_i	6	3	6	12	13	11	8	5	2	3	2

2.12.2 Percentage of Affected Embryos



2.12.3 Haber's Law

Probability on adverse effect equal for each duration $\times$ temperature combination

$$P(d,t) = F(\beta_0 + \beta_1 t \delta_d + \beta_2 dt)$$

$$\delta_d = \begin{cases} 1 & \text{if } d > 37 \\ 0 & \text{otherwise} \end{cases}$$

Implication of Haber's Law : $\beta_1 = 0$

2.12.4 Challenges in Model Development

- Hierarchical structure of developmental outcomes

  in order for a malformation to be observed, the embryo must not have previously suffered a prenatal death or resorption

- Litter effect

  tendency for animals from the same litter to respond more alike than animals from different litters

- Litter size may be related to outcomes among live foetuses

- Multivariate nature of the data

  potential correlations among multiple fetal endpoints

- Combination of continuous and discrete outcomes

Chapter 3
Some Useful Notation and Conventions

- Data organization
- Distance
- Matrix and vector notation
- Normality

3.1 Organization of Data: Scalar Case

- In this section, we will establish some useful notation, as well as some commonly used estimators. The next section is devoted to vectorized notation.

- Let $i = 1, \ldots, N$ denote subjects, individuals, items, experimental units, experimental trials,...

- Replication within units:

  ▷ Multivariate: Let $j = 1, \ldots, p$ denote variables, characteristics, measured for each individual.

  ▷ Repeated measures: Let $j = 1, \ldots, n_i$ denote repeated measures within a subject.

- The jth measurement for the ith individual will be denoted by y_{ij}. Data can be grouped into a matrix

$$\boldsymbol{Y} = (y_{ij})_{i,j} = \begin{pmatrix} y_{11} & \cdots & y_{1p} \\ \vdots & \ddots & \vdots \\ y_{n1} & \cdots & y_{np} \end{pmatrix}.$$

3.1.1 Descriptive statistics

- **Sample Mean**

$$\overline{y}_j = \frac{1}{N} \sum_{i=1}^{N} y_{ij}$$

Mean vector $\overline{\boldsymbol{y}} = (\overline{y}_1, \ldots, \overline{y}_p)^T$

- **Sample Variance**

$$s_j^2 = \frac{1}{N} \sum_{i=1}^{N} (y_{ij} - \overline{y}_j)^2$$

Mind: a very common definition is

$$s_j^2 = \frac{1}{N-1} \sum_{i=1}^{N} (y_{ij} - \overline{y}_j)^2,$$

what would be the reason for this ambiguity?

When sample variances are put along the main diagonal of a matrix, they will be denoted by $s_{jj} = s_j^2$. Standard deviation: $\sqrt{s_j^2}$.

- Linear association:

  ▷ **Sample Covariance**

$$s_{jk} = \frac{1}{N} \sum_{i=1}^{N} (y_{ij} - \overline{y}_j)(y_{ik} - \overline{y}_k)$$

We define the $p \times p$ symmetric covariance matrix

$$\boldsymbol{S}_N = (s_{jk})_{j,k}$$

and the matrix of squares and cross-products as

$$\boldsymbol{W} = N\boldsymbol{S}_N.$$

▷ **Sample Correlation Coefficient.** (Pearson's product moment correlation coefficient)

$$r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}s_{kk}}} = \frac{\sum_{i=1}^{N}(y_{ij} - \overline{y}_j)(y_{ik} - \overline{y}_k)}{\left[\sum_{i=1}^{N}(y_{ij} - \overline{y}_j)^2 \sum_{i=1}^{N}(y_{ik} - \overline{y}_k)^2\right]^{1/2}}.$$

Obvious properties:

* unitless

* covariance of standardized variables

* range: $-1 \leq r_{jk} \leq 1$

* $r_{jk} = 0$: no linear association

* $r_{jk} > 0$: both variables tend to deviate from their respective means in the same direction

* the correlation coefficient is invariant under the following transformations:

$$\begin{aligned} \widetilde{y}_{ij} &= a y_{ij} + b, \\ \widetilde{y}_{ik} &= c y_{ik} + d \end{aligned}$$

  with $ac > 0$. What happens if $ac < 0$?

* DO NOT overestimate the importance of linear association measures:

  * they don't capture other kinds of association (e.g. quadratic relationships between variables);

  * the correlation coefficient is very sensitive to outliers

3.2 Distance

- Consider two points, $P = (y_{11}, \ldots, y_{1p})$ and $Q = (y_{21}, \ldots, y_{2p})$.

- We will discuss several proposals for defining the distance between P and Q and select a statistically relevant alternative.

- **Euclidean distance**

$$\begin{aligned} d(P,Q) &= \sqrt{\sum_{j=1}^{p}(y_{1j} - y_{2j})^2} \\ &= \sqrt{(\boldsymbol{y}_1 - \boldsymbol{y}_2)^T(\boldsymbol{y}_1 - \boldsymbol{y}_2)} \\ &= ||\boldsymbol{y}_1 - \boldsymbol{y}_2||_2. \end{aligned}$$

- The distance between P and the origin O becomes

$$\sqrt{\sum_{j=1}^{p} y_{1j}^2}.$$

- Drawbacks:
  - ▷ All components contribute equally, although there may be random fluctuations of a different magnitude in the components.
  - ▷ Different pairs of components may be correlated differently.

- This calls for the definition of a "statistical distance".

- First attempt: take standard deviation into account:

$$d(O,P) = \sqrt{\sum_{j=1}^{p} \frac{y_{1j}^2}{s_j^2}} = \sqrt{\sum_{j=1}^{p} \frac{y_{1j}^2}{s_{jj}}}.$$

- Contours in the Euclidian case are circles. In our current proposal they change to ellipses, with principal axes along the variables under consideration. The distance between two points is obvious:

$$d(P,Q) = \sqrt{\sum_{j=1}^{p} \frac{(y_{1j} - y_{2j})^2}{s_{jj}}}. \tag{3.1}$$

- Next, we take the correlation into account. The covariance matrix $\boldsymbol{S}_N$ will play a key role.

- In order to apply (3.1), we transform the variables $\boldsymbol{y}_i$ to uncorrelated ones, say $\boldsymbol{x}_i$:

$$\boldsymbol{x}_i = \boldsymbol{C}\boldsymbol{y}_i \qquad \Longrightarrow \qquad \boldsymbol{y}_i = \boldsymbol{C}^{-1}\boldsymbol{x}_i.$$

- The requirement is satisfied if the covariance of the y-variables assumes the form

$$\begin{aligned} \boldsymbol{I} = \boldsymbol{S}_x &= \frac{1}{N}\sum_i (\boldsymbol{x}_i - \overline{\boldsymbol{x}})(\boldsymbol{x}_i - \overline{\boldsymbol{x}})^T \\ &= \frac{1}{N}\sum_i \boldsymbol{C}(\boldsymbol{y}_i - \overline{\boldsymbol{y}})(\boldsymbol{y}_i - \overline{\boldsymbol{y}})^T\boldsymbol{C}^T \\ &= \boldsymbol{C}\boldsymbol{S}_y\boldsymbol{C}^T, \end{aligned}$$

  which is satisfied by

$$\boldsymbol{S} = \boldsymbol{S}_y = (\boldsymbol{C}^T\boldsymbol{C})^{-1}.$$

- Now, measuring the distance between P and Q can be performed in Euclidean distance, using the coordinate system $\boldsymbol{y}$, as they represent uncorrelated variables with unit variance:

$$\begin{aligned} d(P,Q)^2 &= (\boldsymbol{x}_1 - \boldsymbol{x}_2)^T(\boldsymbol{x}_1 - \boldsymbol{x}_2) \\ &= [\boldsymbol{C}(\boldsymbol{y}_1 - \boldsymbol{y}_2)]^T[\boldsymbol{C}(\boldsymbol{y}_1 - \boldsymbol{y}_2)] \\ &= (\boldsymbol{y}_1 - \boldsymbol{y}_2)^T\boldsymbol{C}^T\boldsymbol{C}(\boldsymbol{y}_1 - \boldsymbol{y}_2). \end{aligned}$$

  We define the *Mahalanobis distance* by

$$d(P,Q)^2 = (\boldsymbol{y}_1 - \boldsymbol{y}_2)^T\boldsymbol{S}_y^{-1}(\boldsymbol{y}_1 - \boldsymbol{y}_2). \tag{3.2}$$

- If the variables are uncorrelated but have other than unit variances, then the Mahalanobis distance takes the form (3.1).

3.3 Matrix and Vector Formulation

- When several random variables are considered simultaneously, it is convenient to group them into vectors. Consider the following example: on each unit, we measure two variables, $Y_1 =$height and $Y_2 =$weight.

- In this case, we can conveniently write the data as

$$\boldsymbol{Y} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}.$$

- If we observe the same random vector for N individuals, we can group the result in an $N \times p$ matrix. In general, we will often encounter sets of random variables that are grouped into a matrix scheme

$$\boldsymbol{Y} = (Y_{ij})_{i,j}$$

- Multivariate setting:
  - ▷ Each row corresponds to an experimental unit.
  - ▷ Each column corresponds to a variable.
- Most operations can be defined on matrix variables, e.g. the *expectation* of $\boldsymbol{Y}$ will be

$$E(\boldsymbol{Y}) = \begin{pmatrix} E(X_{11}) & \dots & E(X_{1p}) \\ \vdots & \ddots & \vdots \\ E(X_{n1}) & \dots & E(X_{np}) \end{pmatrix} = \begin{pmatrix} E(\boldsymbol{X}_1^T) \\ \vdots \\ E(\boldsymbol{X}_n^T) \end{pmatrix},$$

where

$$E(X_{ij}) = \int x_{ij} dP_{ij}$$

and P_{ij} is the corresponding probability measure.

- Special cases:
  - ▷ Y_{ij} is a discrete r.v.: the integration w.r.t. P_{ij} assumes the form of a summation:

$$E(Y_{ij}) = \sum_{y_{ij}} y_{ij} p_{ij}(y_{ij}),$$

    with p the probability function.
  - ▷ Y_{ij} is a continuous r.v. on $\mathbb{R}$:

$$E(Y_{ij}) = \int_{-\infty}^{+\infty} y_{ij} f_{ij}(y_{ij}) dy_{ij}$$

    with f the probability density function (p.d.f.).

- Immediate results:

$$E(\boldsymbol{X} + \boldsymbol{Y}) = E(\boldsymbol{X}) + E(\boldsymbol{Y})$$

$$E(\boldsymbol{AXB}) = \boldsymbol{A}E(\boldsymbol{X})\boldsymbol{B}.$$

3.3.1 Notation for Random Vectors

- Let $\boldsymbol{Y}$ be a random vector ($\boldsymbol{Y} = (Y_j)_j$).
- The (marginal) means of $\boldsymbol{Y}$ are defined by

$$\mu_j = E(Y_j)$$

- The population mean vector is given by

$$\boldsymbol{\mu} = E(\boldsymbol{Y}) = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix}.$$

- The (marginal) (co)variances of $\boldsymbol{Y}$ are defined by

$$\sigma_{jk} = E[(Y_j - \mu_j)(Y_k - \mu_k)].$$

- Define the population variance-covariance matrix as

$$\boldsymbol{\Sigma} = \mathrm{Cov}(\boldsymbol{Y}) = \begin{pmatrix} \sigma_{11} & \dots & \sigma_{1p} \\ \vdots & \ddots & \vdots \\ \sigma_{p1} & \dots & \sigma_{pp} \end{pmatrix}$$

- Clearly, $\boldsymbol{\Sigma}$ is a symmetric matrix.

- The population correlations are constructed as follows:

$$\rho_{jk} = \frac{\sigma_{jk}}{\sqrt{\sigma_{jj}\sigma_{kk}}} = \sigma_{jj}^{-1/2}\sigma_{jk}\sigma_{kk}^{-1/2}$$

  which yields

$$\boldsymbol{\Sigma} = \boldsymbol{V}^{1/2}\boldsymbol{\rho}\boldsymbol{V}^{1/2}$$

- Here, $\boldsymbol{V} = \mathrm{diag}(\sigma_{11}, \dots, \sigma_{pp})$.

- We call
  - ▷ $\boldsymbol{\rho}$ the *population correlation matrix*,
  - ▷ $\boldsymbol{V}$ the diagonal *matrix of variances*, and
  - ▷ $\boldsymbol{V}^{1/2}$ the diagonal *matrix of standard deviations.*

3.3.2 Independence

- Explicit expressions for means and covariances for continuous variables are:

$$\begin{aligned} \mu_j &= \int y_j f_j(y_j) dy_j \\ \sigma_{jk} &= \iint (y_j - \mu_j)(y_k - \mu_k) f_{jk}(y_j, y_k) dy_j dy_k \end{aligned}$$

  with f_j the derivative of

$$F_j(y_j) = P(Y_j \leq y_j)$$

- Y_j and Y_k are said to be statistically independent iff

$$P(Y_j \leq y_j, Y_k \leq y_k) = P(Y_j \leq y_j) P(Y_k \leq y_k)$$

  which in the continuous case is equivalent to

$$f_{jk}(y_j, y_k) = f_j(y_j) f_k(y_k) \qquad \text{(a.s.)}$$

- It is easy to generalize this definition to p variables.

- $\boldsymbol{Y}$ has mutually (statistically) independent components iff

$$P(\bigcap_{j=1}^{p} Y_j \leq y_j) = \prod_{j=1}^{p} P(Y_j \leq y_j)$$

  which in the continuous case can be represented as

$$f(y_1, y_2, \ldots, y_p) = f_{12\ldots p}(y_1, y_2, \ldots, x_p) = \prod_{j=1}^{p} f_j(y_j)$$

  (a.s.)

- Y_j and Y_k statistically independent $\Rightarrow$ their covariance is zero

- If $\boldsymbol{Y}$ has mutually independent components, then its covariance matrix has a diagonal structure.

3.3.3 Partitioning Random Variables

- Example: In a given study, 4 physical/medical measurements are recorded $(Y_1, \ldots, Y_4)$, together with 6 socio-economic variables $(Y_5, \ldots, Y_{10})$.

- It is then natural to partition $\boldsymbol{Y}$ into two parts, say

$$\boldsymbol{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_4 \\ \hline Y_5 \\ \vdots \\ Y_{10} \end{pmatrix}.$$

- In general, we will assume that the p dimensional random vector $\boldsymbol{X}$ is partitioned into two subvectors of dimensions q and $p - q$ respectively.

$$\boldsymbol{Y} = \begin{pmatrix} \boldsymbol{Y}_1 \\ \hline \boldsymbol{Y}_2 \end{pmatrix}.$$

- The mean vector and the covariance matrix can be partitioned accordingly as

$$\boldsymbol{\mu} = \left(\frac{\boldsymbol{\mu}_1}{\boldsymbol{\mu}_2} \right) \tag{3.3}$$

and

$$\boldsymbol{\Sigma} = \left(\begin{array}{c|c} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \hline \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{array} \right) \tag{3.4}$$

- The meaning of the component matrices of $\boldsymbol{\Sigma}$ is

$$\begin{aligned} \boldsymbol{\Sigma}_{11} &= \mathsf{cov}(\boldsymbol{Y}_1), \\ \boldsymbol{\Sigma}_{22} &= \mathsf{cov}(\boldsymbol{Y}_2), \\ \boldsymbol{\Sigma}_{12} = \boldsymbol{\Sigma}_{21}^T &= \mathsf{cov}(\boldsymbol{Y}_1, \boldsymbol{Y}_2). \end{aligned}$$

- This partitioning is presented for population quantities. It can in a similar fashion be carried out for their sample analogue.

- If we have a sample of N vectors $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_N$, then the mean $\overline{\boldsymbol{y}}$ and covariance $\boldsymbol{S}_N$ can be partitioned as (3.3) and (3.4) respectively.

- Most of the population derivations have a sample counterpart, and vice versa.

3.4 The Omnipresence of the Normal Distribution

- For a simple random sample with just one outcome variable, the mean and the standard deviation, and/or the standard error of the mean, are easily computed. This action is independent of the true distribution of the data.

- However, for some distributions a mean and standard deviation will be less meaningful:

  ▷ a bimodal distribution may have a mean, but primary interest might be in the two modes;

  ▷ the Cauchy distribution has infinite standard deviation; the sample standard deviation does not lead anywhere...

- The normal distribution has easy interpretations for the mean and standard deviation of samples drawn from it. The usual definitions of mean and standard deviation are the least squares estimators (and maximum likelihood estimators) of the population quantities.

- In addition, under the usual regularity conditions, the sample mean converges to the location parameter of the true distribution, even if it is not normal. This is based on so-called laws of large numbers (central limit theory), which means, roughly speaking, that large samples show a normal spread, even when they are drawn from non-normal distributions.

- In analogy to the univariate setting, the **multivariate normal distribution** is heavily used in multivariate statistics. This has lead to a number of techniques:

  ▷ for the multivariate setting: multivariate regression, multivariate analysis of variance, principal components analysis, factor analysis, canonical correlation analysis, discriminant analysis

  ▷ for the longitudinal setting: the linear mixed-effects model.

  ▷ for the spatial setting: the auto-Gaussian model.

- For the same reasons as above, these techniques can be used as well, with caution, when samples are non-normal, for two reasons:

  ▷ merely for describing, summarizing, reducing data (such as a sample mean);

  ▷ for samples that are large enough (large sample inference).

- However, whether a sample is "large" depends on the discrepancy between the true distribution and the normal distribution.

- Clearly, for binary data the discrepancy might be high. Example: NTP data with response: malformed/non-malformed.

- In such a case, alternative approaches are necessary. Therefore, more recent work focuses on non-normal outcomes.

Part II

Multivariate Methods

Important note:

▷ For this part, see the set of notes provided by Olivier Thas

Part III

Repeated Measures

Chapter 8
Example: Lizard Data

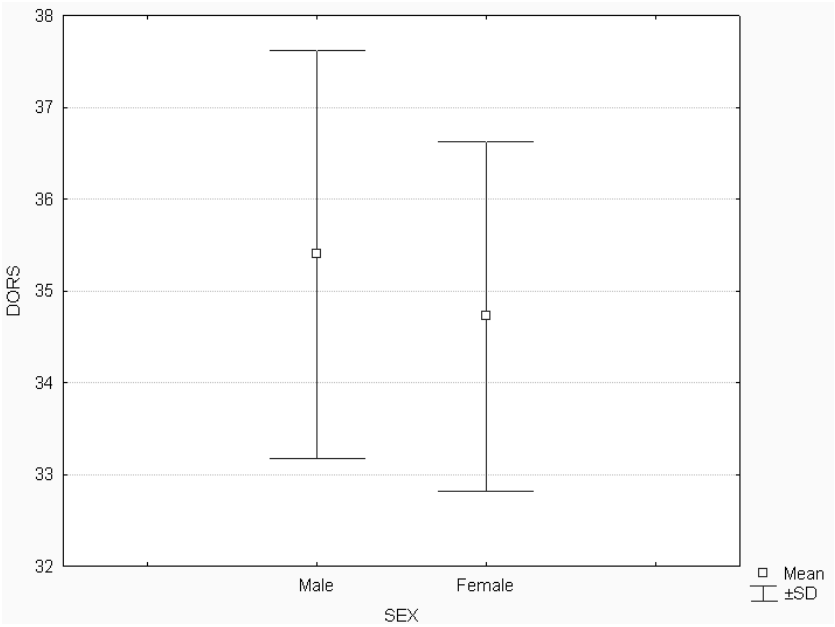
- ▷ Example
- ▷ Two-way ANOVA
- ▷ Mixed models
- ▷ Fitting mixed models in SAS
- ▷ Remarks

8.1 Example

- Data on 102 lizards
- Response of interest: Number of dorsal shells
- Research question:

> Is number of dorsal shells gender-related ?

- Graphically:

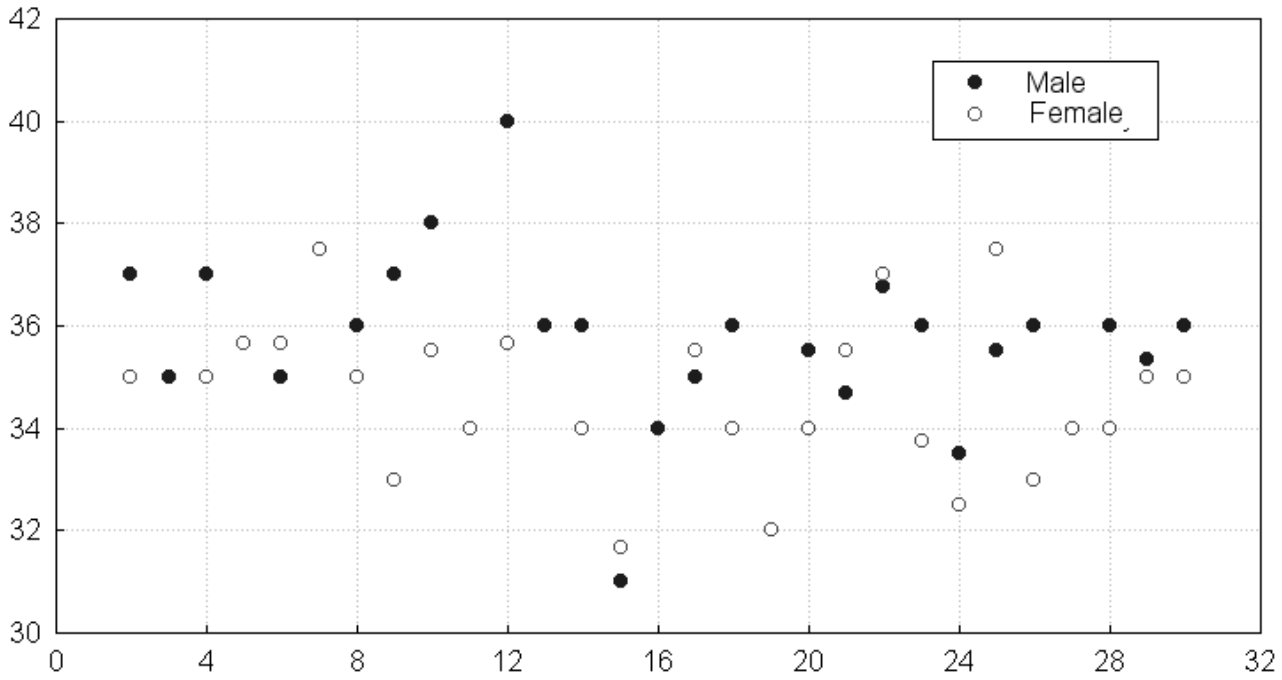


- Two-sample t-test:

Variable	T-tests; Grouping: SEX (schildformengels.STA) Group 1: Male Group 2: Female **Mean Male**	Mean Female	t-value	df	p
DORS	35,39583	34,72222	1,648480	100	0,102393

- Hence, the small observed difference is not significant ($p = 0.1024$).
- A typical aspect of the data is that some animals have the same mother.
- We have 102 lizards from 30 mothers
- Mother effects might be present
- Hence a comparison between male and female animals should be based on within-mother comparisons.

- Graphically:



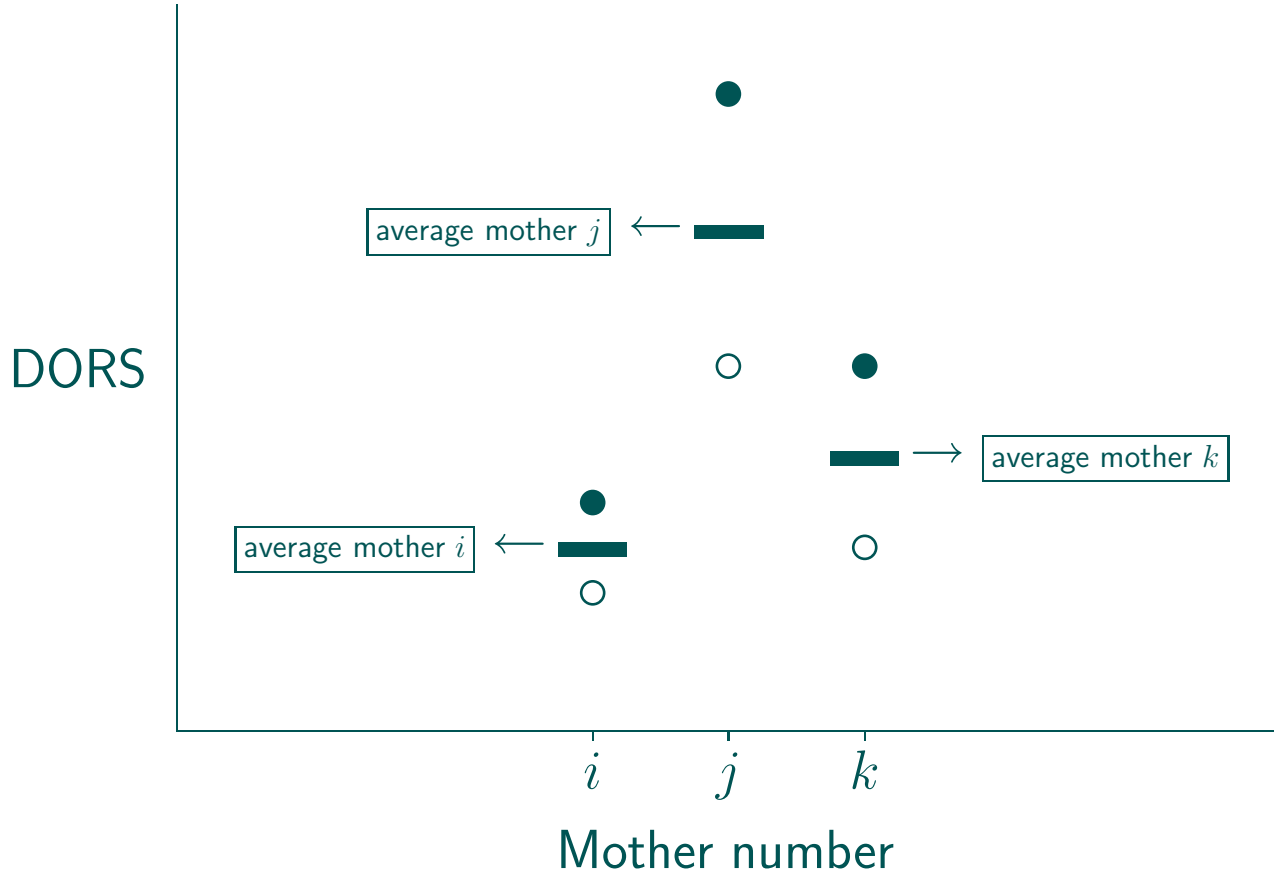
- Observations:
  - ▷ Much between-mother variability
  - ▷ Often, males (considerably) higher than females
  - ▷ In cases where females higher than males, small differences

- Hence the non-significant t-test result may be due to the between-mother variability
- This is an example of **clustered data**: observations are clustered within mothers
- It is to be expected that measurements within mothers are more alike than measurements from different mothers.
- We expect correlated observations within mothers and independent observations between mothers.
- How to correct for differences between mothers ?

8.2 Two-way ANOVA

- An obvious first choice to test for a 'sex' effect, correcting for 'mother' effects, is 2-way ANOVA with factors 'sex' and 'mother'.
- The mother effect then represents the variability between mothers.
- Let Y_{ij} be the jth measurement on the ith mother, and let t_{ij} be 1 for males and 0 for females.
- The model then equals: $Y_{ij} = \mu + \alpha_i + \beta t_{ij} + \varepsilon_{ij}$
- β is the parameter of interest, and we need the usual restrictions on the parameters α_i, e.g., $\Sigma_i \alpha_i = 0$
- Residual distribution: $\varepsilon_{ij} \sim N(0, \sigma^2_{res})$

- Graphically:



- SAS program:

```
proc glm data = lizard;
class sex mothc;
model dors = sex mothc;
run;
```

- Relevant SAS output:

```
                    Class Level Information

Class       Levels  Values

SEX              2  1 2

MOTHC           30  1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
                    19 20 21 22 23 24 25 26 27 28 29 30


Dependent Variable: DORS

                                   Sum of
Source                      DF         Squares     Mean Square     F Value    Pr > F

Model                       29     268.4685062       9.2575347        3.98    <.0001

Error                       72     167.3746310       2.3246477

Corrected Total            101     435.8431373


R-Square     Coeff Var      Root MSE     DORS Mean

0.615975      4.351352      1.524680      35.03922
```

Source	DF	Type III SS	Mean Square
SEX	1	16.7253690	16.7253690
MOTHC	28	256.9378690	9.1763525

Source	F Value	Pr > F
SEX	7.19	0.0091
MOTHC	3.95	<.0001

- Note the highly significant mother effect.
- We now also obtain a significant gender effect.
- Many degrees of freedom are spent to the estimation of the mother effect, which is not even of interest.

8.3 Mixed Models

- Note the different nature of the two factors:
  - ▷ SEX: defines 2 groups of interest
  - ▷ MOTHER: defines 30 groups not of real interest. A new sample would imply other mothers.
- In practice, one therefore considers the factor 'mother' as a **random factor**.
- The factor 'sex' is a **fixed effect**.
- Thus the model is a **mixed model**.
- In general, models can contain multiple fixed and/or random factors.

- The model is still of the form:

$$Y_{ij} = \mu + \alpha_i + \beta t_{ij} + \varepsilon_{ij}$$

- But the fact that mothers can be assumed to be randomly selected from a population of mothers is reflected in the additional assumption

$$\alpha_i \sim N(0, \sigma^2_{moth})$$

- Note that we still have that the α_i have mean zero. Before, we had the restriction $\Sigma_i \, \alpha_i = 0$

8.4 Fitting Mixed Models in SAS

- Mixed model with 'sex' as fixed and 'mother' as random effect:

```
proc mixed data = lizard;
class sex mothc;
model dors = sex;
random mothc;
run;
```

- Fixed effects are specified in the MODEL statement.

- Random effects are specified in the RANDOM statement.

- Relevant SAS-output:

Iteration History

Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	436.17789431	
1	3	407.96849207	0.00072385
2	1	407.88032382	0.00001530
3	1	407.87858406	0.00000001

Convergence criteria met.

Covariance Parameter Estimates

Cov Parm	Estimate
MOTHC	1.7799
Residual	2.2501

Type 3 Tests of Fixed Effects

Effect	Num DF	Den DF	F Value	Pr > F
SEX	1	72	6.63	0.0121

- Covariance parameter estimates:
  - ▷ Total variability, correcting for gender is decomposed as:

$$\sigma^2 \;=\; \sigma^2_{moth} \;+\; \sigma^2_{res}$$

$$4.03 \;=\; 1.78 \;+\; 2.25$$

  - ▷ σ^2_{moth} represents the variability between mothers
  - ▷ σ^2_{res} represents the variability within mothers
  - ▷ The 'mother' factor explains $1.78/4.03 = 44\%$ of the total variability, after correction for gender

- Note the significant difference between male and female animals ($p = 0.0121$)

- With the t-test, ignoring the mother effect, this was $p = 0.1024$.

- The mixed model implies a specific correlation structure:
  - ▷ Observations from different mothers are independent.
  - ▷ Observations within mothers are positively correlated:

$$\text{corr(within mother)} = \frac{\sigma^2_{moth}}{\sigma^2_{moth} + \sigma^2_{res}} = \frac{1.78}{1.78 + 2.25} = 0.44$$

8.5 Remarks

- The simplest example of clustered data are paired observations, typically analyzed using a paired t-test.
- In our example, this would mean that we have exactly one male and one female animal per mother.
- The mixed models can be viewed as an extension of the paired t-test to :
  - ▷ more than 2 observations per cluster
  - ▷ unbalanced data: unequal number of measurements per cluster
  - ▷ models with covariates, e.g., 'sex', or others
  - ▷ models with multiple random effects (see later)

Chapter 9
Example: Growth Curves

- Example
- The model
- ESTIMATE and CONTRAST statements
- Random intercepts model
- Remarks
- The linear mixed model

9.1 Example

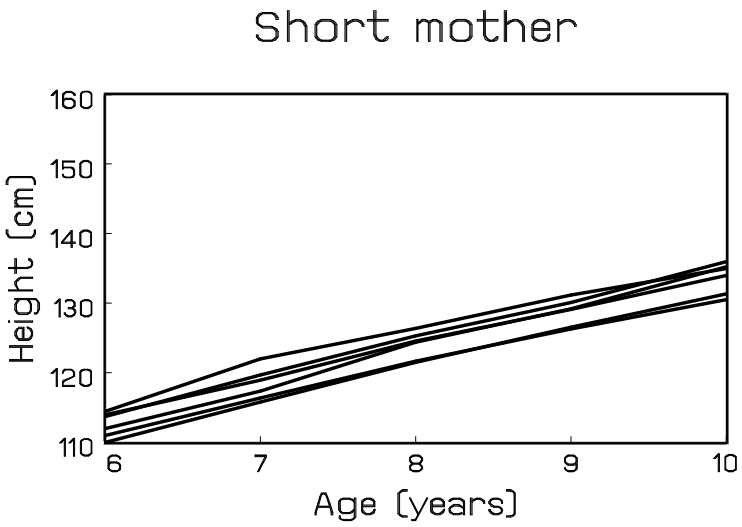
- Taken from Goldstein (1979).
- Research question:

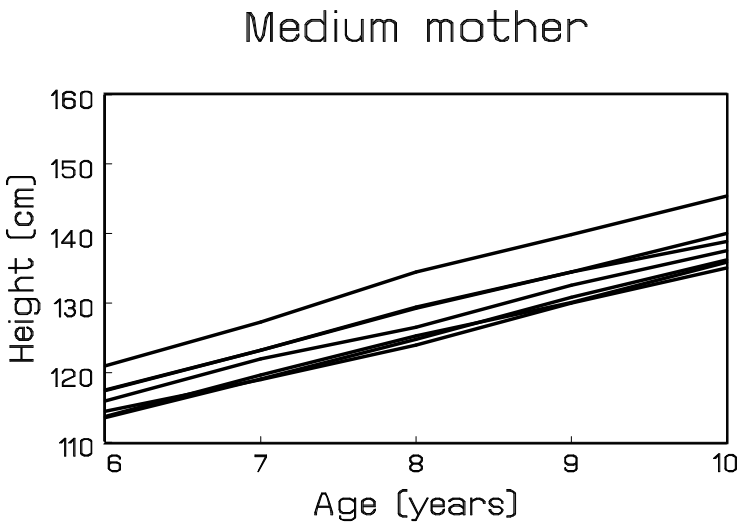
  Is growth related to height of mother ?

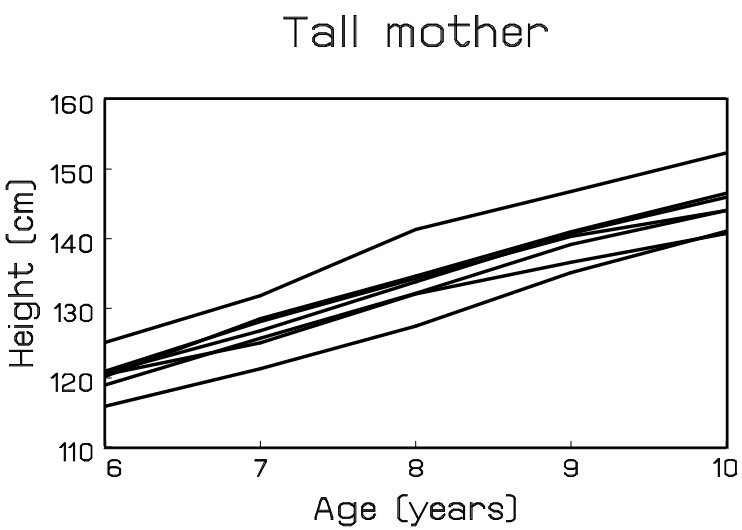
- The height of 20 schoolgirls, with small, medium, or tall mothers, was measured over a 4-year period:

	Mothers height	Children numbers
Small mothers	< 155 cm	$1 \rightarrow 6$
Medium mothers	$[155\text{cm};\ 164\text{cm}]$	$7 \rightarrow 13$
Tall mothers	> 164 cm	$14 \rightarrow 20$

- Individual profiles:







- Remarks:
  - ▷ Almost perfect linear relation between Age and Height
  - ▷ Much variability between girls
  - ▷ Little variability within girls
  - ▷ Fixed number of measurements per subject
  - ▷ Measurements taken at fixed time points

9.2 The Model

- As for the lizard data, the observations are clustered within children.
- Correction for the variability between children is done through a random child effect.
- Further, we will assume a linear relation between Age and Height, possibly different for the different groups.
- Ignoring the clustered nature of the data, the following ANOCOVA could be used:

```
proc glm data = growth;
class group;
model height = age group age*group;
run;
```

- Inclusion of a random child effect is obtained by:

```
proc mixed data = growth;
class group child;
model height = age group age*group / solution;
random child;
run;
```

- As before, let Y_{ij} be the jth measurement of height for the ith cluster (child), taken at time t_{ij} (age). Our model is then of the form:

$$Y_{ij} = \begin{cases} \beta_1 + b_i + \beta_2 t_{ij} + \varepsilon_{ij}, & \text{if short mother} \\ \beta_3 + b_i + \beta_4 t_{ij} + \varepsilon_{ij}, & \text{if medium mother} \\ \beta_5 + b_i + \beta_6 t_{ij} + \varepsilon_{ij}, & \text{if tall mother} \end{cases}$$

- As before, it is assumed that random effects b_i are normal with mean zero and variance σ^2_{child}.

- The errors ε_{ij} are normal with mean zero and variance σ^2_{res}.

- Relevant SAS output:

Covariance Parameter Estimates

Cov Parm	Estimate
CHILD	8.9603
Residual	0.7696

Solution for Fixed Effects

Effect	GROUP	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept		83.1229	1.4162	17	58.69	<.0001
AGE		6.2486	0.1049	77	59.59	<.0001
GROUP	1	-1.8229	2.0846	77	-0.87	0.3846
GROUP	2	-0.1486	2.0028	77	-0.07	0.9411
GROUP	3	0	.	.	.	.
AGE*GROUP	1	-0.9786	0.1543	77	-6.34	<.0001
AGE*GROUP	2	-0.6814	0.1483	77	-4.60	<.0001
AGE*GROUP	3	0	.	.	.	.

Type 3 Tests of Fixed Effects

Num	Den

Effect	DF	DF	F Value	Pr > F
AGE	1	77	8385.15	<.0001
GROUP	2	77	0.46	0.6330
AGE*GROUP	2	77	21.66	<.0001

- Covariance parameter estimates:
  - ▷ Total variability, correcting for age and group is decomposed as:

$$\sigma^2 = \sigma^2_{child} + \sigma^2_{res}$$

$$9.73 = 8.96 + 0.77$$

  - ▷ σ^2_{child} represents the variability between children
  - ▷ σ^2_{res} represents the variability within children
  - ▷ The 'child' factor explains $8.96/9.73 = 92\%$ of the total variability, after correction for group and age

- Note the significant difference in slopes between the groups ($p < 0.0001$)

- The mixed model again implies a specific correlation structure:
  - ▷ Observations from different children are independent.
  - ▷ Observations within children are positively correlated:

$$\text{corr(within child)} = \frac{\sigma^2_{child}}{\sigma^2_{child} + \sigma^2_{res}} = \frac{8.96}{8.96 + 0.77} = 0.92$$

9.3 ESTIMATE and CONTRAST Statements

- As in many other SAS procedures, ESTIMATE and CONTRAST statements can be used to obtain inferences about specific contrasts of the fixed effects.

- Slopes for each group separately, as well as pairwise comparisons are obtained using the following program:

```
proc mixed data=growth;
class child group;
model height = group age*group / noint solution;
random child;
contrast 'small-medium' group*age 1 -1 0;
contrast 'small-tall' group*age 1 0 -1;
contrast 'medium-tall' group*age 0 1 -1;
estimate 'small' group*age 1 0 0 / cl;
estimate 'medium' group*age 0 1 0 / cl;
estimate 'tall' group*age  0 0 1 / cl;
run;
```

- Note the different parameterization for the fixed effects, when compared to the original program:

```
proc mixed data = growth;
class group child;
model height = age group age*group / solution;
random child;
run;
```

- Relevant SAS output:

The Mixed Procedure

Solution for Fixed Effects

Effect	GROUP	Estimate	Standard Error	DF	t Value	Pr > |t|
GROUP	1	81.3000	1.5297	77	53.15	<.0001
GROUP	2	82.9743	1.4162	77	58.59	<.0001
GROUP	3	83.1229	1.4162	77	58.69	<.0001
AGE*GROUP	1	5.2700	0.1133	77	46.53	<.0001
AGE*GROUP	2	5.5671	0.1049	77	53.10	<.0001
AGE*GROUP	3	6.2486	0.1049	77	59.59	<.0001

Type 3 Tests of Fixed Effects

Effect	Num DF	Den DF	F Value	Pr > F
GROUP	3	77	3234.13	<.0001
AGE*GROUP	3	77	2845.30	<.0001

Estimates

Label	Estimate	Standard Error	DF	t Value	Pr > |t|	Alpha	Lower	Upper
small	5.2700	0.1133	77	46.53	<.0001	0.05	5.0445	5.4955
medium	5.5671	0.1049	77	53.10	<.0001	0.05	5.3584	5.7759
tall	6.2486	0.1049	77	59.59	<.0001	0.05	6.0398	6.4574

Contrasts

Label	Num DF	Den DF	F Value	Pr > F
small-medium	1	77	3.71	0.0579
small-tall	1	77	40.20	<.0001
medium-tall	1	77	21.12	<.0001

- The differences in slopes is mainly explained from the difference between the third group on one hand, and the other two groups on the other hand.

9.4 Random Intercepts Model

- Our fitted model was:

$$Y_{ij} = \begin{cases} (\beta_1 + b_i) + \beta_2 t_{ij} + \varepsilon_{ij}, & \text{if short mother} \\ (\beta_3 + b_i) + \beta_4 t_{ij} + \varepsilon_{ij}, & \text{if medium mother} \\ (\beta_5 + b_i) + \beta_6 t_{ij} + \varepsilon_{ij}, & \text{if tall mother} \end{cases}$$

- This can be interpreted as a ANOCOVA model, but with child-specific intercepts b_i

- Such a b_i represents the deviation of the intercept of a specific child from the average intercept in the group to which that child belongs, i.e., deviation from β_1, β_2, or β_3.

- An alternative way to fit a random intercepts model in PROC MIXED is:

```
proc mixed data = growth;
class group child;
model height = age group age*group / solution;
random intercept / subject=child;
run;
```

- The results are identical to those discussed earlier.

9.5 Remarks

- The growth-curve dataset is an example of a longitudinal dataset
- In longitudinal data, there is a natural ordering of the measurements within clusters
- The ordering is of primary interest
- Our random-intercepts model implies very strong assumptions:
  - ▷ Parallel profiles within all 3 groups
  - ▷ Constant variance $\sigma^2 = \sigma^2_{child} + \sigma^2_{res}$
  - ▷ Constant correlation within children: $\sigma^2_{child}/(\sigma^2_{child} + \sigma^2_{res})$
- In many longitudinal settings, these assumptions are too restrictive

9.6 Linear Mixed Models

- One way to extend the random-intercepts model is to allow also the slopes to be subject-specific:

$$Y_{ij} = \begin{cases} (\beta_1 + b_{1i}) + (\beta_2 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if short mother} \\ (\beta_3 + b_{1i}) + (\beta_4 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if medium mother} \\ (\beta_5 + b_{1i}) + (\beta_6 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if tall mother} \end{cases}$$

- This is an example of the general linear mixed model

- As before, the random effects are assumed to be normally distributed with mean zero:

$$\boldsymbol{b_i} = (b_{1i}, b_{2i})' \sim N(\mathbf{0}, D)$$

- D then equals the 2×2 covariance matrix of the random effects:

$$D = \begin{pmatrix} d_{11} & d_{12} \\ d_{12} & d_{22} \end{pmatrix}$$

- Interpretation of the parameters:
  - ▷ d_{11} equals the variance of the intercepts b_{1i}
  - ▷ d_{22} equals the variance of the slopes b_{2i}
  - ▷ d_{12} equals the covariance between the intercepts b_{1i} and the slopes b_{2i}.
  - ▷ The correlation between the intercepts and slopes then equals:

$$\text{Corr}(b_{1i}, b_{2i}) = \frac{d_{12}}{\sqrt{d_{11}}\sqrt{d_{22}}}$$

- Random-intercepts models imply constant variance and constant correlation between any two outcomes of the same cluster (see earlier).

- The above model with random intercepts and slopes implies:

  ▷ Variance function:

$$\text{Var}(\boldsymbol{Y_i}(t)) \;=\; d_{22}t^2 + 2d_{12}t + d_{11} + \sigma^2$$

  ▷ Correlation function:

$$\text{Corr}(\boldsymbol{Y_i}(t_1), \boldsymbol{Y_i}(t_2)) \;=\; \frac{d_{22}t_1t_2 + d_{12}(t_1+t_2) + d_{11}}{\sqrt{d_{22}t_1^2 + 2d_{12}t_1 + d_{11} + \sigma^2}\;\sqrt{d_{22}t_2^2 + 2d_{12}t_2 + d_{11} + \sigma^2}}$$

- More complicated random-effects structures will yield more complicated variance and correlations functions.

- SAS program:

```
proc mixed data=growth;
class child group;
model height=age group age*group;
random intercept age / type=un subject=child g gcorr;
run;
```

- As before, fixed effects are to be specified in the MODEL statement, while random effects are specified in the RANDOM statement.

- Relevant SAS output:

```
 Covariance Parameter Estimates

Cov Parm     Subject    Estimate

UN(1,1)      CHILD        7.6028
UN(2,1)      CHILD       -0.4437
UN(2,2)      CHILD        0.1331
Residual                  0.4758
```

Estimated G Matrix

Row	Effect	CHILD	Col1	Col2
1	Intercept	1	7.6028	-0.4437
2	AGE	1	-0.4437	0.1331

Estimated G Correlation Matrix

Row	Effect	CHILD	Col1	Col2
1	Intercept	1	1.0000	-0.4412
2	AGE	1	-0.4412	1.0000

Type 3 Tests of Fixed Effects

Effect	Num DF	Den DF	F Value	Pr > F
AGE	1	17	3572.36	<.0001
GROUP	2	60	0.60	0.5514
AGE*GROUP	2	60	9.23	0.0003

- Note the differences in test results for the fixed effects, when compared to the random-intercepts model:

Type 3 Tests of Fixed Effects

Effect	Num DF	Den DF	F Value	Pr > F
AGE	1	77	8385.15	<.0001
GROUP	2	77	0.46	0.6330
AGE*GROUP	2	77	21.66	<.0001

Chapter 10
The General Linear Mixed Model

- ▷ The general model
- ▷ Examples

10.1 The General Model

- Let Y_{ij} be response j for cluster i, $i = 1, \ldots, N$, $j = 1, \ldots, n_i$
- Examples:
  - ▷ Y_{ij} is the number of dorsal shells for lizard j within mother i
  - ▷ Y_{ij} is the height of child i at visit j
- The response vector for cluster i equals:

$$\boldsymbol{Y_i} = (Y_{i1}, Y_{i2}, \ldots, Y_{in_i})'$$

- A linear mixed model is a linear regression model for each cluster separately, with fixed as well as random regression coefficients.

- Formally:

$$\begin{cases} \boldsymbol{Y_i} = X_i\boldsymbol{\beta} + Z_i\boldsymbol{b_i} + \boldsymbol{\varepsilon_i} \\ \boldsymbol{b_i} \sim N(\mathbf{0}, D), \\ \boldsymbol{\varepsilon_i} \sim N(\mathbf{0}, \sigma^2 I_{n_i}), \\ \boldsymbol{b_1}, \ldots, \boldsymbol{b_N}, \boldsymbol{\varepsilon_1}, \ldots, \boldsymbol{\varepsilon_N} \text{ independent,} \end{cases}$$

- X_i and Z_i are design matrices
- The vector $\boldsymbol{\beta}$ contains all regression parameters which are the same for all clusters
- The vector $\boldsymbol{b_i}$ contains all cluster-specific parameters
- $\boldsymbol{\beta}$ describes average trends in the population

- $\boldsymbol{b_i}$ describes how a specific cluster deviates from the average trend
- As before, the $\boldsymbol{b_i}$ are normally distributed with mean zero and covariance matrix D
- The vector $\boldsymbol{\varepsilon_i}$ contains the measurement error components which are normally distributed with mean zero and variance σ^2
- Terminology:
  - ▷ **Fixed effects:** $\boldsymbol{\beta}$
  - ▷ **Random effects:** $\boldsymbol{b_i}$
  - ▷ **Variance components:** σ^2 and all elements in D

10.2 Example: The Lizard Data

- Our model was given by:

$$Y_{ij} = \mu + \alpha_i + \beta t_{ij} + \varepsilon_{ij}$$

- Fixed effects μ and β, random effects α_i
- The average response is given by μ for females and $\mu + \beta$ for males
- α_i represents how mother i deviates from the overall mean (the mother-effect).

10.3 Example: The Growth Curves

- Our extended model was given by:

$$Y_{ij} = \begin{cases} (\beta_1 + b_{1i}) + (\beta_2 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if short mother} \\ (\beta_3 + b_{1i}) + (\beta_4 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if medium mother} \\ (\beta_5 + b_{1i}) + (\beta_6 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if tall mother} \end{cases}$$

- Fixed effects $\beta_1, \ldots, \beta_6$, random effects b_{1i} and b_{2i}.

- $\beta_1, \beta_3, \beta_5$ represent the average intercepts

- $\beta_2, \beta_4, \beta_6$ represent the average slopes

- b_{1i} expresses how much the intercept of child i deviates from the average intercept in the group to which this child belongs

- b_{2i} expresses how much the slope of child i deviates from the average slope in the group to which this child belongs

Chapter 11
Example: The Rat Data

- The data
- A linear mixed model
- Fitting the model in SAS

11.1 Example

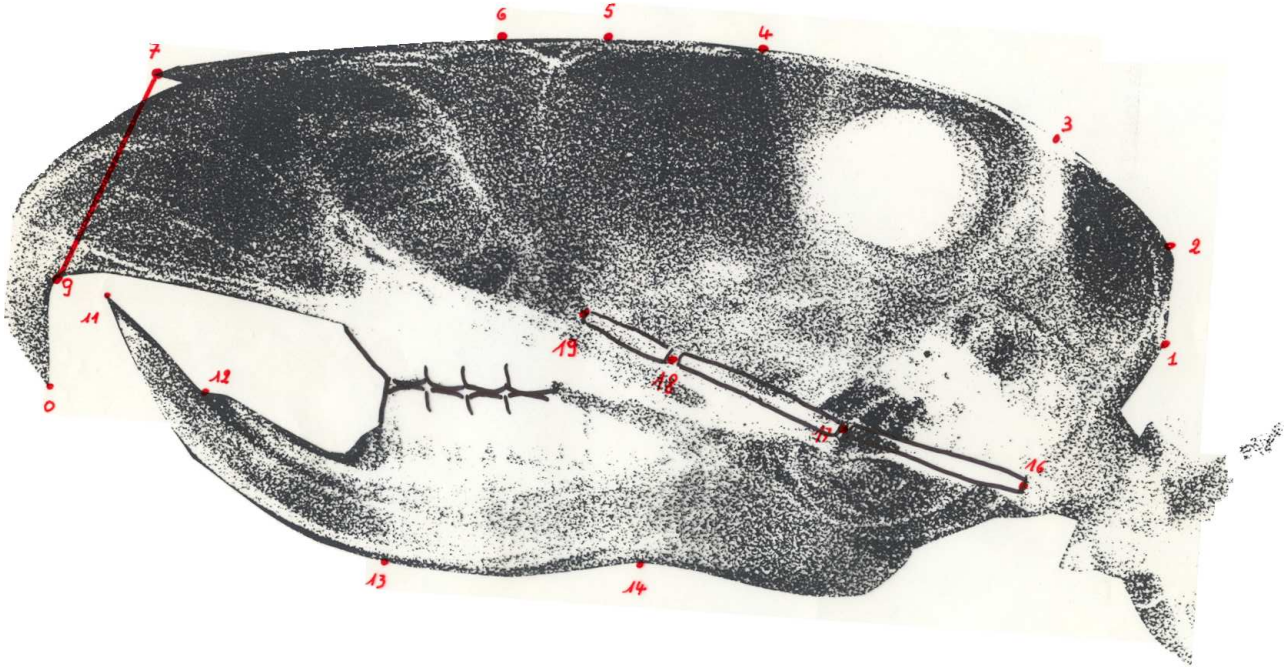
- Research question (Dentistry, K.U.Leuven):

> How does craniofacial growth depend on testosteron production ?

- Randomized experiment in which 50 male Wistar rats are randomized to:
  - ▷ Control (15 rats)
  - ▷ Low dose of Decapeptyl (18 rats)
  - ▷ High dose of Decapeptyl (17 rats)

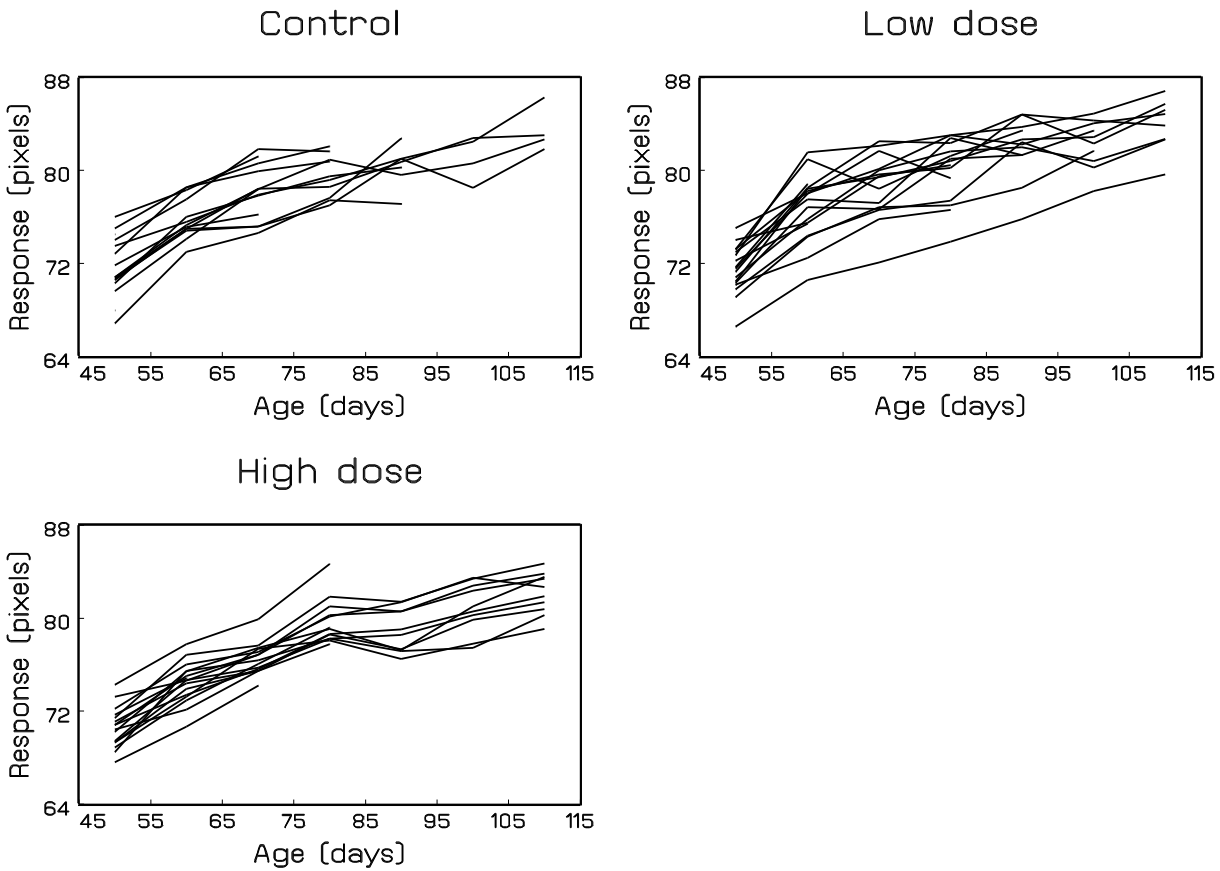
- Treatment starts at the age of 45 days; measurements taken every 10 days, from day 50 on.

- The responses are distances (pixels) between well defined points on x-ray pictures of the skull of each rat:



- Measurements with respect to the roof, base and height of the skull. Here, we consider only one response, reflecting the height of the skull.

- Individual profiles:



- Complication: Dropout due to anaesthesia (56%):

	# Observations			
Age (days)	Control	Low	High	Total
50	15	18	17	50
60	13	17	16	46
70	13	15	15	43
80	10	15	13	38
90	7	12	10	29
100	4	10	10	24
110	4	8	10	22

- Remarks:
  - ▷ Much variability between rats, much less variability within rats
  - ▷ Fixed number of measurements scheduled per subject, but not all measurements available due to dropout, for known reason.
  - ▷ Measurements taken at fixed time points

11.2 A Linear Mixed Model

- Since linear mixed models assume a linear regression for each cluster separately, they can also be used for unbalanced data, i.e., data with unequal number of measurements per cluster.

- Note that this was also the case for the lizard data.

- Individual profiles show very similar evolutions for all rats (apart from measurement error)

- This suggests a random-intercepts model

- Non-linearity can be accounted for by using a logarithmic transformation of the time scale:

$$\text{Age}_{ij} \longrightarrow t_{ij} = \ln[1 + (\text{Age}_{ij} - 45)/10)]$$

- We then get the following model:

$$Y_{ij} = (\beta_0 + b_i) + (\beta_1 L_i + \beta_2 H_i + \beta_3 C_i) t_{ij} + \varepsilon_{ij}$$

$$= \begin{cases} \beta_0 + b_i + \beta_1 t_{ij} + \varepsilon_{ij}, & \text{if low dose} \\ \beta_0 + b_i + \beta_2 t_{ij} + \varepsilon_{ij}, & \text{if high dose} \\ \beta_0 + b_i + \beta_3 t_{ij} + \varepsilon_{ij}, & \text{if control.} \end{cases}$$

- L_i, H_i, and C_i are indicator variables:

$$L_i = \begin{cases} 1 & \text{if low dose} \\ 0 & \text{otherwise} \end{cases} \qquad H_i = \begin{cases} 1 & \text{if high dose} \\ 0 & \text{otherwise} \end{cases} \qquad C_i = \begin{cases} 1 & \text{if control} \\ 0 & \text{otherwise} \end{cases}$$

- Parameter interpretation:
  - ▷ β_0: average response at the start of the treatment (independent of treatment)
  - ▷ β_1, β_2, and β_3: average time effect for each treatment group
  - ▷ b_i: subject-specific intercepts

11.3 Fitting the Model in SAS

- The following SAS program can be used:

```
data rats;
set rats;
t=log(1+(age-45)/10);
run;
```

```
proc mixed data = rats ;
class treat rat;
model y = treat*t / solution;
random intercept / type=un subject=rat g;
contrast 'treatment effect' treat*t 1 -1 0, treat*t 1 0 -1;
run;
```

- Note the parameterization of the fixed effects

- Relevant SAS output:

```
 Covariance Parameter Estimates

Cov Parm     Subject    Estimate

UN(1,1)      RAT          3.5649
Residual                  1.4448
```

Solution for Fixed Effects

Effect	TREAT	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept		68.6074	0.3312	49	207.13	<.0001
t*TREAT	con	7.3138	0.2808	199	26.05	<.0001
t*TREAT	hig	6.8711	0.2276	199	30.19	<.0001
t*TREAT	low	7.5069	0.2252	199	33.34	<.0001

Type 3 Tests of Fixed Effects

Effect	Num DF	Den DF	F Value	Pr > F
t*TREAT	3	199	734.11	<.0001

Contrasts

Label	Num DF	Den DF	F Value	Pr > F
treatment effect	2	199	2.32	0.1013

- Note the difference between the test for 't*treat' and the test for the treatment effect

- A lot of variability between rats, while little variability within rats:
  - ▷ $\sigma^2_{rat} = 3.565$ represents the variability between rats
  - ▷ $\sigma^2_{res} = 1.445$ represents the variability within rats

- No significant difference between the treatment groups with respect to the average evolution over time ($p = 0.1013$)

Chapter 12
Example: The BLSA Prostate Data

- The data
- A linear mixed model
- Fitting the model in SAS

12.1 Example

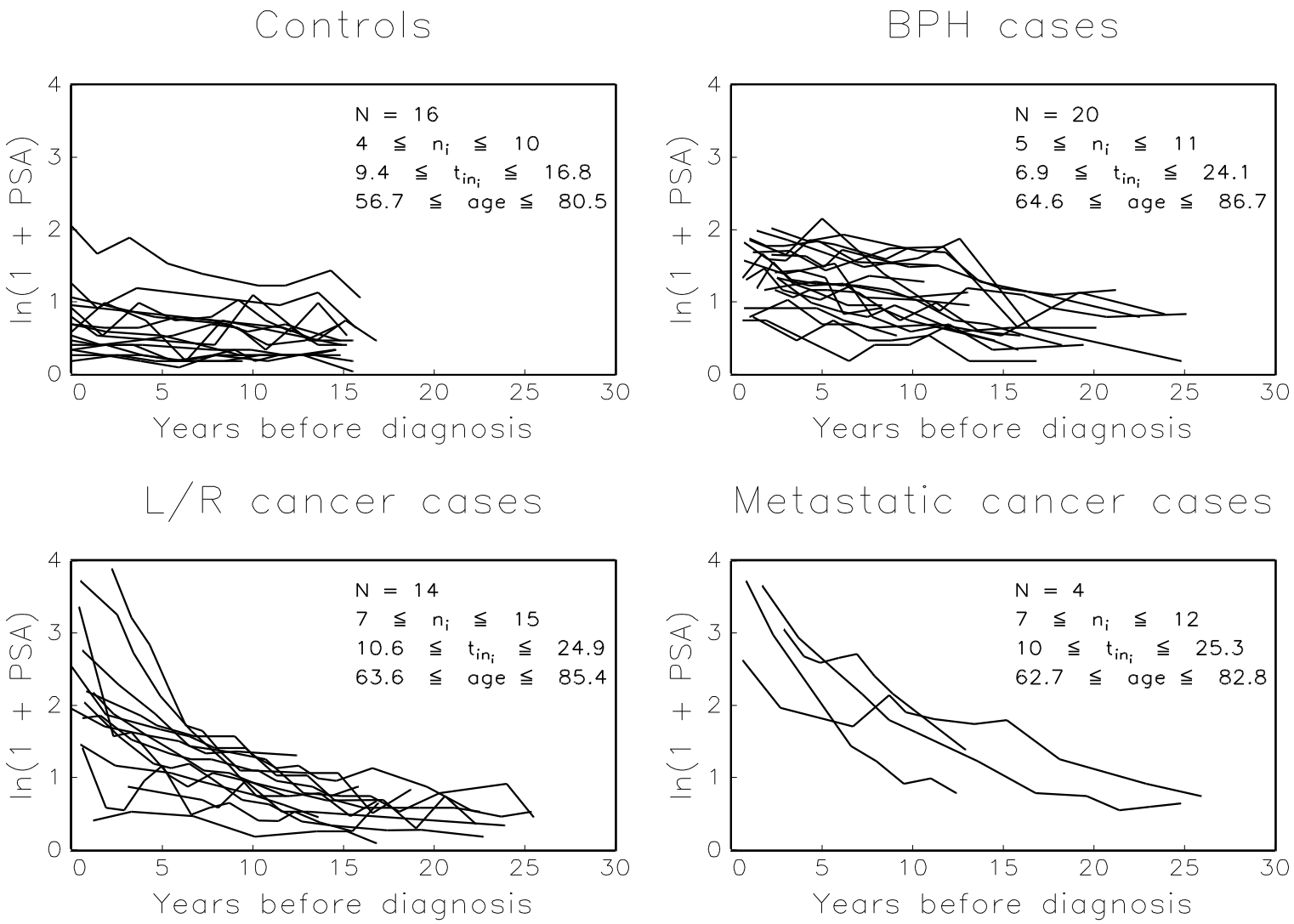
- References:
  - ▷ Carter *et al* (1992, Cancer Research).
  - ▷ Carter *et al* (1992, Journal of the American Medical Association).
  - ▷ Morrell *et al* (1995, Journal of the American Statistical Association).
  - ▷ Pearson *et al* (1994, Statistics in Medicine).
- Prostate disease is one of the most common and most costly medical problems in the United States
- Important to look for markers which can detect the disease at an early stage
- **P**rostate-**S**pecific **A**ntigen is an enzyme produced by both normal and cancerous prostate cells

- PSA level is related to the volume of prostate tissue.
- Problem: Patients with **B**enign **P**rostatic **H**yperplasia also have an increased PSA level
- Overlap in PSA distribution for cancer and BPH cases seriously complicates the detection of prostate cancer.
- Research question (hypothesis based on clinical practice):

> Can longitudinal PSA profiles be used to detect prostate cancer in an early stage ?

- A retrospective case-control study based on frozen serum samples:
  - ▷ 16 control patients
  - ▷ 20 BPH cases
  - ▷ 14 local cancer cases
  - ▷ 4 metastatic cancer cases
- Complication: No perfect match for age at diagnosis and years of follow-up possible
- Hence, analyses will have to correct for these age differences between the diagnostic groups.

- Individual profiles:



- Remarks:
  - ▷ Much variability between subjects
  - ▷ Little variability within subjects
  - ▷ Highly unbalanced data

12.2 A Linear Mixed Model

- A model for the prostate data:

$$\begin{aligned}\ln(\text{PSA}_{ij}+1) \;=\; & \beta_1\text{Age}_i + \beta_2 C_i + \beta_3 B_i + \beta_4 L_i + \beta_5 M_i \\ & + (\beta_6\text{Age}_i + \beta_7 C_i + \beta_8 B_i + \beta_9 L_i + \beta_{10} M_i)\, t_{ij} \\ & + (\beta_{11}\text{Age}_i + \beta_{12} C_i + \beta_{13} B_i + \beta_{14} L_i + \beta_{15} M_i)\, t_{ij}^2 \\ & + b_{1i} \;+\; b_{2i} t_{ij} \;+\; b_{3i} t_{ij}^2 \;+\; \varepsilon_{ij}.\end{aligned}$$

- C_i, B_i, L_i, M_i are indicators for the 4 diagnostic groups.

- Parameter interpretation:
  - ▷ Average age-corrected quadratic profiles for all groups, modeled through the fixed effects in $\boldsymbol{\beta}$
  - ▷ Random effects b_{1i}, b_{2i}, and b_{3i} allowing subject-specific evolutions to differ from the average in that diagnostic group, even correcting for age differences

12.3 Fitting the Model in SAS

- SAS program:

```
proc mixed data=prostate;
class id group;
model lnpsa = group age group*time age*time group*time2 age*time2 / noint solution;
random intercept time time2 / type=un subject=id g gcorr ;
run;
```

- Note again the particular parameterization for the fixed effects

- Relevant SAS output:

Covariance Parameter Estimates

Cov Parm	Subject	Estimate
UN(1,1)	XRAY	0.4518
UN(2,1)	XRAY	-0.5178
UN(2,2)	XRAY	0.9153
UN(3,1)	XRAY	0.1625
UN(3,2)	XRAY	-0.3356
UN(3,3)	XRAY	0.1308
Residual		0.02820

Estimated G Matrix

Effect	XRAY	Col1	Col2	Col3
Intercept	19	0.4518	-0.5178	0.1625
time	19	-0.5178	0.9153	-0.3356
time2	19	0.1625	-0.3356	0.1308

Estimated G Correlation Matrix

Effect	XRAY	Col1	Col2	Col3
Intercept	19	1.0000	-0.8053	0.6686
time	19	-0.8053	1.0000	-0.9700
time2	19	0.6686	-0.9700	1.0000

Solution for Fixed Effects

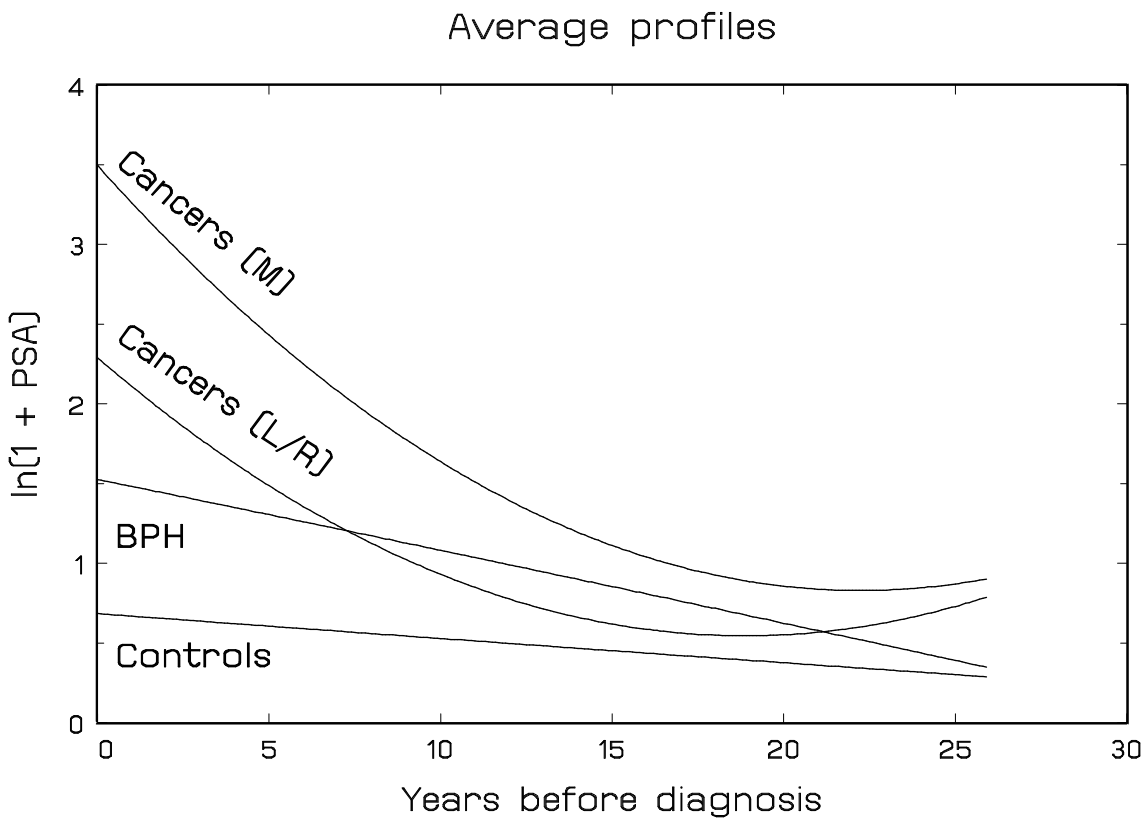
Effect	group	Estimate	Standard Error	DF	t Value	Pr > |t|
group	1	-1.0984	0.9763	299	-1.13	0.2615
group	2	-0.5228	1.0895	299	-0.48	0.6317
group	3	0.2964	1.0587	299	0.28	0.7797
group	4	1.5494	1.0856	299	1.43	0.1546
AGEDIAG		0.02655	0.01423	299	1.87	0.0631
time*group	1	0.5681	1.4725	299	0.39	0.6999
time*group	2	0.3956	1.6377	299	0.24	0.8093
time*group	3	-1.0359	1.5928	299	-0.65	0.5159
time*group	4	-1.6049	1.6258	299	-0.99	0.3244
AGEDIAG*time		-0.01117	0.02142	299	-0.52	0.6026
time2*group	1	-0.1295	0.6100	299	-0.21	0.8320
time2*group	2	-0.1585	0.6723	299	-0.24	0.8138
time2*group	3	0.3419	0.6563	299	0.52	0.6028
time2*group	4	0.3951	0.6660	299	0.59	0.5535
AGEDIAG*time2		0.002259	0.008829	299	0.26	0.7982

Type 3 Tests of Fixed Effects

Effect	Num DF	Den DF	F Value	Pr > F
group	4	299	15.90	<.0001
AGEDIAG	1	299	3.48	0.0631
time*group	4	299	7.85	<.0001
AGEDIAG*time	1	299	0.27	0.6026
time2*group	4	299	4.44	0.0017
AGEDIAG*time2	1	299	0.07	0.7982

- Note the very strong correlations between random effects
- CONTRAST statements can be used to test for group differences

- Based on the fixed effects, fitted average profiles can be plotted (at median age at diagnosis):



Chapter 13
Estimation of Random Effects

- ▷ Empirical Bayes estimation
- ▷ Example: Growth curves
- ▷ Example: Prostate data
- ▷ Average versus cluster-specific prediction

13.1 Empirical Bayes Estimation

- Random effects $\boldsymbol{b_i}$ reflect how the ith cluster deviates from the population average
- For example, for the growth curves:

$$Y_{ij} = \begin{cases} (\beta_1 + b_{1i}) + (\beta_2 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if short mother} \\ (\beta_3 + b_{1i}) + (\beta_4 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if medium mother} \\ (\beta_5 + b_{1i}) + (\beta_6 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if tall mother} \end{cases}$$

- b_{1i} and b_{2i} express how much the intercept and slope of child i deviate from the average intercept and slope in the group to which this child belongs
- Estimation of the $\boldsymbol{b_i}$ helpful for detecting outlying profiles or clusters
- Since the parameters $\boldsymbol{b_i}$ are assumed to be stochastic, Bayesian methods are

applied.

- Posterior means:

$$\widehat{\boldsymbol{b_i}} = E(\boldsymbol{b_i} \mid \boldsymbol{Y_i} = \boldsymbol{y_i})$$

- The so-obtained estimates $\widehat{\boldsymbol{b_i}}$ are called Empirical Bayes (EB) estimates.

- In practice histograms and/or scatterplots of EB estimates are used to detect outlying clusters

13.2 Example: Growth Curves

- We re-consider the extended model:

$$Y_{ij} = \begin{cases} (\beta_1 + b_{1i}) + (\beta_2 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if short mother} \\ (\beta_3 + b_{1i}) + (\beta_4 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if medium mother} \\ (\beta_5 + b_{1i}) + (\beta_6 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if tall mother} \end{cases}$$

- SAS program for calculation of EB estimates:

```
proc mixed data=growth;
class child group;
model height=age group age*group;
random intercept age / type=un subject=child solution;
ods listing exclude solutionr;
ods output solutionr=out;
run;
```

- The ODS statements are used to write the EB estimates into a SAS output data set, and to prevent SAS from printing them in the output window.

- SAS data management steps:

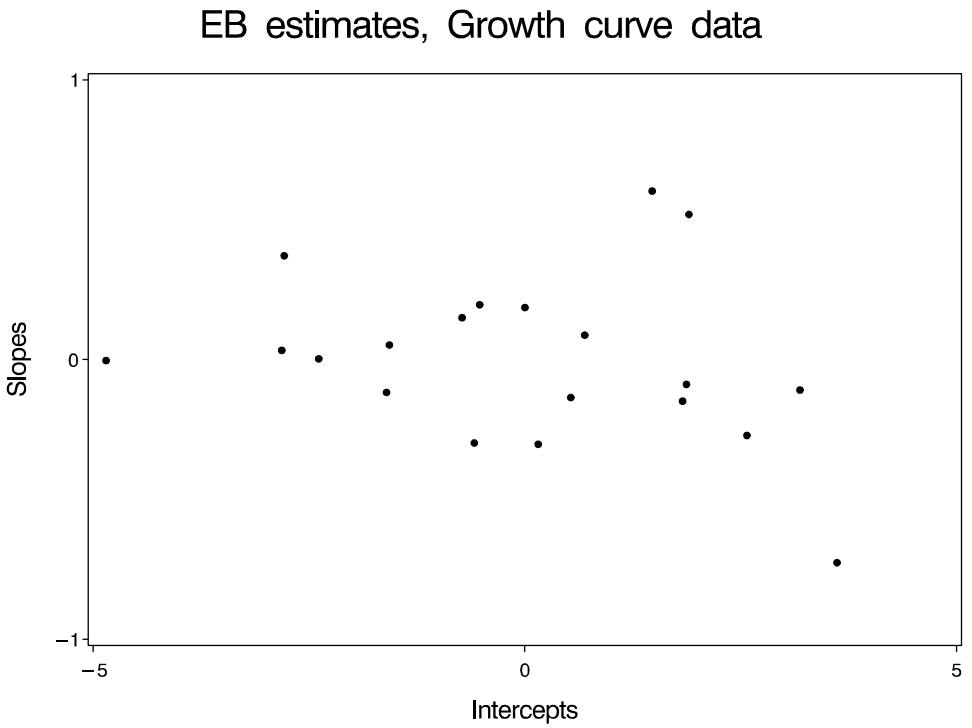
```
data int;set out;if effect='Intercept';
         int=estimate;keep int child;
data slope;set out;if effect='AGE';
         slope=estimate;keep slope child;
data eb;merge int slope; by child;
proc print;run;
```

- SAS program for scatterplot of slopes versus intercepts:

```
proc gplot data=eb;
plot slope*int  / haxis=axis1 vaxis=axis2;
symbol c=black value=dot w=2 i=none;
axis1 label=(h=2  'Intercepts') value=(h=1.5) order=(-5 to 5 by 5) minor=none;
axis2 label=(h=2 A=90 'Slopes') value=(h=1.5) order=(-1 to 1 by 1) minor=none;
title h=3 'EB estimates, Growth curve data';
run;quit;
```

- EB estimates (and scatterplot):

Obs	CHILD	int	slope
1	1	0.15504	-0.30272
2	2	-2.38577	0.00284
3	3	0.00204	0.18604
4	4	1.82784	-0.14857
5	5	3.18613	-0.10875
6	6	-2.78528	0.37117
7	7	0.53293	-0.13570
8	8	1.87116	-0.08839
9	9	1.90073	0.51893
10	10	-0.58431	-0.29783
11	11	0.69442	0.08735
12	12	-1.60125	-0.11740
13	13	-2.81368	0.03303
14	14	3.61566	-0.72571
15	15	-0.52214	0.19608
16	16	-1.56764	0.05223
17	17	-0.72546	0.14969
18	18	2.57185	-0.27115
19	19	-4.84795	-0.00349
20	20	1.47568	0.60234

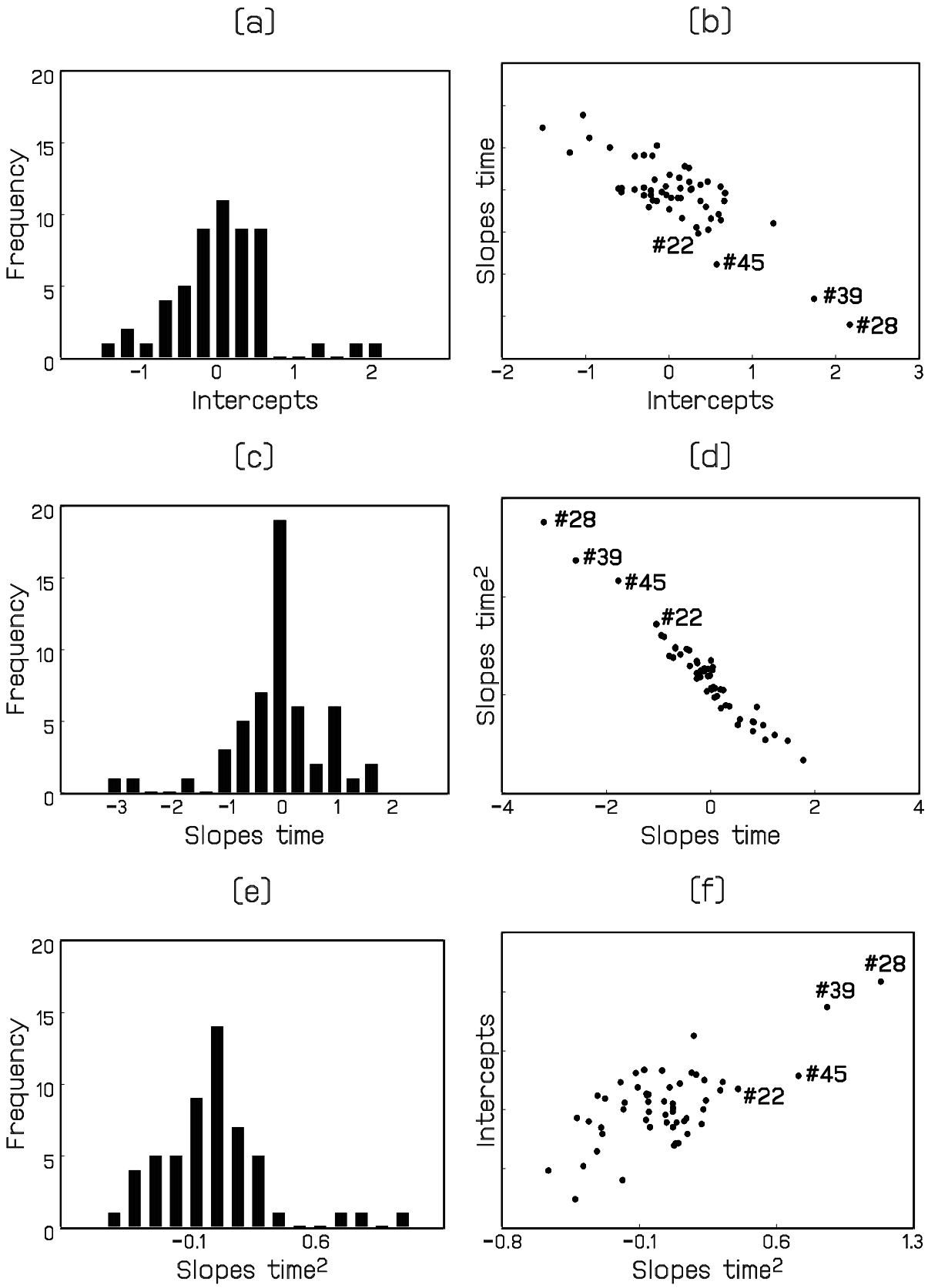


13.3 Example: Prostate Data

- We re-consider the model

$$\begin{aligned}
&\ln(\mathsf{PSA}_{ij}+1)\\
&= \beta_1\mathsf{Age}_i + \beta_2 C_i + \beta_3 B_i + \beta_4 L_i + \beta_5 M_i\\
&\quad + (\beta_6\mathsf{Age}_i + \beta_7 C_i + \beta_8 B_i + \beta_9 L_i + \beta_{10} M_i)\, t_{ij}\\
&\quad + (\beta_{11}\mathsf{Age}_i + \beta_{12} C_i + \beta_{13} B_i + \beta_{14} L_i + \beta_{15} M_i)\, t_{ij}^2\\
&\quad + b_{1i} + b_{2i} t_{ij} + b_{3i} t_{ij}^2 + \varepsilon_{ij}.
\end{aligned}$$

- Again, histograms and scatterplots of components of $\widehat{\boldsymbol{b_i}}$ can be used to detect model deviations or subjects with 'exceptional' evolutions over time


(a)
Frequency
Intercepts
(b)
Slopes time
Intercepts
#22
#45
#39
#28
(c)
Frequency
Slopes time
(d)
Slopes time2
Slopes time
#28
#39
#45
#22
(e)
Frequency
Slopes time2
(f)
Intercepts
Slopes time2
#28
#39
#45
#22

- Strong negative correlations in agreement with correlation matrix corresponding to fitted D:

$$\widehat{D}_{\text{corr}} = \begin{pmatrix} 1.000 & -0.805 & 0.669 \\ -0.805 & 1.000 & -0.970 \\ 0.669 & -0.970 & 1.000 \end{pmatrix}$$

- Histograms and scatterplots show outliers

- Subjects #22, #28, #39, and #45, have highest four slopes for time^2 and smallest four slopes for time, i.e., with the strongest (quadratic) growth.

- Subjects #22, #28 and #39 have been further examined and have been shown to be metastatic cancer cases which were misclassified as local cancer cases.

- Subject #45 is the metastatic cancer case with the strongest growth

13.4 Average versus Cluster-specific Prediction

- Once the EB estimates have been calculated, predictions can be obtained both at the cluster level, as well as on the population average level.

- Re-consider the general linear mixed model:

$$\boldsymbol{Y_i} = X_i\boldsymbol{\beta} + Z_i\boldsymbol{b_i} + \boldsymbol{\varepsilon_i}$$

- Predictions:
  - ▷ On population average level:

$$E(\boldsymbol{Y_i}) = X_i\widehat{\boldsymbol{\beta}}$$

  - ▷ On cluster level:

$$\widehat{\boldsymbol{Y_i}} = X_i\widehat{\boldsymbol{\beta}} + Z_i\widehat{\boldsymbol{b_i}}$$

13.5 Example: Growth Curves

- We re-consider the extended model:

$$Y_{ij} = \begin{cases} (\beta_1 + b_{1i}) + (\beta_2 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if short mother} \\ (\beta_3 + b_{1i}) + (\beta_4 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if medium mother} \\ (\beta_5 + b_{1i}) + (\beta_6 + b_{2i})t_{ij} + \varepsilon_{ij}, & \text{if tall mother} \end{cases}$$

- SAS program for predictions:

```
proc mixed data=growth;
class child group;
model height= group age*group / noint solution
      outpm=predmean outp=pred;
random intercept age / type=un subject=child solution;
id child age height;
run;
```

```
proc print data=predmean;
proc print data=pred;
run;
```

- Table of predicted means:

Obs	CHILD	AGE	HEIGHT	Pred	StdErr Pred
1	1	6	111.0	112.920	1.10709
2	1	7	116.4	118.190	1.15855
3	1	8	121.7	123.460	1.23249
4	1	9	126.3	128.730	1.32516
5	1	10	130.5	134.000	1.43293
6	2	6	110.0	112.920	1.10709
98	20	8	141.3	133.111	1.14107
99	20	9	146.8	139.360	1.22686
100	20	10	152.3	145.609	1.32664

- Table with predictions on subject level:

Obs	CHILD	AGE	HEIGHT	Pred	StdErr Pred
1	1	6	111.0	111.259	0.47163
2	1	7	116.4	116.226	0.35258
3	1	8	121.7	121.193	0.30673
4	1	9	126.3	126.161	0.36296
5	1	10	130.5	131.128	0.48713
6	2	6	110.0	110.551	0.47163
98	20	8	141.3	139.406	0.30668
99	20	9	146.8	146.257	0.36253
100	20	10	152.3	153.108	0.48571

- Components needed to calculate predictions:

Solution for Fixed Effects

Effect	GROUP	Estimate	Standard Error	DF	t Value	Pr > |t|
GROUP	1	81.3000	1.3381	60	60.76	<.0001
GROUP	2	82.9743	1.2388	60	66.98	<.0001
GROUP	3	83.1229	1.2388	60	67.10	<.0001
AGE*GROUP	1	5.2700	0.1735	60	30.37	<.0001
AGE*GROUP	2	5.5671	0.1606	60	34.66	<.0001
AGE*GROUP	3	6.2486	0.1606	60	38.90	<.0001

Solution for Random Effects

Effect	CHILD	Estimate	Std Err Pred	DF	t Value	Pr > |t|
Intercept	1	0.1550	1.7264	60	0.09	0.9287
AGE	1	-0.3027	0.2195	60	-1.38	0.1730

- Population average prediction for Child #1 (in group 1), at the age of 6 years old:

$$E(Y_{11}) \;=\; 81.30 + 6 \times 5.27 \;=\; 112.92$$

- Subject-specific prediction for Child #1 (in group 1), at the age of 6 years old:

$$\widehat{Y_{11}} \;=\; (81.30 + 0.1550) + 6 \times (5.27 - 0.3027) \;=\; 111.259$$

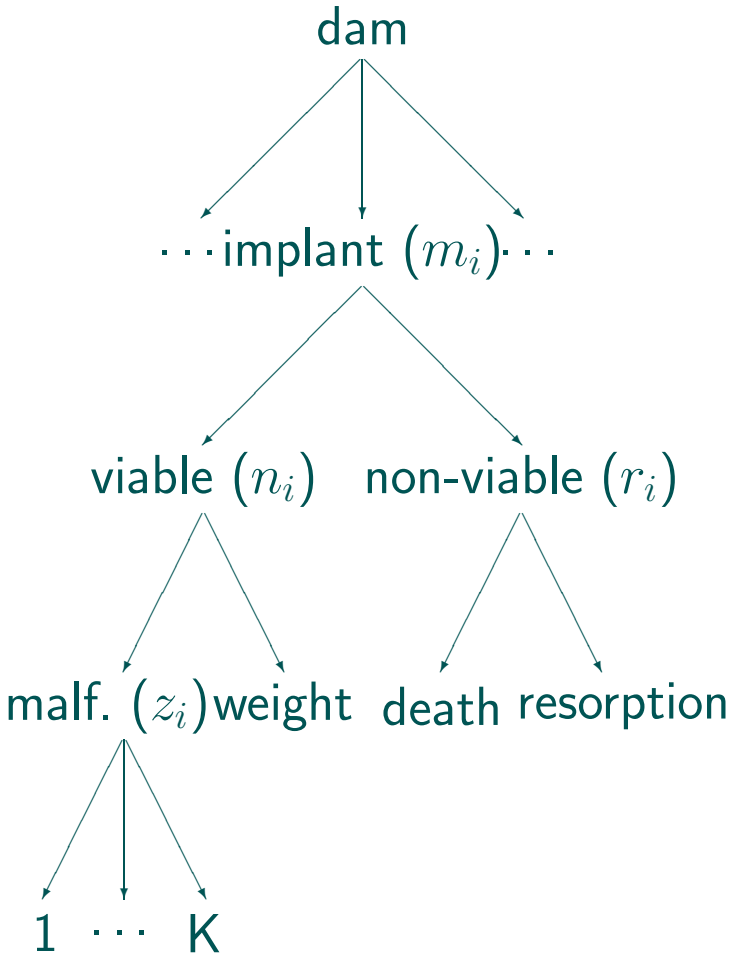
Part IV

Clustered Data

Chapter 14
Clustered Data

- Developmental toxicity studies
- Analysis of a continuous outcome
- Model families for discrete outcomes
- Marginal models
- Conditional models
- Random-effecs models

14.1 Data Structure of Developmental Toxicity Studies



14.2 The NTP Studies

Developmental Toxicity Studies

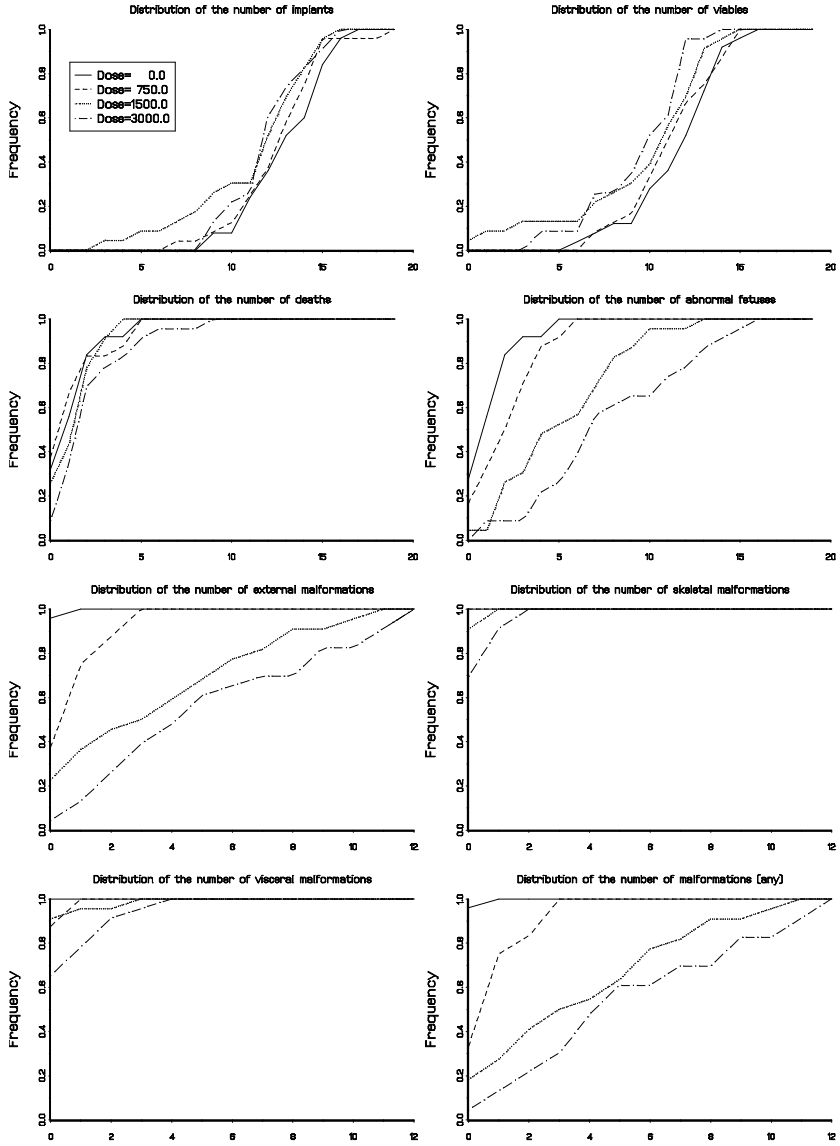
- Research Triangle Institute US National Toxicology Program
- Segment II studies
- The effect in mice of 3 chemicals:
  - ▷ DEHP: di(2-ethyhexyl)-phtalate
  - ▷ EG: ethylene glycol
  - ▷ DYME: diethylene glycol dimethyl ether

14.2.1 EG Study in Mice

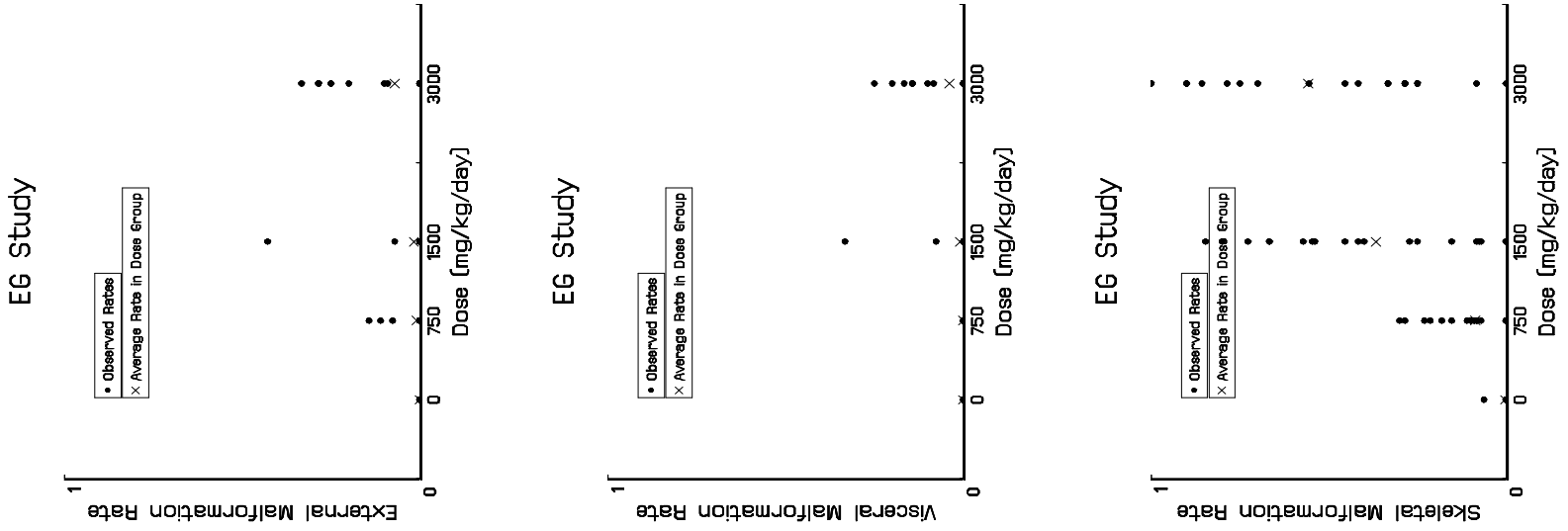
- Timed-pregnant CD-1 mice were dosed by gavage with EG in distilled water.
- Dosing occurred during the period of organogenesis and structural development of the foetuses (gestational days 8 through 15).
- Doses 0, 750, 1500, and 3000 mg/kg/day.
- Clusters consisting of 10–15 implants occur frequently.

Dose	Dams	Live	Litter Size		Malformation								Weight	
			Mean	SD	Ext.		Visc.		Skel.		Coll.		Mean	SD
					Nr.	%	Nr.	%	Nr.	%	Nr.	%		
0.00	25	297	11.9	2.45	0	0.0	0	0.0	1	0.0	1	0.3	0.97	0.09
0.75	24	276	11.5	2.38	3	1.0	0	0.0	24	8.7	26	9.4	0.88	0.10
1.50	22	229	10.4	3.46	4	1.7	2	0.8	84	36.7	89	38.9	0.76	0.10
3.00	23	226	9.8	2.69	16	7.1	9	3.9	126	55.8	129	57.1	0.70	0.12

14.2.2 EG Study in Mice: Cumulative Relative Frequencies



14.2.3 EG Study in Mice: Observed and Averaged Malformation Rates



EG Study in Rats

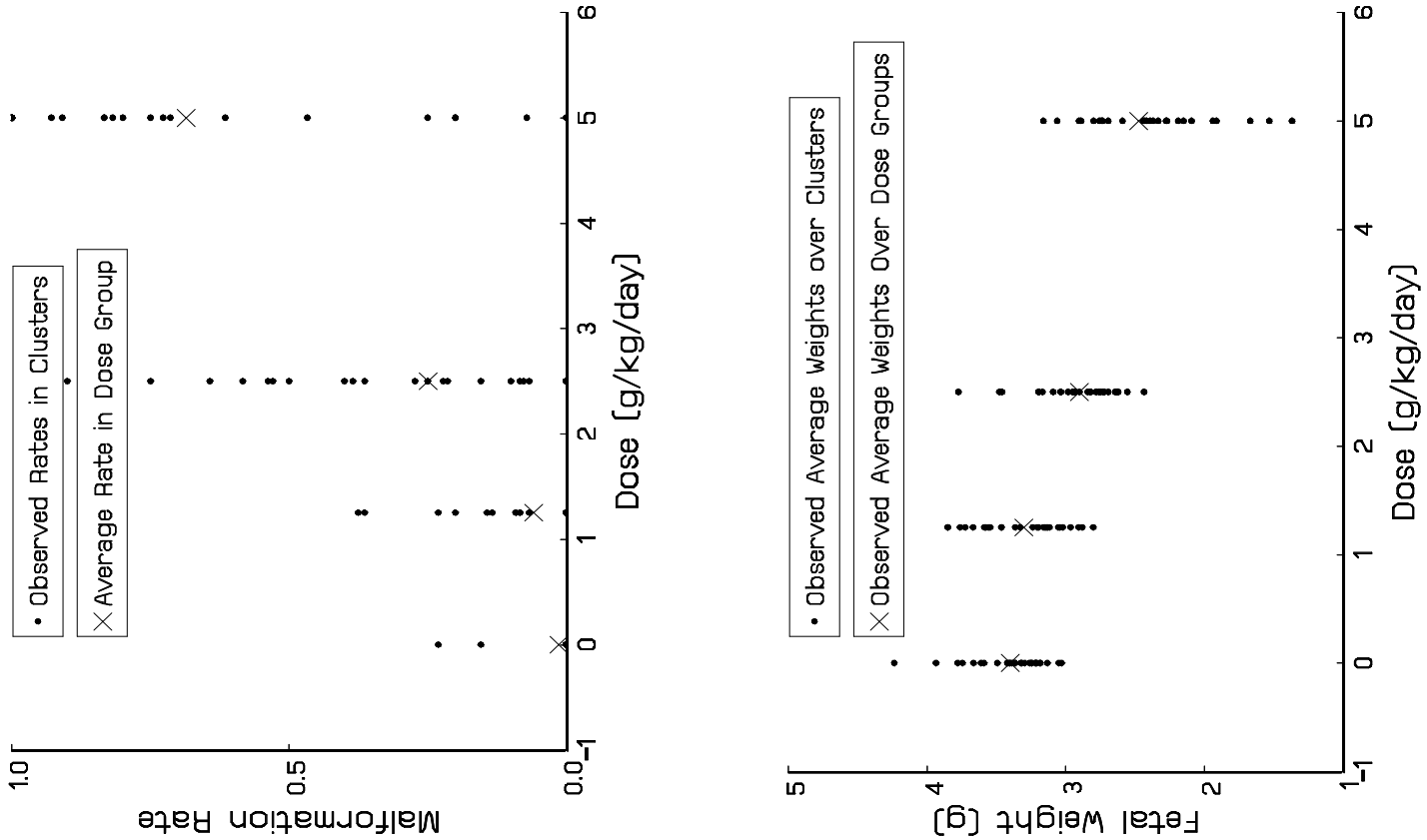
- Another experiment, investigating the effect of EG in rats.
- Doses selected: 0, 1.25, 2.50, and 5.0 g/kg/day.
- $N = 1368$ live rat foetuses were examined for low birth weight (continuous) or defects (binary).
- This joint occurrence of continuous and binary outcomes will provide additional challenges in model development.

Ethylene Glycol Study in Rats

Dose	Dams	Live	Litter Size		Malf.		Weight		ρ
			Mean	SD	#	%	Mean	SD	
0.00	28	379	13.50	1.50	5	1.3	3.40	0.38	0.07
1.25	28	357	12.75	2.03	21	5.8	3.30	0.37	0.00
2.50	29	345	11.89	3.26	86	24.9	2.90	0.36	-0.29
5.00	26	287	11.04	4.01	197	68.6	2.48	0.46	-0.70

14.2.4 EG Study in Rats

Observed malformation rates and average weights for all clusters.



14.3 Analysis of the Weight Outcome

- Four analyses will be conducted:
  - ▷ Ordinary linear regression
  - ▷ A random-intercepts mixed model
  - ▷ A compound symmetry marginal model
  - ▷ An independence marginal model (equivalent to the first model)

14.3.1 Ordinary Linear Regression

- Model formulation (d_i represents dose):

$$Y_i = \beta_0 + \beta_1 d_i + \varepsilon_i$$

- There is only one index i since littermates are assumed to be independent
- Two parameters: intercept β_0 and linear dose effect β_1
- The error term is assumed to follow a normal distribution with variance σ^2
- SAS code (using PROC MIXED):

```
proc mixed data=help method=reml;
title 'ordinary regression model for weight (DEHP2)';
model weight=dose / solution;
run;
```

- Relevant portions of the output:

ordinary regression model for weight (DEHP2)

Covariance Parameter Estimates

Cov Parm	Estimate
Residual	0.01270

Solution for Fixed Effects

Effect	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept	0.9641	0.004714	1080	204.51	<.0001
DOSE	-0.1085	0.01301	1080	-8.34	<.0001

14.3.2 Random-intercepts Model

- Model formulation:

$$
\begin{aligned}
Y_{ij} &= \beta_0 + b_{0i} + \beta_1 d_i + \varepsilon_{ij} \\
b_{0i} &\sim N(0, d) \\
\varepsilon_{ij} &\sim N(0, \sigma^2)
\end{aligned}
$$

- SAS code:

```
proc mixed data=help method=reml;
title 'linear mixed model for weight (DEHP2)';
title2 'random intercept';
class litter;
id litter dose;
model weight=dose / solution outpredm=m.prmwt outpred=m.predwt ddfm=satterth;
random intercept / subject=litter solution v g;
ods listing exclude solutionr;
ods output solutionr=m.solrwt;
run;
```

- Apart from the fixed-effects parameters, the Empirical Bayes estimators of the random effects $\hat{b}_{0i}$ are stored, as well as unconditional and conditional predictions:

  outpredm: $$E(Y_{ij}|d_i) = \widehat{\beta}_0 + \widehat{\beta}_1 d_i$$

  outpred: $$E(Y_{ij}|d_i, \hat{b}_{0i}) = \widehat{\beta}_0 + \hat{b}_{0i} + \widehat{\beta}_1 d_i$$

  Note that the expectation of the second quantity over the random intercepts produces the first one.

- SAS statements have been added to this effect.

- Relevant portions of the output:
  - ▷ The overall covariance matrix is of the form:

$$V_i = d\, J_{n_i} + \sigma^2\, I_{n_i}$$

Estimated V Matrix for LITTER 38

Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	0.01306	0.005917	0.005917	0.005917	0.005917	0.005917	0.005917
2	0.005917	0.01306	0.005917	0.005917	0.005917	0.005917	0.005917
3	0.005917	0.005917	0.01306	0.005917	0.005917	0.005917	0.005917
4	0.005917	0.005917	0.005917	0.01306	0.005917	0.005917	0.005917
5	0.005917	0.005917	0.005917	0.005917	0.01306	0.005917	0.005917
6	0.005917	0.005917	0.005917	0.005917	0.005917	0.01306	0.005917
7	0.005917	0.005917	0.005917	0.005917	0.005917	0.005917	0.01306

  - ▷ The two variance components are given by

Covariance Parameter Estimates

Cov Parm	Subject	Estimate
Intercept	LITTER	0.005917
Residual		0.007148

▷ The regression parameters are:

Solution for Fixed Effects

Effect	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept	0.9665	0.01106	101	87.41	<.0001
DOSE	-0.1312	0.02681	112	-4.89	<.0001

- We will discuss the various estimates of the regression parameters once they have all been introduced

14.3.3 Compound-symmetry Model

- Given that the marginal model, implied by the above random-intercepts model, is of a compound-symmetry type, such a model can be fitted directly.

- The model formulation has now changed to

$$Y_{ij} = \beta_0 + \beta_1 d_i + \varepsilon_{ij}$$

$$\boldsymbol{\varepsilon}_i \sim N(\mathbf{0}, V_i = d\, J_{n_i} + \sigma^2\, I_{n_i})$$

- Thus, the effect of the random intercept has been absorbed in the marginal covariance structure.

- SAS code:

```
proc mixed data=help method=reml;
title 'reduced multivariate model for weight (DEHP2)';
title2 'compound symmetry';
class litter;
model weight=dose / solution ddfm=satterth;
repeated / subject=litter type=cs r rcorr;
run;
```

- The random-intercepts model implies CS, the reverse is not true.

- This is seen in the fact that no empirical Bayes inference is possible.

- The remaining part of the output is equivalent:

Estimated R Matrix for LITTER 38

Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	0.01306	0.005917	0.005917	0.005917	0.005917	0.005917	0.005917
2	0.005917	0.01306	0.005917	0.005917	0.005917	0.005917	0.005917
3	0.005917	0.005917	0.01306	0.005917	0.005917	0.005917	0.005917
4	0.005917	0.005917	0.005917	0.01306	0.005917	0.005917	0.005917
5	0.005917	0.005917	0.005917	0.005917	0.01306	0.005917	0.005917
6	0.005917	0.005917	0.005917	0.005917	0.005917	0.01306	0.005917
7	0.005917	0.005917	0.005917	0.005917	0.005917	0.005917	0.01306

Covariance Parameter Estimates

Cov Parm	Subject	Estimate
CS	LITTER	0.005917
Residual		0.007148

Solution for Fixed Effects

Effect	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept	0.9665	0.01106	101	87.41	<.0001
DOSE	-0.1312	0.02681	112	-4.89	<.0001

- The option 'rcorr' in the REPEATED statement also produces the correlation matrix:

Estimated R Correlation Matrix for LITTER 38

Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	1.0000	0.4529	0.4529	0.4529	0.4529	0.4529	0.4529
2	0.4529	1.0000	0.4529	0.4529	0.4529	0.4529	0.4529
3	0.4529	0.4529	1.0000	0.4529	0.4529	0.4529	0.4529
4	0.4529	0.4529	0.4529	1.0000	0.4529	0.4529	0.4529
5	0.4529	0.4529	0.4529	0.4529	1.0000	0.4529	0.4529
6	0.4529	0.4529	0.4529	0.4529	0.4529	1.0000	0.4529
7	0.4529	0.4529	0.4529	0.4529	0.4529	0.4529	1.0000

- This matrix produces the intraclass correlation:

$$\widehat{\text{ICC}} = \widehat{\rho} = 0.4529$$

capturing the effect of clustering.

- (See the lizards example.)

14.3.4 Independence Model

- By replacing the compound-symmetry structure in the above code by the 'simple' or independence structure, we are back to the original OLS model:

```
proc mixed data=help method=reml;
title 'independence multivariate model for weight (DEHP2)';
title2 'equivalent to OLS';
class litter;
model weight=dose / solution ddfm=satterth;
repeated / subject=litter type=simple r;
run;
```

- The corresponding output is:

Estimated R Matrix for LITTER 38

Row	Col1	Col2	Col3	Col4	Col5	Col6	Col7
1	0.01270						
2		0.01270					
3			0.01270				
4				0.01270			
5					0.01270		
6						0.01270	
7							0.01270

Covariance Parameter Estimates

Cov Parm	Subject	Estimate
Residual	LITTER	0.01270

Solution for Fixed Effects

Effect	Estimate	Standard Error	DF	t Value	Pr > |t|
Intercept	0.9641	0.004714	1080	204.51	<.0001
DOSE	-0.1085	0.01301	1080	-8.34	<.0001

- While parameter estimates do not seem to change much, there is a strong impact on standard errors and degrees of freedom.

14.4 EB Estimates, Predicted Means, and Predicted Values

Litter	Dose	n_i	Obs. Mean	Fitted Mean	Rand. Int.	Pred. Val.
38	0.000	9	0.821	0.967	-0.129	0.838
49	0.000	15	0.950	0.967	-0.016	0.951
50	0.000	8	0.994	0.967	0.024	0.991
61	0.000	6	0.920	0.967	-0.039	0.927
62	0.000	7	0.778	0.967	-0.161	0.806
73	0.000	9	0.857	0.967	-0.096	0.870
85	0.000	16	0.902	0.967	-0.060	0.906
86	0.000	12	0.957	0.967	-0.008	0.958
97	0.000	16	0.927	0.967	-0.037	0.929
98	0.000	13	1.023	0.967	0.052	1.018
109	0.000	7	0.971	0.967	0.004	0.971
110	0.000	13	1.088	0.967	0.111	1.077
119	0.000	15	0.889	0.967	-0.072	0.895
120	0.000	9	1.045	0.967	0.069	1.036
...						
160	0.000	12	0.863	0.967	-0.094	0.873
169	0.000	13	0.968	0.967	0.001	0.968
170	0.000	11	0.875	0.967	-0.083	0.884
179	0.000	14	0.807	0.967	-0.147	0.820
180	0.000	12	1.053	0.967	0.079	1.045
189	0.000	3	0.903	0.967	-0.045	0.921
190	0.000	9	1.015	0.967	0.043	1.009
199	0.000	8	0.920	0.967	-0.041	0.926
200	0.000	7	0.987	0.967	0.017	0.984

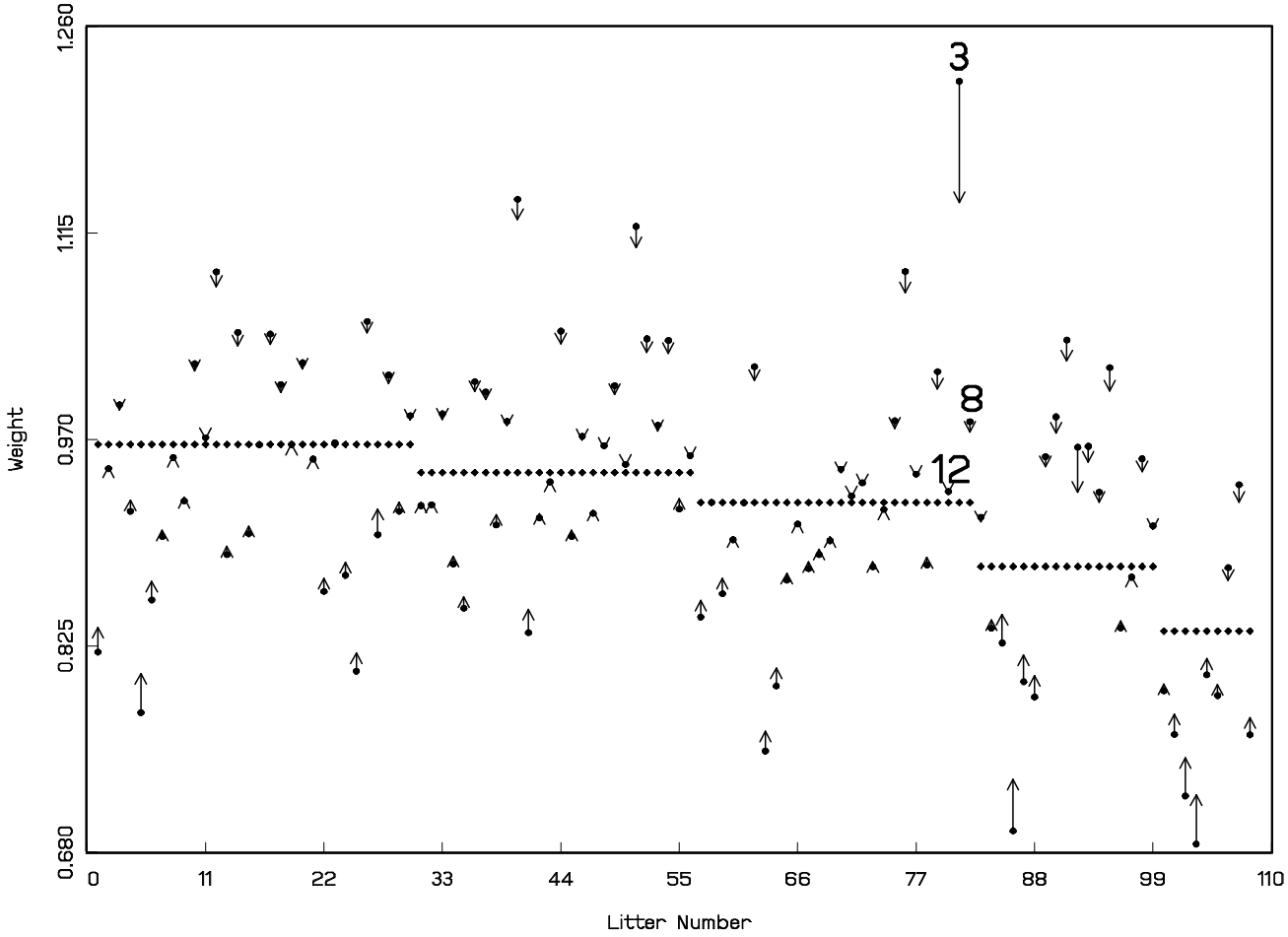
Litter	Dose	n_i	Obs. Mean	Fitted Mean	Rand. Int.	Pred. Val.
39	0.151	10	0.924	0.947	-0.021	0.926
60	0.151	14	0.924	0.947	-0.021	0.926
63	0.151	12	0.988	0.947	0.038	0.984
72	0.151	13	0.882	0.947	-0.059	0.888
75	0.151	13	0.851	0.947	-0.087	0.860
84	0.151	10	1.011	0.947	0.057	1.004
87	0.151	12	1.003	0.947	0.051	0.998
96	0.151	5	0.910	0.947	-0.030	0.917
99	0.151	13	0.983	0.947	0.033	0.980
108	0.151	15	1.139	0.947	0.178	1.124
118	0.151	7	0.834	0.947	-0.096	0.851
121	0.151	11	0.915	0.947	-0.029	0.918
128	0.151	12	0.940	0.947	-0.006	0.941
131	0.151	12	1.046	0.947	0.090	1.037
138	0.151	11	0.902	0.947	-0.041	0.906
141	0.151	12	0.972	0.947	0.023	0.970
148	0.151	14	0.918	0.947	-0.026	0.920
151	0.151	11	0.966	0.947	0.017	0.964
158	0.151	11	1.008	0.947	0.055	1.002
161	0.151	14	0.953	0.947	0.005	0.952
171	0.151	13	1.120	0.947	0.158	1.105
181	0.151	11	1.041	0.947	0.085	1.031
188	0.151	9	0.980	0.947	0.029	0.976
191	0.151	11	1.040	0.947	0.084	1.030
198	0.151	3	0.921	0.947	-0.018	0.928
201	0.151	9	0.959	0.947	0.010	0.957

Litter	Dose	n_i	Obs. Mean	Fitted Mean	Rand. Int.	Pred. Val.
40	0.312	7	0.845	0.926	-0.069	0.857
52	0.312	12	0.926	0.926	0.000	0.926
64	0.312	6	0.862	0.926	-0.053	0.872
71	0.312	13	0.900	0.926	-0.024	0.902
76	0.312	12	0.925	0.926	-0.000	0.925
83	0.312	11	1.021	0.926	0.086	1.012
88	0.312	14	0.751	0.926	-0.161	0.765
100	0.312	11	0.797	0.926	-0.116	0.809
107	0.312	12	0.871	0.926	-0.050	0.876
112	0.312	11	0.911	0.926	-0.014	0.912
117	0.312	10	0.879	0.926	-0.042	0.884
122	0.312	10	0.889	0.926	-0.033	0.893
127	0.312	11	0.899	0.926	-0.024	0.901
137	0.312	10	0.949	0.926	0.021	0.947
142	0.312	11	0.930	0.926	0.004	0.930
147	0.312	13	0.939	0.926	0.013	0.938
152	0.312	14	0.880	0.926	-0.042	0.884
157	0.312	11	0.921	0.926	-0.004	0.921
162	0.312	12	0.983	0.926	0.052	0.978
167	0.312	12	1.088	0.926	0.148	1.073
172	0.312	14	0.946	0.926	0.018	0.944
177	0.312	9	0.882	0.926	-0.039	0.887
182	0.312	8	1.018	0.926	0.080	1.006
187	0.312	12	0.933	0.926	0.007	0.933
192	0.312	3	1.222	0.926	0.211	1.137
197	0.312	8	0.983	0.926	0.049	0.975

Litter	Dose	n_i	Obs. Mean	Fitted Mean	Rand. Int.	Pred. Val.
53	0.654	12	0.915	0.881	0.032	0.912
70	0.654	8	0.837	0.881	-0.038	0.843
82	0.654	2	0.827	0.881	-0.034	0.847
89	0.654	5	0.695	0.881	-0.150	0.731
94	0.654	4	0.800	0.881	-0.062	0.819
101	0.654	6	0.789	0.881	-0.076	0.804
106	0.654	12	0.958	0.881	0.070	0.951
126	0.654	10	0.986	0.881	0.094	0.975
136	0.654	12	1.040	0.881	0.144	1.025
153	0.654	2	0.964	0.881	0.052	0.933
163	0.654	8	0.965	0.881	0.073	0.954
166	0.654	8	0.933	0.881	0.045	0.926
173	0.654	9	1.020	0.881	0.123	1.004
186	0.654	9	0.837	0.881	-0.038	0.843
193	0.654	8	0.873	0.881	-0.006	0.874
196	0.654	9	0.956	0.881	0.067	0.948
203	0.654	13	0.909	0.881	0.026	0.907

Litter	Dose	n_i	Obs. Mean	Fitted Mean	Rand. Int.	Pred. Val.
57	1.000	9	0.793	0.835	-0.037	0.798
66	1.000	5	0.763	0.835	-0.058	0.777
78	1.000	4	0.719	0.835	-0.089	0.746
90	1.000	4	0.686	0.835	-0.115	0.720
93	1.000	2	0.804	0.835	-0.019	0.816
105	1.000	6	0.790	0.835	-0.038	0.798
124	1.000	5	0.880	0.835	0.036	0.871
154	1.000	9	0.938	0.835	0.091	0.926
184	1.000	6	0.762	0.835	-0.061	0.775

14.4.1 Observed, Fitted, and Predicted Means



- Solid circles are observed means of weight per cluster

- $E(Y_{ij}|d_i) = \widehat{\beta}_0 + \widehat{\beta}_1 d_i$ is given by diamonds: one level per dose level
- Predicted means $E(Y_{ij}|d_i, \hat{b}_{0i}) = \widehat{\beta}_0 + \hat{b}_{0i} + \widehat{\beta}_1 d_i$ are arrows pointing away from observed means:
  - ▷ Shrinkage towards the fitted means
  - ▷ Amount of shrinkage is related to the cluster size

14.4.2 Parameter Estimates, Standard Errors, Degrees of Freedom

- Summary of the estimates (s.e.) and degrees of freedom for the uncorrelated (OLS) and correlated (R.I./C.S.) analyses:

		OLS / Independence		R.I. / C.S.	
Effect	Par.	Estimate (s.e.)	D.F.	Estimate (s.e.)	D.F.
Int.	β_0	0.9641 (0.0047)	1080	0.9665 (0.0111)	101
Dose	β_1	-0.1085 (0.0130)	1080	-0.1312 (0.0268)	112

- The points estimates are reasonably close to each other.

- Standard errors in the correlated case are about twice those in the uncorrelated case. This is because, in the uncorrelated case, the data points are assumed to be independent.

- This is also reflected in the degrees of freedom. In the uncorrelated case, these are equal to the number of data points (foetuses) minus the number of mean parameters estimates. The corresponding numbers are much lower in the correlated data case.

- One can get a feel for this phenomenon, based on the **effective sample size**.

14.5 Effective Sample Size

- For simplicity, we first focus on binary data.

- Let us focus on the information contributed by a cluster, regarding the average (probability of success) within a cluster:

$$\begin{aligned}
\text{var}\left(\frac{\sum_{j=1}^{n_i} y_{ij}}{n_i}\right) &= \frac{1}{n_i^2}\left[\sum_{j=1}^{n_i} \text{var}(y_{ij}) + 2\sum_{j<j'} \text{cov}(y_{ij}, y_{ij'})\right] \\
&= \frac{1}{n_i^2}\left[n_i\pi_i(1-\pi_i) + n_i(1-n_i)\rho\pi_i(1-\pi_i)\right] \\
&= \frac{1}{n_i}\pi_i(1-\pi_i)[1+\rho(n_i-1)]
\end{aligned}$$

- In case the same experiment was conducted with independence within clusters, the corresponding quantity would be:

$$\frac{1}{\widetilde{n}_i}\pi_i(1-\pi_i)$$

- Equating both yields

$$\widetilde{n}_i = \frac{n_i}{1+\rho(n_i-1)}$$

- Hence, for the entire sample, we obtain:

$$N_{\text{eff}} = \sum_{i=1}^{N} \widetilde{n}_i = \sum_{i=1}^{N} \frac{n_i}{1+\rho(n_i-1)}$$

- When the outcomes are continuous and normally distributed, and the compound-symmetry model applies,

$$\rho \leftarrow \frac{d}{d + \sigma^2}$$

  and exactly the same formula is obtained.

- Two special cases:

$$N_{\text{eff}}(\rho = 0) = \sum_i n_i = \#\text{ fetuses}$$

$$N_{\text{eff}}(\rho = 1) = \sum_i \frac{n_i}{n_i} = \#\text{ litters}$$

- Applied to the NTP data ($\widehat{\rho} = 0.4529$):

ρ	N_{eff}
0.0000	1082
0.4529	208.85
1.0000	108

- Thus, the information obtained from this experiment is similar to what would be obtained from 209 independent foetuses. Note that the D.F. obtained from SAS are different. SAS uses the so-called Satterthwaite approximation, using similar logic, but directly on the degrees-of-freedom of the approximate distribution for the Wald tests for the individual parameter estimates. This method is less accurate in many cases, but is broadly applicable, throughout a wide range of mixed-model applications.

- **Information limit.** The expression for the effective sample size shows asymptotic behavior in ρ:

$$\lim_{n_i \to +\infty} \frac{n_i}{1 + \rho(n_i - 1)} = \frac{1}{\rho}$$

  Hence, the information contribution per cluster is finite, except when $\rho = 0$ (independence).

- For the NTP (weight) data, the limit is equal to 2.208; hence, a litter can never contribute more information for the average per cluster than would be obtained from about 2.2 independent littermates.

14.6 Model Families

14.6.1 Continuous Outcomes

- **Marginal Models**

$$E(Y_{ij}|\boldsymbol{x}_{ij}) = \boldsymbol{x}'_{ij}\boldsymbol{\beta}$$

- **Random-Effects Models**

$$E(Y_{ij}|\boldsymbol{b}_i, \boldsymbol{x}_{ij}) = \boldsymbol{x}'_{ij}\boldsymbol{\beta} + \boldsymbol{z}'_{ij}\boldsymbol{b}_i$$

- **Transition Models**

$$E(Y_{ij}|Y_{i,j_1}, \ldots, Y_{i1}, \boldsymbol{x}_{ij}) = \boldsymbol{x}'_{ij}\boldsymbol{\beta} + \alpha Y_{i,j-1}$$

14.6.2 Binary Outcomes

Marginal Models

- Specify $E(Y_{ij}|\boldsymbol{x}_{ij})$.

  The probability of each outcome (or set of outcomes) is directly modeled (integrating or summing the other outcomes away).

- Sometimes called: population averaged method.

- **Non-likelihood method:**

  ▷ **Generalized estimating equations (GEE)**

- **Likelihood methods:**
  - ▷ Multivariate probit model
  - ▷ **Bahadur model**
  - ▷ Odds ratio models

Random-effects Models

- Specify $E(Y_{ij}|\boldsymbol{b}_i, \boldsymbol{x}_{ij})$.
  The outcome(s), conditional on an unobserved (latent) random effect or set of random effects.

- **Beta-binomial model**

- **Generalized linear mixed models**

- Hierarchical generalized linear models

14.6.3 Conditional Models

- Specify $E(Y_{ij}|\{Y_{ik}\}, \boldsymbol{x}_{ij})$.

  An outcome or set of outcomes is modeled, conditional on the other outcomes or at least a set of other outcomes.

- In a longitudinal setting, a very relevant family is the so-called set of **transition models**:

$$E(Y_{ij}|\{Y_{i1}, \ldots, Y_{i,j-1}\}, \boldsymbol{x}_{ij}),$$

  (e.g., Markov models.)

- Rather than integrating over other outcomes, they have to be conditioned upon.

- **multivariate exponential family:**

  this model forms the basis for the well known **loglinear model**.

14.7 Generalized Linear Models

1. $E(Y_i) = \mu_i$

2. $\eta(\mu_i) = \boldsymbol{x}_i^T \boldsymbol{\beta}$ with $\eta(.)$ the **link function**

3. $\text{Var}(Y_i) = \phi v(\mu_i)$, where
   - ▷ $v(.)$ is a known **variance function**
   - ▷ ϕ is a scale (overdispersion) parameter

4. exponential family p.d.f.

$$f(y|\theta_i, \phi) = \exp\left\{\phi^{-1}[y\theta_i - \psi(\theta_i)] + c(y, \phi)\right\}$$

with θ_i the natural parameter and $\psi(.)$ a function satisfying

$$\mu_i = \psi'(\theta_i) \quad \longleftrightarrow \quad v(\mu_i) = \psi''(\theta_i)$$

14.7.1 The Bernoulli Model

$$Y \sim \text{Bernoulli}(\pi)$$

- Density function:

$$\begin{aligned} f(y|\theta,\phi) &= \pi^y(1-\pi)^{1-y} \\ &= \exp\{y\ln\pi + (1-y)\ln(1-\pi)\} \\ &= \exp\left\{y\ln\left(\frac{\pi}{1-\pi}\right) + \ln(1-\pi)\right\} \end{aligned}$$

- Exponential family:
  - $\theta = \ln\left(\frac{\pi}{1-\pi}\right)$
  - $\psi(\theta) = \ln(1-\pi) = \ln(1+\exp(\theta))$
  - $\phi = 1$
  - $c(y, \phi) = 0$

- Mean and variance function:

$$\mu = \frac{\exp\theta}{1+\exp\theta} = \pi \quad \longleftrightarrow \quad v(\mu) = \frac{\exp\theta}{(1+\exp\theta)^2} = \pi(1-\pi)$$

- Note that, under this model, the mean and variance are related:

$$\phi v(\mu) \;=\; \mu(1-\mu)$$

- The link function is the logit link:

$$\theta = \ln\left(\frac{\mu}{1-\mu}\right)$$

14.7.2 Generalized Linear Models (GLM)

- Suppose a sample $Y_1, \ldots, Y_N$ of independent observations is available
- All Y_i have densities $f(y_i|\theta_i, \phi)$ which belong to the exponential family

Linear predictor: $\theta_i = \boldsymbol{x_i}'\boldsymbol{\beta}$

Log-likelihood: $\ell(\beta, \phi) = \frac{1}{\phi}\sum_i [y_i\theta_i - \psi(\theta_i)] + \sum_i c(y_i, \phi)$

Score equations: $S(\boldsymbol{\beta}) = \sum_i \frac{\partial \mu_i}{\partial \boldsymbol{\beta}} v_i^{-1} (y_i - \mu_i) = 0$

14.8 A Conditional Model

No Clustering

- $i = 1, \ldots, N$ individuals $\qquad$ $j = 1, \ldots, M$ evaluations
- $Y_{ij} = 1$ if subject i exhibits response j and 0 otherwise

  Cox (1972)

$$f_{\boldsymbol{Y}}(\boldsymbol{y}_i; \boldsymbol{\Theta}_i) = \exp\left\{\sum_{j=1}^{M} \theta_{ij} y_{ij} + \sum_{j<j'} \omega_{ijj'} y_{ij} y_{ij'} + \ldots + \omega_{i12\ldots M} y_{i1} y_{i2} \ldots y_{iM} - A(\boldsymbol{\Theta}_i)\right\}$$

- θ_{ij}: main effect parameters: conditional logits
- $\omega_{ijj'}$: pairwise interactions: conditional log odds ratios

- **Quadratic simplification:**

  Zhao and Prentice (1990)

$$f_{\boldsymbol{Y}}(\boldsymbol{y}_i; \boldsymbol{\Theta}_i) \;=\; \exp\left\{ \sum_{j=1}^{M} \theta_{ij} y_{ij} + \sum_{j<j'} \omega_{ijj'} y_{ij} y_{ij'} - A(\boldsymbol{\Theta}_i) \right\},$$

- **Linear simplification** $\equiv$ logistic regression

$$f_{\boldsymbol{Y}}(\boldsymbol{y}_i; \boldsymbol{\Theta}_i) \;=\; \exp\left\{ \sum_{j=1}^{M} \theta_{ij} y_{ij} - A(\boldsymbol{\Theta}_i) \right\},$$

14.8.1 Clustering

- $i = 1, \ldots, N$ clusters $\quad k = 1, \ldots, n_i$ individuals $\quad j = 1, \ldots, M$ malformations

- Outcome:

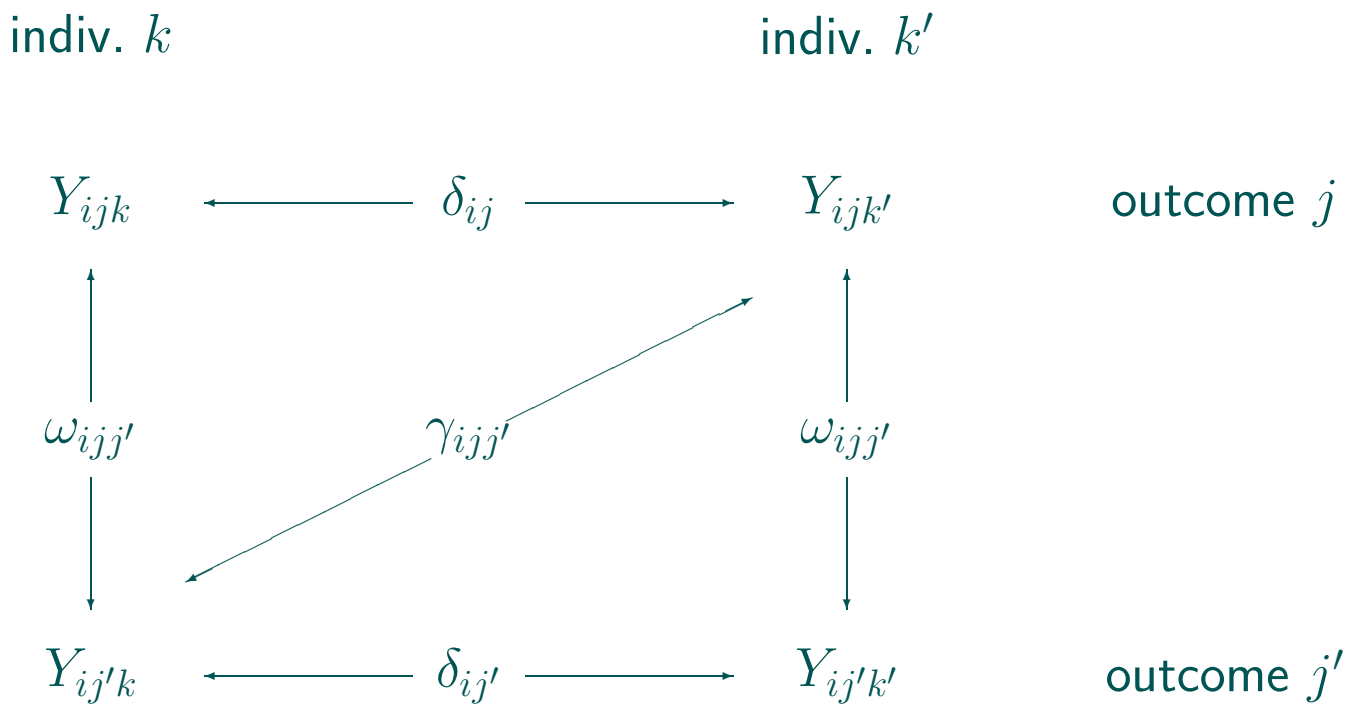
$$Y_{ijk} = \begin{cases} 1 & \text{if } k\text{th indiv. of } i\text{th cluster exhibits malf. } j \\ \text{-1} & \text{otherwise.} \end{cases}$$

Molenberghs and Ryan (1997)

$$f_{\boldsymbol{Y}_i}(\boldsymbol{y}_i; \boldsymbol{\Theta}_i) = \exp\left\{ \sum_{j=1}^{M} \sum_{k=1}^{n_i} \theta_{ij} y_{ijk} + \sum_{j=1}^{M} \sum_{k<k'} \delta_{ij} y_{ijk} y_{ijk'} \right.$$

$$\left. + \sum_{j<j'} \sum_{k=1}^{n_i} \omega_{ijj'} y_{ijk} y_{ij'k} + \sum_{j<j'} \sum_{k \neq k'} \gamma_{ijj'} y_{ijk} y_{ij'k'} - A(\boldsymbol{\Theta}_i) \right\}$$

14.8.2 Association Structure



14.8.3 A Single Clustered Binary Outcome

- NTP data
- Y_{ij} is malformation indicator for fetus j in litter i
- Code Y_{ij} as -1 or 1
- d_i is dose level at which litter i is exposed
- Model:

$$\theta_{ij} = \beta_0 + \beta_d d_i \qquad\qquad \omega_{ijj'} = \beta_a$$

- Using Z_i: the number of "successes" we obtain

$$f_Y(\boldsymbol{y}_i) \;=\; \exp\left\{\theta_i z_i - \beta_a z_i(n_i - z_i) - A(\boldsymbol{\theta}_i)\right\}$$

14.8.4 The NTP Data

Maximum Likelihood Estimates (Standard Errors) for the Conditional Model.

Outcome	Parameter	DEHP	EG	DYME
External	β_0	-2.81(0.58)	-3.01(0.79)	-5.78(1.13)
	β_d	3.07(0.65)	2.25(0.68)	6.25(1.25)
	β_a	0.18(0.04)	0.25(0.05)	0.09(0.06)
Visceral	β_0	-2.39(0.50)	-5.09(1.55)	-3.32(0.98)
	β_d	2.45(0.55)	3.76(1.34)	2.88(0.93)
	β_a	0.18(0.04)	0.23(0.09)	0.29(0.05)
Skeletal	β_0	-2.79(0.58)	-0.84(0.17)	-1.62(0.35)
	β_d	2.91(0.63)	0.98(0.20)	2.45(0.51)
	β_a	0.17(0.04)	0.20(0.02)	0.25(0.03)
Collapsed	β_0	-2.04(0.35)	-0.81(0.16)	-2.90(0.43)
	β_d	2.98(0.51)	0.97(0.20)	5.08(0.74)
	β_a	0.16(0.03)	0.20(0.02)	0.19(0.03)

14.9 Marginal Models

- $E(\boldsymbol{Y}_i) = \boldsymbol{\mu}_i$
- $\boldsymbol{\eta}_i(\boldsymbol{\mu}_i) = \{\eta_{i1}(\mu_{i1}), \ldots, \eta_{in}(\mu_{in})\}$
- $\boldsymbol{\eta}_i(\boldsymbol{\mu}_i) = \boldsymbol{X}_i\boldsymbol{\beta}$
- var$(\boldsymbol{Y}_i) = \phi\boldsymbol{v}(\boldsymbol{\mu}_i)$ where $\boldsymbol{v}(.)$ is a known variance function
- corr$(\boldsymbol{Y}_i) = \boldsymbol{R}(\boldsymbol{\alpha})$
- **Degree of modelling:**
  - ▷ joint distribution fully specified $\Rightarrow$ likelihood procedures
  - ▷ only a limited number of moments $\Rightarrow$ generalized estimating equations (GEE)

14.9.1 Univariate Link Functions

- The logit link:

$$\eta_{ij} = \ln(\mu_{ij}) - \ln(1 - \mu_{ij}) = \text{logit}(\mu_{ij}).$$

- The probit link:

$$\eta_{ij} = \Phi_1^{-1}(\mu_{ij}),$$

- The complementary log-log link:

$$\eta_{ij} = \log(-\log(1 - \mu_{ij}))$$

14.9.2 Marginal Correlation Coefficient

- Expression and model

$$\rho_{ijk} = \frac{\mu_{ijk} - \mu_{ij}\mu_{ik}}{\sqrt{\mu_{ij}(1-\mu_{ij})\mu_{ik}(1-\mu_{ik})}}$$

$$\eta_{ijk} = \ln(1+\rho_{ijk}) - \ln(1-\rho_{ijk})$$

- Bahadur (1961)

- **Advantages:**
  - ▷ Computation of correlations is easy.
  - ▷ Many believe that a correlation is easily understood
  - ▷ Intuition drawn from multivariate normal models

- **Disadvantages:**

  ▷ Constraints, already in the bivariate case:

$$\begin{array}{cc|c} 0 & \mu & \mu \\ \mu & 1-2\mu & 1-\mu \\ \hline \mu & 1-\mu & 1 \end{array}$$

  The smallest achievable correlation with these margins is:

$$\frac{-\mu}{1-\mu}$$

14.9.3 Marginal Odds Ratio

- Dale (1986) & Molenberghs and Lesaffre (1994)
- Expression and model:

$$\psi_{ijk} = \frac{(\mu_{ijk})(1-\mu_{ij}-\mu_{ik}+\mu_{ijk})}{(\mu_{ik}-\mu_{ijk})(\mu_{ij}-\mu_{ijk})} = \left(\frac{\mathsf{Pr}(Y_{ij}=1,Y_{ik}=1)\mathsf{Pr}(Y_{ij}=0,Y_{ik}=0)}{\mathsf{Pr}(Y_{ij}=0,Y_{ik}=1)\mathsf{Pr}(Y_{ij}=1,Y_{ik}=0)}\right)$$

$$\eta_{ijk} = \ln(\psi_{ijk})$$

- **Advantages:**
  - ▷ No constraints in bivariate case
  - ▷ Mild constraints in higher orders
- **Disadvantages:**
  - ▷ No closed form for higher order probabilities
  - ▷ Computationally more intensive

14.10 The Bahadur Model

- Using higher order correlations instead $\Rightarrow$ Bahadur model
- Define residuals:

$$R_{ij} = \frac{Y_{ij} - \mu_{ij}}{\sqrt{\mu_{ij}(1 - \mu_{ij})}}$$

- Define correlations of all orders:

$$\begin{aligned}
\rho_{ijk} &= \text{Corr}(Y_{ij}, Y_{ik}) = E(R_{ij}R_{ik}) \\
\rho_{ijk\ell} &= E(R_{ij}R_{ik}R_{i\ell}) \\
&\cdots \\
\rho_{i1\ldots n_i} &= E(R_{i1}\ldots R_{in_i}).
\end{aligned}$$

- Using
  - ▷ definition of correlations
  - ▷ relationship between μ_{ijk} and ρ_{ijk},
  - ▷ relationship between $\mu_{ijk\ell}$ and $\rho_{ijk\ell}$,
  - ▷ etc.
- a closed form probability mass function is obtained:

$$f(\boldsymbol{y}_i, \boldsymbol{\mu}_i, \boldsymbol{\rho}_i) = \prod_{j=1}^{n} \mu_{ij}^{y_{ij}} (1-\mu_{ij})^{(1-y_{ij})} \left(1 + \sum_{j_1<j_2} \rho_{ij_1j_2} r_{ij_1} r_{ij_2} + \right.$$

$$\left. \sum_{j_1<j_2<j_3} \rho_{ij_1j_2j_3} r_{ij_1} r_{ij_2} r_{ij_3} + \ldots + \rho_{i1\ldots n} r_{i1} \ldots r_{in} \right)$$

14.10.1 A Single Clustered Binary Outcome

- Simplifications:
  - cluster level covariates only
  - pairwise correlations only
- Expression:

$$f(z_i|\mu_i,\rho_i) = \binom{n_i}{z_i}\mu_i^{z_i}(1-\mu_i)^{n_i-z_i} \times \left[1+\rho_i\left\{\binom{z_i}{2}\frac{1-\mu_i}{\mu_i}+\binom{n_i-z_i}{2}\frac{\mu_i}{1-\mu_i}-z_i(n_i-z_i)\right\}\right].$$

- Parameterization:

$$\text{logit}(\mu_i) = \beta_0+\beta_d d_i,$$

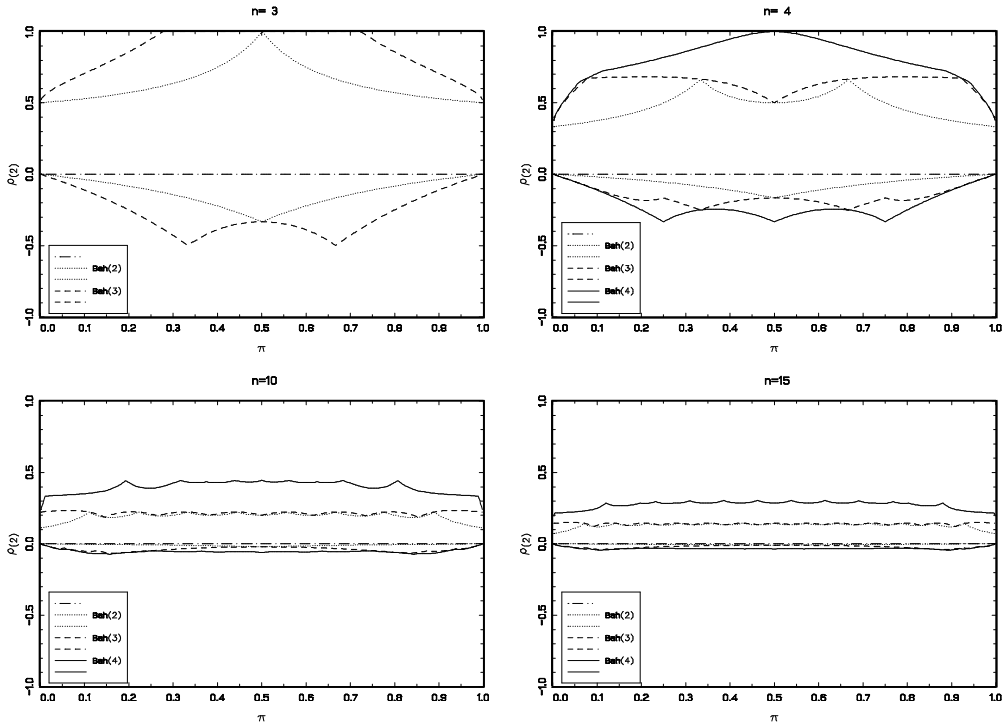
$$\ln\left(\frac{1+\rho_i}{1-\rho_i}\right) = \beta_a.$$

14.10.2 Disadvantages Revisited

- Restrictions on second order correlation when higher order associations are left out:

$$-\frac{2}{n(n-1)}\min\left(\frac{\mu}{1-\mu},\frac{1-\mu}{\mu}\right)\leq\rho_{(2)}\leq\frac{2\mu(1-\mu)}{(n-1)\mu(1-\mu)+0.25-\gamma_0},$$

$$\gamma_0=\min_{z=0}^{n}\left\{[z-(n-1)\mu-0.5]^2\right\}.$$



14.10.3 The NTP Data

Maximum Likelihood Estimates (Standard Errors) for the Bahadur Model.

Outcome	Parameter	DEHP	EG	DYME
External	β_0	-4.93(0.39)	-5.25(0.66)	-7.25(0.71)
	β_d	5.15(0.56)	2.63(0.76)	7.94(0.77)
	β_a	0.11(0.03)	0.12(0.03)	0.11(0.04)
Visceral	β_0	-4.42(0.33)	-7.38(1.30)	-6.89(0.81)
	β_d	4.38(0.49)	4.25(1.39)	5.49(0.87)
	β_a	0.11(0.02)	0.05(0.08)	0.08(0.04)
Skeletal	β_0	-4.67(0.39)	-2.49(0.11)	-4.27(0.61)
	β_d	4.68(0.56)	2.96(0.18)	5.79(0.80)
	β_a	0.13(0.03)	0.27(0.02)	0.22(0.05)
Collapsed	β_0	-3.83(0.27)	-2.51(0.09)	-5.31(0.40)
	β_d	5.38(0.47)	3.05(0.17)	8.18(0.69)
	β_a	0.12(0.03)	0.28(0.02)	0.12(0.03)

14.11 Generalized Estimating Equations

- Liang and Zeger (1986)
- Start from classical score equations

$$S(\boldsymbol{\beta}) = \sum_{i=1}^{N} [D_i]^T [V_i(\boldsymbol{\alpha})]^{-1} (\boldsymbol{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$$

- **BUT**
  - ▷ suppose $\boldsymbol{V}_i(.)$ is not the true variance of $\boldsymbol{Y}_i$ but only a plausible guess
  - ▷ the score equations are solved in a standard way

The asymptotic normality results:

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \sim N(\mathbf{0}, I_0^{-1} I_1 I_0^{-1})$$

$$I_0 = \sum_{i=1}^{N} D_i^T [V_i(\boldsymbol{\alpha})]^{-1} D_i$$

$$I_1 = \sum_{i=1}^{N} D_i^T [V_i(\boldsymbol{\alpha})]^{-1} \mathsf{Var}(\boldsymbol{Y}_i) [V_i(\boldsymbol{\alpha})]^{-1} D_i.$$

14.11.1 Fitting GEE

1. Compute initial estimates for $\boldsymbol{\beta}$, using a univariate GLM (i.e., assuming independence).
2. Compute:
   - Pearson residuals e_{ij}.
   - estimates for $\boldsymbol{\alpha}$.
   - $R_i(\boldsymbol{\alpha})$.
   - estimate for ϕ.
   - $V_i(\boldsymbol{\beta}, \boldsymbol{\alpha}) = \phi A_i^{1/2}(\boldsymbol{\beta}) R_i(\boldsymbol{\alpha}) A_i^{1/2}(\boldsymbol{\beta})$.
3. Update estimate for $\boldsymbol{\beta}$:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \left[\sum_{i=1}^{N} D_i^T V_i^{-1} D_i\right]^{-1} \left[\sum_{i=1}^{N} D_i^T V_i^{-1} (\boldsymbol{y}_i - \boldsymbol{\mu}_i)\right].$$

4. Iterate 2.–3. until convergence.

Estimates of precision by means of I_0^{-1} and $I_0^{-1} I_1 I_0^{-1}$.

14.11.2 Applied to the NTP Data

- SAS code for data manipulation (defining 0/1 variable and a collapsed outcome):

```
data help;
set m.dehp2;
dose=dose/292;
collaps = ((visceral-1) or (skeletal-1) or (external-1));
if visceral=. then delete;
skeletal=skeletal-1;
visceral=visceral-1;
external=external-1;
run;
```

- The mean structure is specified by

$$\text{logit}(\mu_{ij}) = \beta_0 + \beta_1 d_i$$

- Two GENMOD programs, one for **independence** working assumptions:

```
proc genmod data=help descending;
title 'DEHP2 - GEE for external outcome';
class litter;
model external = dose / dist=binomial;
repeated subject=litter / type=ind modelse;
run;
```

- and one for **exchangeable** working assumptions:

```
proc genmod data=help descending;
title 'DEHP2 - GEE for external outcome';
class litter;
model external = dose / dist=binomial;
repeated subject=litter / type=exch corrw modelse;
run;
```

- The REPEATED statements makes the difference between an ordinary generalized linear model and GEE. In this case, omitting the REPEATED statement would turn the procedure into fitting ordinary logistic regression.

- 'modelse' requests model-based standard errors, next to the empirically-corrected ones

- The working assumptions:

  - ▷ **Independence working assumption.** Two foetuses within the same litter have zero correlation.

  - ▷ **Exchangeable working assumption.** Two foetuses within the same litter exhibit a constant correlation.

  - ▷ Note that these assumptions may be incorrect. Nevertheless, point estimates are consistent, and the sandwich estimator (or robust estimator, or empirically corrected) estimator of the standard errors provides correct precision measures as well. This is opposed to the model-based estimator (or naive estimator) of the standard error.

- Output for the independence working assumption case:
  - ▷ There is some general book-keeping information:

GEE Model Information

Correlation Structure	Independent
Subject Effect	LITTER (108 levels)
Number of Clusters	108
Correlation Matrix Dimension	16
Maximum Cluster Size	16
Minimum Cluster Size	2

▷ Three sets of parameter estimates (standard errors) are provided:

Analysis Of Initial Parameter Estimates

Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Chi-Square	Pr > ChiSq
Intercept	1	-4.7729	0.3032	-5.3672	-4.1786	247.75	<.0001
DOSE	1	4.8175	0.4552	3.9254	5.7097	112.02	<.0001

Analysis Of GEE Parameter Estimates
Empirical Standard Error Estimates

Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > |Z|
Intercept	-4.7729	0.4469	-5.6487	-3.8970	-10.68	<.0001
DOSE	4.8175	0.6254	3.5918	6.0433	7.70	<.0001

Analysis Of GEE Parameter Estimates
Model-Based Standard Error Estimates

Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > |Z|
Intercept	-4.7729	0.3032	-5.3672	-4.1786	-15.74	<.0001
DOSE	4.8175	0.4552	3.9254	5.7097	10.58	<.0001

- ▷ All point estimates are the same and the initial and model-based standard errors are the same as well. This is only true for the independence case. The initial estimates are based on logistic regression, but independence working assumptions have the same effect as fitting a logistic regression.
- ▷ However, the empirical standard errors are different, and larger. They account for the fact that, due to within-cluster correlation, the amount of information is less than when littermates would be independent.
- ▷ The empirical standard errors are often larger than the model-based ones, but this is not always the case. A general result is that

$$I_0^{-1} I_1 I_0^{-1} \geq I_0^{-1}$$

in the sense that the matrix

$$I_0^{-1} I_1 I_0^{-1} - I_0^{-1}$$

is positive definite.

- We now turn to the exchangeable case:
  - ▷ The parameter estimates change to:

Analysis Of Initial Parameter Estimates

Parameter	DF	Estimate	Standard Error	Wald 95% Confidence Limits		Chi-Square
Intercept	1	-4.7729	0.3032	-5.3672	-4.1786	247.75
DOSE	1	4.8175	0.4552	3.9254	5.7097	112.02

Analysis Of GEE Parameter Estimates
Empirical Standard Error Estimates

Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > |Z|
Intercept	-4.7207	0.4259	-5.5554	-3.8860	-11.08	<.0001
DOSE	4.7526	0.6118	3.5534	5.9517	7.77	<.0001

Analysis Of GEE Parameter Estimates
Model-Based Standard Error Estimates

Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > |Z|
Intercept	-4.7207	0.4453	-5.5935	-3.8480	-10.60	<.0001
DOSE	4.7526	0.6377	3.5027	6.0024	7.45	<.0001

- ▷ The initial and final parameter estimates are now different, i.e., the impact of the working assumption.
- ▷ All standard errors are different. The gap between the empirical and model-based standard errors is smaller (and of the opposite magnitude!) than under independence. This is because an exchangeable working assumption may be close to the truth.
- ▷ In addition, a 16×16 matrix, displaying the working correlation, is produced (only a portion shown; 16 is the maximum cluster size):

Working Correlation Matrix

	Col1	Col2	Col3	Col4	Col5	Col6
Row1	1.0000	0.1302	0.1302	0.1302	0.1302	0.1302
Row2	0.1302	1.0000	0.1302	0.1302	0.1302	0.1302
Row3	0.1302	0.1302	1.0000	0.1302	0.1302	0.1302
Row4	0.1302	0.1302	0.1302	1.0000	0.1302	0.1302
Row5	0.1302	0.1302	0.1302	0.1302	1.0000	0.1302
Row6	0.1302	0.1302	0.1302	0.1302	0.1302	1.0000

- ▷ The working correlation is equal to 0.13, but one should not (over)interpret these quantities; they are merely a technical device to produce correct inference (parameter estimates and standard errors).

14.11.3 Results for the NTP Data

Outcome	Parameter	DEHP	EG	DYME
		Independence Working Assumptions		
External	β_0	-4.77 (0.45;0.30)	-3.02 (0.25;0.17)	-5.31 (0.47;0.29)
	β_d	4.82 (0.63;0.46)	3.55 (0.42;0.24)	7.08 (0.82;0.45)
Visceral	β_0	-4.94 (0.37;0.32)	-5.73 (0.68;0.59)	-7.42 (0.74;0.57)
	β_d	5.20 (0.56;0.47)	3.19 (0.79;0.68)	8.12 (0.85;0.64)
Skeletal	β_0	-4.38 (0.35;0.27)	-7.59 (1.06;1.25)	-7.27 (0.86;0.77)
	β_d	4.34 (0.57;0.42)	4.45 (1.14;1.34)	5.84 (1.02;0.84)
Collapsed	β_0	-3.88 (0.29;0.21)	-2.95 (0.24;0.17)	-5.72 (0.51;0.34)
	β_d	5.51 (0.60;0.40)	3.55 (0.42;0.24)	8.77 (1.05;0.64)
		Exchangeable Working Assumptions		
External	β_0	-4.72 (0.43;0.45)	-3.02 (0.25;0.32)	-5.28 (0.47;0.53)
	β_d	4.75 (0.61;0.64)	3.55 (0.41;0.45)	7.13 (0.78;0.78)
	ρ	0.13	0.25	0.26
Visceral	β_0	-4.88 (0.36;0.39)	-5.59 (0.69;0.72)	-7.49 (0.74;0.74)
	β_d	5.20 (0.54;0.57)	3.09 (0.79;0.83)	8.24 (0.85;0.81)
	ρ	0.06	0.06	0.06
Skeletal	β_0	-4.40 (0.35;0.34)	-7.57 (1.06;1.27)	-7.28 (0.86;0.80)
	β_d	4.46 (0.58;0.52)	4.44 (1.14;1.36)	5.87 (1.01;0.87)
	ρ	0.07	0.004	0.008
Collapsed	β_0	-3.94 (0.30;0.30)	-2.89 (0.24;0.31)	-5.76 (0.50;0.45)
	β_d	5.76 (0.60;0.55)	3.45 (0.41;0.44)	8.96 (0.99;0.91)
	ρ	0.10	0.25	0.11

- Results are reasonably in agreement when comparing both sets of working assumptions:
  - ▷ For parameter estimates
  - ▷ For empirically-corrected standard errors
- They are dissimilar when the model-based standard errors are compared. This confirms that such standard errors should not be trusted (and usually they do not need to be reported!)

14.12 The Beta-binomial Model

Random effects on the probability scale

- Useful for clustered data, e.g., NTP data, with covariates at cluster level only.
- Cluster i consists of:
  - ▷ n_i littermates
  - ▷ z_i of which are successes
  - ▷ the dose level is d_i
- Assume each cluster has random success probability π_i.

- Intracluster correlation is assumed to arise from natural heterogeneity in the success probability across litters.

- In contrast, marginal models specify marginal means and association separately.

- Building blocks:
  - ▷ **The beta part:** The π_i are drawn from a beta distribution

$$\pi_i \sim \frac{p^{\alpha_i - 1}(1-p)^{\beta_i - 1}}{B(\alpha_i, \beta_i)}, \qquad 0 \leq p \leq 1,$$

    where $B(.,.)$ denotes the beta function.

  - ▷ **The binomial part:** Conditional on the success probability π_i in cluster i, the responses $Y_{i1}, \ldots, Y_{in_i}$ are independent with common probability π_i. The marginal mean μ_i, with probability P_i follows from:

$$\mu_i = n_i P_i = n_i \frac{\alpha_i}{\alpha_i + \beta_i}$$

- Further, variance and within-cluster correlation are:

$$\sigma_i^2 = n_i P_i (1 - P_i) \frac{1 + n_i \theta_i}{1 + \theta_i}$$

$$\rho_i = \frac{1}{\alpha_i + \beta_i + 1}$$

with

$$\theta_i = \frac{1}{(\alpha_i + \beta_i)}$$

- The moments can be written as:

$$E(Z_i) = n_i P_i$$

$$\text{Var}(Z_i) = n_i P_i (1 - P_i)[1 + (n_i - 1)\rho_i]$$

where indeed

$$\rho_i = \frac{\theta_i}{1+\theta_i}$$

- The marginal distribution of Z_i is then beta-binomial with

$$f(z_i \mid P_i, \rho) = \frac{B(P_i(\rho^{-1}-1)+z_i, (1-P_i)(\rho^{-1}-1)+(n_i-z_i))}{B(P_i(\rho^{-1}-1), (1-P_i)(\rho^{-1}-1))}$$

- Williams (1975)

- Likelihood contribution for cluster i:

$$\ln f(z_i|P_i, \rho_i, n_i) \equiv \ell_i$$

can be written as

$$\ell_i = \sum_{r=0}^{z_i-1} \ln\left(P_i + \frac{r\rho_i}{1-\rho_i}\right) + \sum_{r=0}^{n_i-z_i-1} \ln\left(1 - P_i + \frac{r\rho_i}{1-\rho_i}\right) - \sum_{r=0}^{n_i-1} \ln\left(1 + \frac{r\rho_i}{1-\rho_i}\right)$$

- Thus, BB model is a random-effects model but parameters have a maginal interpretation as well:

  ▷ Shared with the linear mixed model

  ▷ This is *different* in generalized linear mixed models!

14.12.1 The NTP Data

Maximum Likelihood Estimates (Standard Errors) for the Beta-Binomial Model.

Outcome	Parameter	DEHP	EG	DYME
External	β_0	-4.91(0.42)	-5.32(0.71)	-7.27(0.74)
	β_d	5.20(0.59)	2.78(0.81)	8.01(0.82)
	β_a	0.21(0.09)	0.28(0.14)	0.21(0.12)
Visceral	β_0	-4.38(0.36)	-7.45(1.17)	-6.21(0.83)
	β_d	4.42(0.54)	4.33(1.26)	4.94(0.90)
	β_a	0.22(0.09)	0.04(0.09)	0.45(0.21)
Skeletal	β_0	-4.88(0.44)	-2.89(0.27)	-5.15(0.47)
	β_d	4.92(0.63)	3.42(0.40)	6.99(0.71)
	β_a	0.27(0.11)	0.54(0.09)	0.61(0.14)
Collapsed	β_0	-3.83(0.31)	-2.51(0.09)	-5.42(0.45)
	β_d	5.59(0.56)	3.05(0.17)	8.29(0.79)
	β_a	0.32(0.10)	0.28(0.02)	0.33(0.10)

14.13 Generalized Linear Mixed Models

Random effects in the linear predictor

- $E(Y_{ij}|\boldsymbol{b}_i) = \mu_{ij}$
- $\eta(\mu_{ij}) = \boldsymbol{x}_{ij}^T\boldsymbol{\beta} + \boldsymbol{z}_{ij}^T\boldsymbol{b}_i$
- var$(Y_{ij}|\boldsymbol{b}_i) = \phi v(\mu_{ij})$
- Marginal likelihood contribution:

$$f(\boldsymbol{y}_i) = \int \prod_{j=1}^{n_i} f(y_{ij}|\boldsymbol{b}_i) g(\boldsymbol{b}) d\boldsymbol{b}$$

- Can be fitted using NLMIXED in SAS, based on numerical integration

14.13.1 Applied to the NTP Data

- Using a scalar random effect ("random intercept"):

$$E(Y_{ij}|b_i) = \mu_{ij}$$

- and using a logit link:

$$\text{logit}(\mu_{ij}) = \beta_0 + b_i + \beta_1 d_i$$

or, equivalently

$$\mu_{ij} = \frac{e^{\beta_0 + b_i + \beta_1 d_i}}{1 + e^{\beta_0 + b_i + \beta_1 d_i}}$$

- Distributional assumption for the random effect:

$$b_i \sim N(0, d)$$

- Ideas from linear mixed models and from generalized linear models are combined

- SAS code:

```
proc nlmixed data=help;
title 'DEHP2 - GLMM for collapsed outcome';
theta = beta0 + b0 + beta1*dose;
exptheta = exp(theta);
p = exptheta/(1+exptheta);
model collaps ~ binary(p);
random b0 ~ normal(0,d) subject=litter;
run;
```

- The procedure is different in spirit from classical procedures such as REG, ANOVA, LOGISTIC, GLM, GENMOD, MIXED

- Programming statements are used to construct the probability p
- Variables in the dataset are automatically recognized (collaps, dose)
- The rest are automatically considered to be parameters (beta0, b0, beta1, d)
- Out of these, b0 is recognized to be a random effect, due to the use of the RANDOM statement
- It is possible to define starting values
- A number of numerical-algorithmic options are available

- Relevant output:
  - ▷ Standard book keeping information:

Specifications

Data Set	WORK.HELP
Dependent Variable	collaps
Distribution for Dependent Variable	Binary
Random Effects	b0
Distribution for Random Effects	Normal
Subject Variable	LITTER
Optimization Technique	Dual Quasi-Newton
Integration Method	Adaptive Gaussian Quadrature

Dimensions

Observations Used	1082
Observations Not Used	0
Total Observations	1082
Subjects	108
Max Obs Per Subject	16
Parameters	3
Quadrature Points	5

  - ▷ If no starting values are supplied, the iteration is started using ones. It may be prudent to specify starting values, e.g., from ordinary logistic regression:

Parameters			
beta0	beta1	d	NegLogLike
1	1	1	658.614199

▷ An overview of the iteration history:

Iteration History

Iter	Calls	NegLogLike	Diff	MaxGrad	Slope
1	3	401.098454	257.5157	69.75335	-2039.21
2	5	327.82828	73.27017	20.73818	-81.971
3	7	315.303102	12.52518	17.33259	-14.477
4	9	280.790465	34.51264	6.824566	-12.417
5	11	273.774685	7.01578	2.060826	-6.83407
6	13	273.59353	0.181155	1.63775	-0.19738
7	14	273.404154	0.189376	1.939517	-0.04804
8	16	272.649859	0.754295	2.994915	-0.26405
9	18	272.013393	0.636466	1.151541	-0.75296
10	20	271.991963	0.021429	0.565793	-0.09417
11	22	271.982555	0.009408	0.038957	-0.01939
12	24	271.982511	0.000045	0.019247	-0.00032
13	25	271.982462	0.000048	0.00228	-0.0001
14	27	271.982461	1.003E-6	0.000406	-2.11E-6

▷ Parameter estimates, standard errors,. . . for all three parameters in the model specification:

Parameter Estimates

Parameter	Estimate	Standard Error	DF	t Value	Pr > |t|	Alpha
beta0	-4.9062	0.4204	107	-11.67	<.0001	0.05
beta1	7.3602	0.8279	107	8.89	<.0001	0.05
d	1.7974	0.6251	107	2.88	0.0049	0.05

Parameter	Lower	Upper	Gradient
beta0	-5.7397	-4.0728	-0.00041
beta1	5.7190	9.0013	9.763E-7
d	0.5582	3.0365	-0.00007

Results

Maximum Likelihood Estimates (Standard Errors) for the Generalized Linear Mixed Model.

Outcome	Parameter	DEHP	EG	DYME
External	β_0	-6.05(0.63)	-4.45(0.44)	-8.37(0.94)
	β_d	6.08(0.91)	5.28(0.66)	11.42(1.41)
	d	2.06(0.88)	2.44(0.66)	3.61(1.29)
Visceral	β_0	-5.84(0.56)	-7.28(1.11)	-8.59(0.95)
	β_d	6.35(0.83)	3.82(1.11)	9.56(1.14)
	d	1.30(0.63)	3.05(1.78)	1.37(0.87)
Skeletal	β_0	-5.27(0.48)	-7.85(1.40)	-8.73(1.20)
	β_d	5.42(0.75)	4.53(1.39)	6.88(1.26)
	d	1.47(0.64)	0.47(1.04)	2.77(1.65)
Collapsed	β_0	-4.91(0.42)	-4.35(0.44)	-7.23(0.66)
	β_d	7.36(0.83)	5.34(0.66)	11.12(1.13)
	d	1.80(0.63)	2.47(0.66)	1.87(0.71)

14.14 Overview of Methods

- NTP: External outcome in DEHP study
  - ▷ β_0: intercept
  - ▷ β_d: dose effect
  - ▷ β_a: association parameters (Fisher's z transformed correlation)
  - ▷ ρ: correlation

- Summary of parameter estimates

Model	β_0	β_d	Association	
Conditional Model Approach				
Quadratic loglinear	-2.81(0.58)	3.07(0.65)	LOG OR	0.18(0.04)
Marginal Model Approaches				
Lik. Bahadur	-4.93(0.39)	5.15(0.56)	β_a	0.11(0.03)
GEE (exch)	-4.98(0.37)	5.33(0.55)	ρ	0.11
GEE (ind)	-5.06(0.38)	5.31(0.57)		
Random-effects Model Approaches				
Beta-binomial	-4.91(0.42)	5.20(0.59)	β_a	0.21(0.09)
GLMM	-6.05(0.63)	6.08(0.91)		

- General observation for parameters estimates in absolute value:

$$\text{conditional} < \text{marginal} < \text{mixed model}$$

- Three expectations:

  ▷ **Conditional:** $E(Y_{ij}|d_i, y_{ij'} = 0, j \neq j')$

  Success probability as function of dose, and given all other littermates exhibit a failure. This is more difficult to achieve than a marginal success, and hence the parameters are smaller.

  ▷ **Marginal:** $E(Y_{ij}|d_i)$

  Success probability as function of dose.

  ▷ **Random-effects:** $E(Y_{ij}|d_i, b_i)$

  Success probability as function of dose, and given the level of the random effects remain the same. This one operates within the fine slice of littermates that share the same random-effects level, and hence this probability is higher.

- In summary:

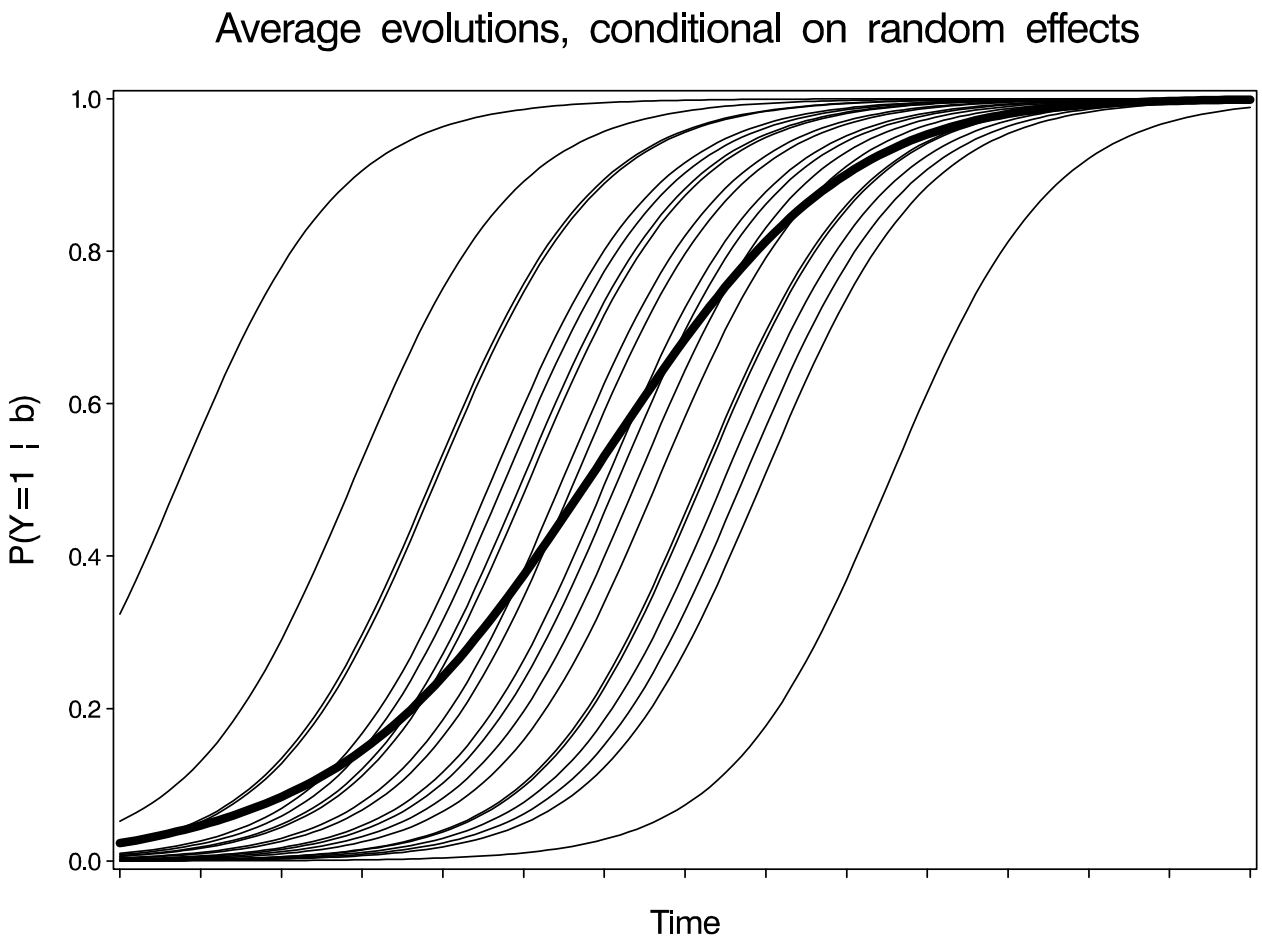
$$
\begin{array}{ccc}
E(Y_{ij}|d_i) & & \\
\neq & & \neq \\
E(Y_{ij}|d_i, y_{ij'} = 0, j \neq j') & \neq & E(Y_{ij}|d_i, b_i)
\end{array}
$$

- For example, the marginal expectation derived from the random-effects model:

$$
E(Y_{ij}|d_i) = E[E(Y_{ij}|d_i, b_i)] = E\left[\frac{\exp(\beta_0 + b_i + \beta_1 d_i)}{1 + \exp(\beta_0 + b_i + \beta_1 d_i)}\right]
$$

$$
\neq \frac{\exp(\beta_0 + \beta_1 d_i)}{1 + \exp(\beta_0 + \beta_1 d_i)}
$$

- Graphical representation:



Chapter 15
Spatial or Serial Structure

- Spatially organized data
- An extended linear mixed-effects model
- Intraclass correlation

15.1 Spatially Organized Data: The Heat Shock Study

- Effects of temperature and duration of exposure on viable foetuses:
  - ▷ embryos are explanted from the uterus of maternal dam
  - ▷ cultured in vitro
  - ▷ each embryo exposed to short period of heat stress:
    - ∗ 3 to 5C
    - ∗ 5 to 60 minutes
  - ▷ these are *individual-level* covariates
  - ▷ examined 24 hours later for impaired/accelerated development

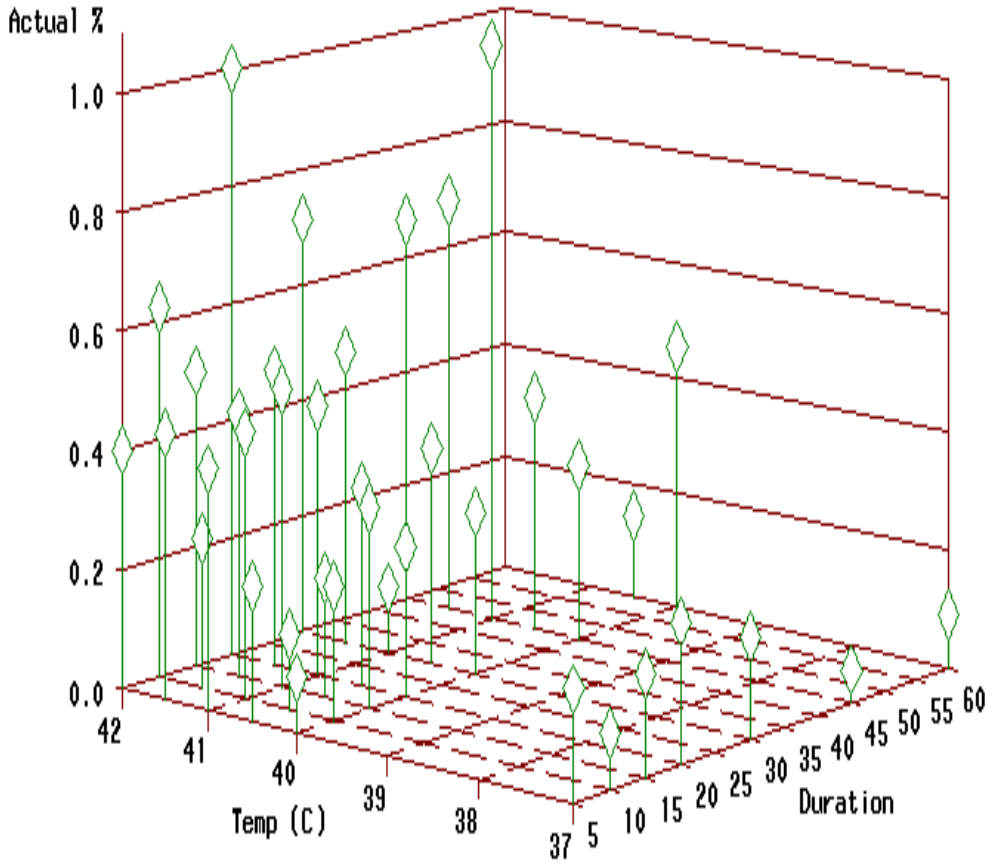
- Alternative Segment II study
- Advantages:
  - ▷ Exposure directly administered to the embryo
  - ▷ Exposure patterns easy to control
  - ▷ Effects of exposure are quickly obtained $\leftrightarrow$ more than a week in the standard Segment II design
  - ▷ Mechanism to assess the joint effects of **duration of exposure** & **exposure level**

15.1.1 Design and Modelling

- 327 embryos:
  - ▷ 50 did not survive heat stress
  - ▷ 277 embryos with complete data
- Haber's Law: duration×exposure
- Continuous outcomes:
  - ▷ YSD: yolk sack
  - ▷ CRL: crown rump length
  - ▷ HEAD: head circumference

Distribution of Cluster Sizes for the Heat Shock Data

cluster size n_i	1	2	3	4	5	6	7	8	9	10	11
number of clusters of size n_i	6	3	6	12	13	11	8	5	2	3	2



Study Design for Heat Shock Experiments: Number of Viable Embryos

Temperature	Duration of Exposure							Total
	5	10	15	20	30	45	60	
37.0	11	11	12	13	12	18	11	88
40.0	11	9	9	8	11	10	11	69
40.5	9	8	10	9	11	10	7	64
41.0	10	9	10	11	9	6	0	55
41.5	9	8	9	10	10	7	0	53
42.0	10	8	10	5	7	6	0	46
Total	60	53	60	56	60	57	29	375

15.1.2 Haber's Law

Probability on adverse effect equal for each duration $\times$ temperature combination

$$P(d_{ij}, t_{ij}) = F(\beta_0 + \beta_1 t_{ij}(\delta_d)_{ij} + \beta_2 d_{ij} t_{ij})$$

$$(\delta_d)_{ij} = \begin{cases} 1 & \text{if } d_{ij} > 37 \\ 0 & \text{otherwise} \end{cases}$$

- $(dt)_{ij} \equiv d_{ij} t_{ij}$ referred to as "durtemp"
- $(pd)_{ij} \equiv t_{ij}(\delta_d)_{ij} = t_{ij} \cdot I(d_{ij} > 37)$ referred to as "posdur"
- Implication of Haber's Law: $\beta_1 = 0$

15.1.3 An Extended Mixed-effects Model

- We will formulate a linear mixed-effects model that accounts for:

  - ▷ The effects of durtemp and posdur in the mean structure
  - ▷ Random (litter-specific) intercepts and slopes in durtemp
  - ▷ The effect of durtemp on the association structure:

    - ∗ The random effects influence the cluster as a whole
    - ∗ The measurement error act at the level of the individual fetus
    - ∗ The correlation between two foetuses may be affected by the "distance" between their values for durtemp. For example, the correlation might increase when their durtemp values are closer to each other.

  - ▷ The latter is a **spatial** effect, influenced by the distance between two littermates on the duration by temperature grid. Here we assume, for simplicity and consistent with Haber's law, that the influence is only through the product.

- This model can be formalized as follows:

$$Y_{ij} = (\beta_1 + b_{i1}) + (\beta_2 + b_{i2})(dt)_{ij}$$

$$+ \beta_3(pd)_{ij} + \varepsilon_{(1)ij} + \varepsilon_{(2)ij}$$

$$\boldsymbol{b_i} = \begin{pmatrix} b_{i1} \\ b_{i2} \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, D = \begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix}\right]$$

$$\boldsymbol{\varepsilon}_{(1)i} \sim N(\mathbf{0}, \sigma^2 I_{n_i})$$

$$\boldsymbol{\varepsilon}_{(2)i} \sim N(\mathbf{0}, \tau^2 H_i)$$

- The random effects $\boldsymbol{b_i}$ and the measurement errors $\boldsymbol{\varepsilon}_{(1)i}$ are in line with what we have seen before in the context of linear mixed models.

- In addition, the so-called serial or spatial components expresses the component of variability of which the correlation decays as the "durtemp lag" between two measurements decreases.

- In general, its elements can be expressed as

$$h_{ijk} = f[(dt)_{ij}, (dt)_{ik}]$$

- Two specific forms are:

  ▷ **Gaussian decay**: $h_{ijk} = \exp\left\{-\phi[(dt)_{ij} - (dt)_{ik}]^2\right\}$

  ▷ **exponential decay**: $h_{ijk} = \exp\left\{-\phi[(dt)_{ij} - (dt)_{ik}]^1\right\}$

- ▷ Note that both of these functions are 1 when the durtemp lag is zero and decrease to 0 as the durtemp lag approaches $+\infty$.

- ▷ Measurement error captures the variability arising from the fact that when two measurements are taken within the same unit (litter), at exactly the same 'spatial location' (durtemp level), these would still be different. In our case this is because we might be measuring two different foetuses. Even when the same fetus is re-measured we may find a difference in response, because of natural variability or because of limitations of the measuring equipment. This may be the case for something like plasma levels in blood, but less so for weight rounded to kilograms.

- Let us apply this model to CRL (crown rump length), using the SAS procedure MIXED.

- First, the data will be read from an input text file:

```
data m.heat4;
infile 'c:\bartsas\gent\heat4.asc';
input id tempc duration wec dam vial heartbt yolkcirc
      ysd crl head pros rsomite rdiv lsomite ldiv ysc
      ols tor hrt hbn mbn fbn otc opt olf brb max frl;run;
```

- The unique ID of a cluster is given by the combination of WEC and DAM. It is no problem to use two (nested) identifiers, but they have to be properly ordered:

```
proc sort data=m.heat4;
by wec dam;
run;
```

- As can be seen from the variables present in the input dataset, DURTEMP and POSDUR, being compound variables, are not present and need to be created. Out of caution, they are written to a new permanent dataset. On the side, a single cluster identifier IDNR is created. This could be used as an alternative to the WEC and DAM combination.

```
data m.heat41;
set m.heat4;
idnr=(wec+dam/10)*10;
durtemp=duration*(tempc-37);
posdur=0;
if tempc>37 then posdur=duration;
run;
```

- It is cautious to ensure the ranges for both outcome variables and covariates are not to extreme (very large or very small). While mathematically and statistically not a problem, there could be numerical problems when PROC MIXED is called. As a rule of thumb, it is good practice to ensure that relevant variables have magnitudes and ranges of the order of unity:

```
data m.heat42;
set m.heat41;
durtemp2=durtemp/225;
posdur2=posdur/60;
run;
```

- Another sort operation, so that durtemp is ordered within every unit:

```
proc sort data=m.heat42;
by wec dam durtemp;
run;
```

- Our model, formulated above, can now be fitted using the following PROC MIXED code:

```
proc mixed data=m.heat42 method=ml covtest nobound;
title 'outcome=crl - initial model';
class wec dam;
model crl=durtemp2 posdur / solution ddfm=satterth;
random intercept durtemp2/ subject=wec(dam) solution type=un g v;
repeated / subject=wec(dam) local type=sp(gau)(durtemp) r;
parms (0.01) (0.07) (-0.03) (0.04) (4.26) (0.09);
run;
```

- the 'type=sp(·)(·)' structure has two arguments:
  - ▷ The first argument is the type of spatial decay, e.g., GAU for Gaussian or EXP for exponential.
  - ▷ The second argument is the coordinate system (durtemp, time, x and y coordinates, x and y and z coordinates).

- The various model parameters are produced by the MODEL, RANDOM, and REPEATED statements and the LOCAL option:

Effects	Parameters	Statements / options
Fixed effects	$\beta_1, \beta_2, \beta_3$	'solution' within MODEL
Random-effects var.-cov.	D	RANDOM
Random effects	$(b_{i1}, b_{i2})_{i=1,\ldots,N}$	'solution' within RANDOM
Measurement error variance	σ^2	'local' within REPEATED
Serial/spatial	ϕ, τ^2	TYPE=sp(·)(durtemp)

- The random effects are assumed to be correlated, and random intercepts and random slopes are allowed to have a different variance. This means 'type=un' is used for the D matrix.

- Without the 'local' option, no separate measurement error is allowed for, and all residual variance structure (i.e., everything that is not captured by random effects) is confined to the serial structure.

- Since fitting all three components of variability together is a tricky job, starting values are provided using the PARMS statement. The result of evaluating the objective function at the initial parameters is shown in the following panel:

```
                    Parameter Search

   CovP1     CovP2     CovP3     CovP4     CovP5     CovP6  Variance

 0.01000   0.07000  -0.03000   0.04000    4.2600   0.09000   0.09677

    Parameter Search

  Log Like    -2 Log Like

 -184.0070       368.0140
```

- The 'covtest' option in the PROC MIXED statements requests standard errors and p values associated with the variance components. Under a hierarchical view, these should be approached with caution. Under a purely marginal view, they could be of use.

Solution for Fixed Effects

Effect	Estimate	Standard Error	DF	t Value	Pr > |t|
beta1 Intercept	3.6223	0.03398	48.5	106.61	<.0001
beta2 DURTEMP2	-1.5581	0.3758	45	-4.15	0.0001
beta3 POSDUR	0.01916	0.006212	55.9	3.08	0.0032

Covariance Parameter Estimates

Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr Z
d11 UN(1,1)	WEC(DAM)	0.01044	0.01426	0.73	0.4643
d12 UN(2,1)	WEC(DAM)	0.07113	0.03154	2.25	0.0241
d22 UN(2,2)	WEC(DAM)	-0.03764	0.06489	-0.58	0.5619
tau2 Variance	WEC(DAM)	0.04402	0.01690	2.61	0.0092
phi SP(GAU)	WEC(DAM)	4.2681	5.0523	0.84	0.1991
sigma2 Residual		0.09684	0.01417	6.83	<.0001

- The 'nobound' option in the PROC MIXED statement allows D to be fully general, and not restricted to have positive variances or to be positive definite. This implies we take a marginal view on the model. The variance of the random slopes is indeed seen to be negative in this analysis.

- A portion of the 'solutions for random effects' panel (empirical Bayes estimates):

Solution for Random Effects

Effect	WEC	DAM	Estimate	Std Err Pred	DF	t Value	Pr > |t|
Intercept	167	1	-0.07334	0	371	-Infty	<.0001
DURTEMP2	167	1	-0.4352	0	371	-Infty	<.0001
Intercept	168	1	0.07575	0.07509	374	1.01	0.3137
DURTEMP2	168	1	0.2668	0	371	Infty	<.0001
Intercept	169	1	-0.03511	0	371	-Infty	<.0001
DURTEMP2	169	1	0.03764	0	371	Infty	<.0001
Intercept	171	1	-0.04560	0.06330	374	-0.72	0.4718
DURTEMP2	171	1	-0.07370	0	371	-Infty	<.0001
Intercept	172	1	0.1433	0	371	Infty	<.0001
...							

One should not be surprised about these 'nonsensical' results: a fully marginal view has been adopted, with D not positive-definite in this case. Hence, the conditions needed to apply empirical Bayes estimation are **not** fulfilled.

- Three variance-covariance matrices are requested:
  - ▷ The random-effects covariance matrix: $G = D$

Estimated G Matrix

Row	Effect	WEC	DAM	Col1	Col2
1	Intercept	167	1	0.01044	0.07113
2	DURTEMP2	167	1	0.07113	-0.03764

▷ The residual covariance matrix:

$$R = \Sigma_i = \sigma^2 I_{n_i} + \tau^2 H_i$$

Estimated R Matrix for WEC(DAM) 167 1

Row	Col1	Col2	Col3	Col4	Col5	Col6
1	0.1409	0.04402				
2	0.04402	0.1409				
3			0.1409	0.01116	0.01116	
4			0.01116	0.1409	0.04402	2.2E-9
5			0.01116	0.04402	0.1409	2.2E-9
6				2.2E-9	2.2E-9	0.1409
7				2.2E-9	2.2E-9	0.04402
8						0.002007
9						0.002007
10						
11						

Row	Col7	Col8	Col9	Col10	Col11
1					
2					
3					
4	2.2E-9				
5	2.2E-9				
6	0.04402	0.002007	0.002007		
7	0.1409	0.002007	0.002007		
8	0.002007	0.1409	0.04402	1.903E-7	1.903E-7
9	0.002007	0.04402	0.1409	1.903E-7	1.903E-7
10		1.903E-7	1.903E-7	0.1409	0.04402
11		1.903E-7	1.903E-7	0.04402	0.1409

▷ The overall covariance matrix:

$$V = Z_i D Z_i' + \Sigma_i = Z_i D Z_i' + \sigma^2 I_{n_i} + \tau^2 H_i$$

Estimated V Matrix for WEC(DAM) 167 1

Row	Col1	Col2	Col3	Col4	Col5	Col6
1	0.1513	0.05445	0.02466	0.02624	0.02624	0.03178
2	0.05445	0.1513	0.02466	0.02624	0.02624	0.03178
3	0.02466	0.02466	0.1782	0.04995	0.04995	0.04374
4	0.02624	0.02624	0.04995	0.1810	0.08421	0.04507
5	0.02624	0.02624	0.04995	0.08421	0.1810	0.04507
6	0.03178	0.03178	0.04374	0.04507	0.04507	0.1906
7	0.03178	0.03178	0.04374	0.04507	0.04507	0.09374
8	0.03415	0.03415	0.04586	0.04716	0.04716	0.05373
9	0.03415	0.03415	0.04586	0.04716	0.04716	0.05373
10	0.03889	0.03889	0.05010	0.05135	0.05135	0.05571
11	0.03889	0.03889	0.05010	0.05135	0.05135	0.05571

Row	Col7	Col8	Col9	Col10	Col11
1	0.03178	0.03415	0.03415	0.03889	0.03889
2	0.03178	0.03415	0.03415	0.03889	0.03889
3	0.04374	0.04586	0.04586	0.05010	0.05010
4	0.04507	0.04716	0.04716	0.05135	0.05135
5	0.04507	0.04716	0.04716	0.05135	0.05135
6	0.09374	0.05373	0.05373	0.05571	0.05571
7	0.1906	0.05373	0.05373	0.05571	0.05571
8	0.05373	0.1945	0.09769	0.05758	0.05758
9	0.05373	0.09769	0.1945	0.05758	0.05758
10	0.05571	0.05758	0.05758	0.2022	0.1053
11	0.05571	0.05758	0.05758	0.1053	0.2022

- A version of the program where the random intercept and the random slope are uncorrelated, would be found by adapting 'type=' option in the RANDOM statement:

```
random intercept durtemp2/ subject=wec(dam) g type=simple v;
```

- When the serial/spatial structure is removed, and only the random-effects and measurement error is retained, the REPEATED statement changes to:

```
repeated / subject=wec(dam)  r;
```

  Note that the 'local' option is not needed. Since no 'type=' is specified, 'type=simple' is fitted by default. This is equivalent to pure measurement error and hence 'local' is unnecessary.

- Model simplification, using likelihood ratio test statistics, leads to a model with all three variance components, but no random slopes:

```
proc mixed data=m.heat42 method=ml covtest nobound;
title 'outcome=crl - final model';
class wec dam;
model crl=durtemp2  / solution ddfm=satterth;
random intercept / subject=wec(dam) g type=simple v;
repeated / subject=wec(dam) local type=sp(gau)(durtemp) r;
parms 0.032 0.043 2.015 0.094;
run;
```

- The panel describing the covariance parameter estimates has now changed to:

Covariance Parameter Estimates

	Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr Z
d11	Intercept	WEC(DAM)	0.04298	0.01353	3.18	0.0015
tau2	Variance	WEC(DAM)	0.04569	0.01758	2.60	0.0094
phi	SP(GAU)	WEC(DAM)	3.8094	3.5565	1.07	0.1421
sigma2	Residual		0.09764	0.01464	6.67	<.0001

The elements d_{12} and d_{22}, corresponding to the random slopes, have gone. The remaining variance is nonnegative and hence a hierarchical interpretation is meaningful *provided one is prepared to put belief in a hierarchical structure of the model*.

- This is reflected in the empirical Bayes estimates:

Solution for Random Effects

Effect	WEC	DAM	Estimate	Std Err Pred	DF	t Value	Pr > |t|
Intercept	167	1	-0.2439	0.1131	107	-2.16	0.0334
Intercept	168	1	0.1224	0.1459	83.6	0.84	0.4041
Intercept	169	1	-0.03830	0.1173	147	-0.33	0.7445
Intercept	171	1	-0.06329	0.1820	35.1	-0.35	0.7301
Intercept	172	1	0.08645	0.1410	95.5	0.61	0.5414
...							

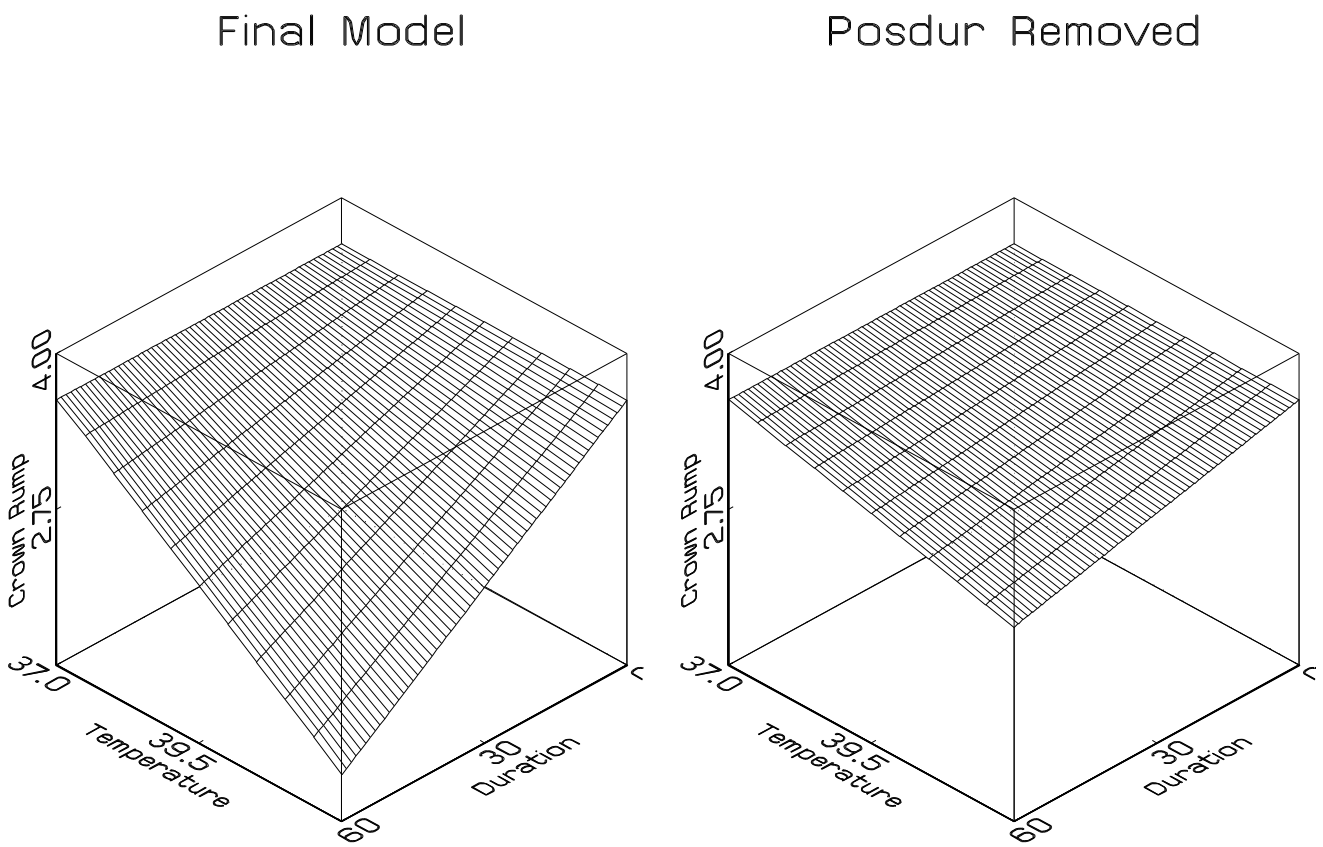
Initial and Final Models

Effect	Parameter	Initial	Final
Fixed effects:			
Intercept	β_1	3.622 (0.034)	3.627 (0.042)
Durtemp $(dt)_{ij}$	β_2	-1.558 (0.376)	-1.331 (0.353)
Posdur $(pd)_{ij}$	β_3	0.019 (0.006)	0.015 (0.006)
Random-effects parameters:			
var(b_{1i})	d_{11}	0.010 (0.014)	0.046 (0.014)
var(b_{12})	d_{12}	-0.038 (0.065)	
cov(b_{1i}, b_{2i})	d_{22}	0.071 (0.032)	
Residual variance parameters:			
var$(\varepsilon_{(1)ij})$	σ^2	0.097 (0.014)	0.097 (0.014)
var$(\varepsilon_{(2)ij})$	τ^2	0.044 (0.017)	0.042 (0.017)
Spatial corr. parameter	ϕ	4.268 (5.052)	4.143 (3.772)

- About 25% of the variability can be ascribed to the random intercepts
- About 25% can be ascribed to the serial process
- The remaining 50% is due to measurement error

15.1.4 Graphical Representation

Fixed-effects structure for (a) the final model and (b) a model with *posdur* removed.



15.2 Intraclass Correlation

- Calculating the intraclass correlation is not straightforward due to the presence of serial correlation.

- One solution is to remove this variance component and estimate the intraclass correlation in a straightforward way:

$$\frac{d}{\sigma^2 + d}$$

- This solution is not satisfactory since it is based upon an inadequate model.

- Progress is possible based on the following extension:

  ▷ Assume first two littermates have exposures infinitely far apart. In that case, the serial process disappears from their correlation, which can then be expressed as:

$$\begin{aligned} IC(\infty) &= \frac{d}{\sigma^2 + \tau^2 + d} \\ &= \frac{0.0456}{0.0972 + 0.0416 + 0.0456} = 0.247 \end{aligned}$$

  ▷ For two littermates at the same level of exposure, we obtain:

$$\begin{aligned} IC(0) &= \frac{\tau^2 + d}{\sigma^2 + \tau^2 + d} \\ &= \frac{0.0416 + 0.0456}{0.0972 + 0.0416 + 0.0456} = 0.473 \end{aligned}$$

▷ In general, we can calculate:

$$IC(\gamma) = \frac{\tau^2 e^{-\phi\gamma^2} + d}{\sigma^2 + \tau^2 + d}$$

where γ is the distance between two levels of durtemp exposure.

▷ This expression leads to the fitted correlation function:

